BRIDGING THE DIGITAL GAP

Evidence from a Multi-Country Pilot on Supporting MSME Digitalization in the EU

WORLD BANK GROUP

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ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interfaces
ATE	Average Treatment Effect
BSMEPA	Bulgarian Small and Medium Enterprise Promotion Agency
CEE	Central and Eastern Europe
CIS	Community Innovation Survey
CRM	Customer Relationship Management
DESI	Digital Economy and Society Index
DG REGIO	Directorate-General for Regional and Urban Policy
EDIH	European Digital Innovation Hub
ERP	Enterprise Resource Planning
EU	European Union
FAT	Firm-level Technology Adoption
FCI	Finance, Competitiveness, and Investment
GBF	General Business Function
ICT	Information and Communication Technology
MDE	Minimum Detectable Impact
MIRRI	Ministry of Investments, Regional Development and Informatization
MSME	Micro, Small and Medium Enterprises
PARP	Polish Agency for Enterprise Development
RCT	Randomized Control Trial
RPA	Robotic Process Automation
SaaS	Software as a Service
SRM	Supplier Relationship Management
WMS	World Management Survey
WBG	World Bank Group

EXECUTIVE SUMMARY

The Digitrans pilot demonstrates that providing MSMEs with technical assistance alongside financial subsidies yields better digital transformation outcomes than financial support alone. This novel approach tested across Bulgaria, Poland, and Slovakia produced evidence that complementary capabilities at firm level–not just funding or technology access–are a critical constraint for small business digitalization. Firms equipped with both financial resources and capability-building support achieved significantly higher rates of digital tool adoption and usage, and implemented operational changes almost twice as often as those receiving only financial subsidies. These findings challenge conventional support programs, widely implemented across the EU and in other regions, that primarily address financial barriers while overlooking the managerial and organizational constraints that prevent firms from effectively implementing and utilizing digital technologies.

The three-country pilot rigorously evaluated different support approaches using a randomized controlled trial methodology with 629 participating MSMEs. Firms were allocated into three groups: those receiving financial subsidies (€2,000), those receiving both financial subsidies and technical assistance (10-week program with assessment, training, and advisory services), and a comparison group receiving only benchmarking reports. This design enabled measurement of both isolated financial subsidy effects and the additional impact of technical assistance through a standardized implementation process. The consistent results across all three countries strengthen external validity of the findings throughout the EU, where MSMEs face similar digitalization challenges.

Key Findings

- **Firms Invest More When Support Includes Technical Assistance**: Firms receiving complementary support invested additional funds beyond the voucher amount. In contrast, firms receiving only financial support typically limited spending to the subsidy value.
- **Technical Assistance Is Key to Adoption**: Firms receiving combined significantly expanded sophistication of digital tools they had adopted. In contrast, firms with financial subsidies alone showed no meaningful increase in the advancement of tools available to them.
- **Digital Tool Usage Is 2.5x Higher with Combined Support**: Firms that received both types of support increased their digital tool usage intensity more than twice as much as those with financial subsidies alone.

- **Managerial Changes Are More Common with Technical Assistance**: Firms in the combined support group were twice as likely to introduce internal organizational and managerial changes–key enablers for effective digitalization.

- **Combined Support Leads to Deeper Use of Digital Tools**: Firms with technical assistance made better use of advanced software functions–82% used CRM systems for client communication (vs. 67% in the subsidy-only group), and 56% used automation features in ERP systems (vs. 42%).

- **Support Effectiveness Varies by Firm Type**: The benefits of combined support were especially strong among smaller firms, those with low initial digital capabilities, service-sector firms, and women-led businesses.

The pilot's results provide evidence that effective digitalization support for MSMEs needs to go beyond subsidizing technology purchases. Programs that combine financial support with targeted technical assistance help firms achieve deeper digital transformation by ensuring that digital investments translate into measurable business improvements. Firms that received technical assistance made higher-quality investment choices, selected more sophisticated tools, and integrated digital solutions into their everyday business operations. In comparison, firms receiving only financial subsidies often limited their investments to the subsidy amount and made less progress in capabilities to use them effectively. These results point to the need for program designs that pair funding with capacity-building support to unlock the full potential of digital investments.

Targeting support based on firm characteristics has potential for driving intervention effectiveness. The pilot revealed that the impact of digitalization support varied significantly across firms, depending on structural features (such as size, sector, and initial digital readiness) and owner attributes (including confidence in digital skills, locus of control, and gender). These differences suggest that a one-size-fits-all model is inefficient. Instead, segmenting firms and adjusting the type and intensity of support accordingly would likely improve both impact and cost-effectiveness. For program design, this means designing interventions that respond to firm profiles–rather than applying uniform support–can better target resources and maximize returns.

The pilot implementation process generated operational insights for designing scalable, effective public support instruments for enhancing firms' digitalization. A simplified, low-barrier application process attracted over 1,500 applicants in just two months, with targeted email outreach and social media proving the most cost-effective recruitment channels. The delivery model–combining automated benchmarking, standardized but adaptable technical assistance, and streamlined financial subsidy administration–ensured high engagement: 90% of participating firms received reimbursements, and over 75% completed the technical assistance. These operational components were tested and validated through the pilot, demonstrating their feasibility

within a public support instrument. As such, they offer a proven foundation that can be integrated into various interventions aiming to support private sector development.

The report organizes findings from the pilot instrument implementation into five chapters. Chapter 1 explains why and how the team designed this instrument, building on adapting methodologies from promising instruments to needs of firms in Bulgaria, Poland, and Slovakia. Chapter 2 describes firm outreach and selection. Chapter 3 analyzes the characteristics of participating firms, discussing similarities and differences between pilot countries. Chapter 4 details how respective components of support were delivered. The final chapter measures preliminary impacts on firm productivity and employment, using data from approx. 40 pilot participants in Slovakia.

In all chapters, lessons learned at this stage are highlighted in dedicated boxes. These insights can be leveraged not only for public support instruments promoting the digitalization of MSMEs but also for a broader range of policies aimed at supporting private sector development.

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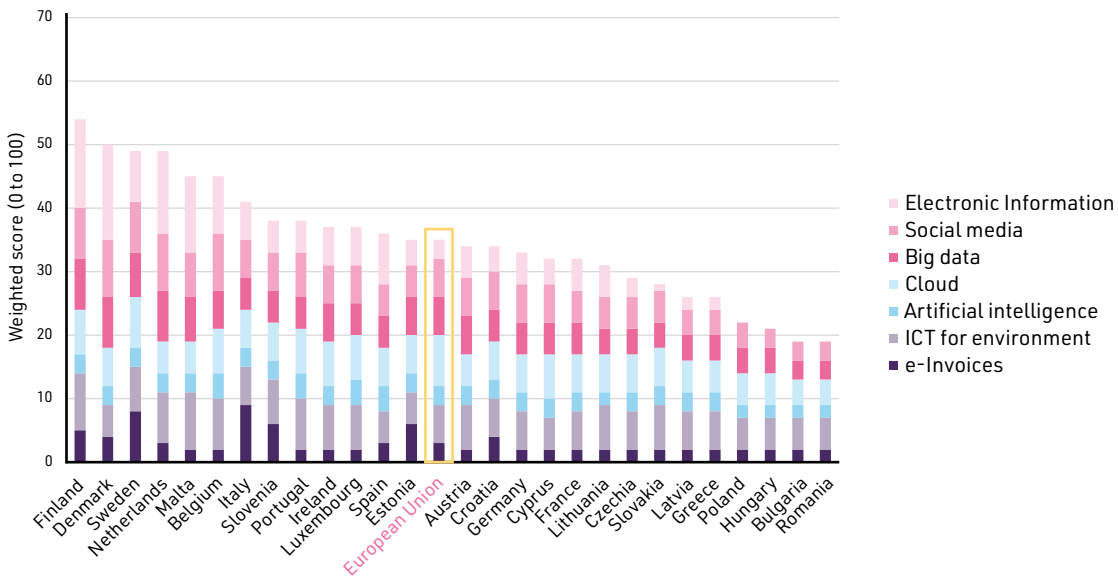
INTRODUCTION AND PROGRAM OVERVIEW

The pilot instrument addresses digitalization challenges of MSMEs in the European Union through a novel approach combining capability building with digital transformation support. The pilot design emerged from extensive analysis of digitalization gaps among EU small firms, which revealed similar patterns of digital adoption barriers across member states. Research findings guided the development of intervention components, focusing particularly on managerial capabilities that were identified as critical drivers of adoption and usage of digital tools. The design process incorporated evidence from promising global digitalization programs, adapting their elements to the EU context. These insights formed the basis for the program's core mechanism of action, which hypothesizes that addressing managerial capability gaps alongside technical support will yield better digitalization outcomes than traditional approaches focused solely on technology adoption.

1.1 POLICY CONTEXT AND RATIONALE

Small businesses in Central and Eastern Europe show limited adoption of digital tools beyond those required by regulations or provided by the financial sector. The 2022 Digital Economy and Society Index places Bulgaria, Poland, and Slovakia in the bottom quartile of EU members, with low scores in business digitalization components (Figure 1.1). Firms in these countries maintain manual processes across most business functions, with digital tools primarily used only where mandated by fiscal regulations or enabled by the banking sector's digital infrastructure.[1] Analysis of technology usage patterns shows that internally driven adoption of digital solutions remains limited, with most MSMEs not advancing into digitalization of areas requiring proactive implementation decisions (Figure 1.2). These functions play a critical role in driving competitiveness and improving productivity, as evidenced by higher-performing firms that actively use digital tools.

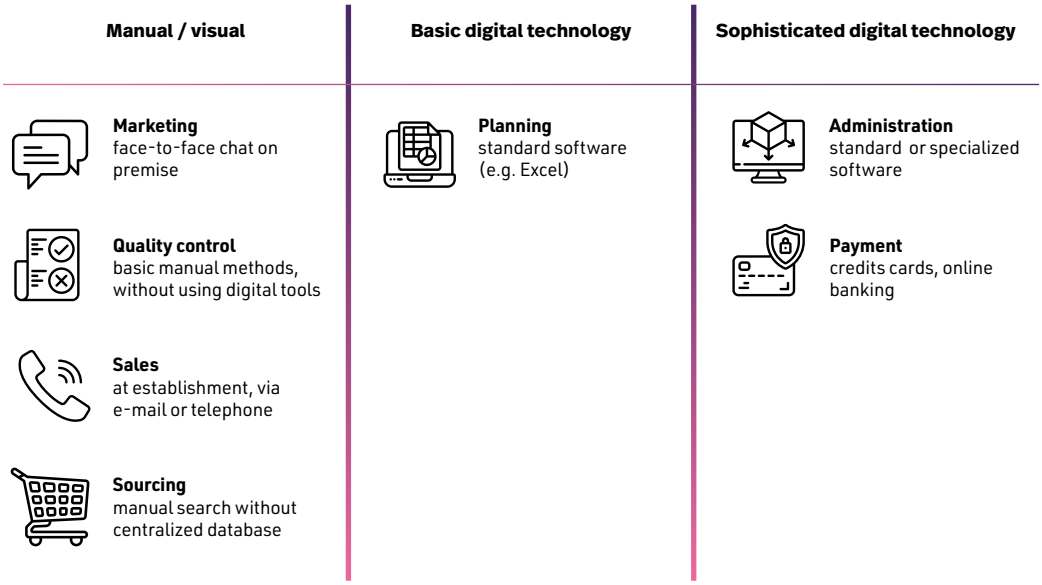
FIGURE 1.1 Digital Technologies for Business Index per country, DESI Index 2022



Source: Eurostat.

1. European Banking Authority. (2021). *Report on the Use of Digital Platforms in the EU Banking and Payments Sector*. EBA/REP/2021/26

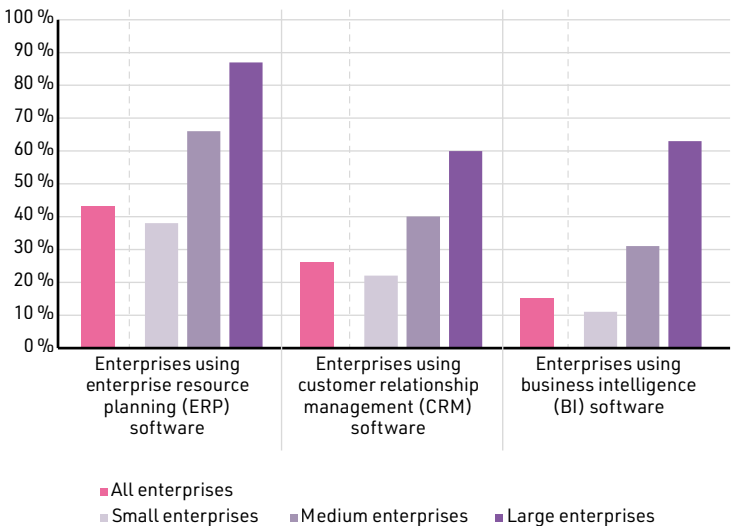
FIGURE 1.2 Technologies most likely used per general business functions among MSMEs in Bulgaria, Poland, and Slovakia, 2023-2024



Source: World Bank team.

Complementary capabilities, rather than access to technology or financial constraints, emerge as the primary barrier to digital adoption among MSMEs in Bulgaria, Poland, and Slovakia. The post-pandemic period has seen an increasing digitalization gap between MSMEs and large firms across Central and Eastern European countries, with larger firms showing faster adoption rates of new technologies (Figure 1.3). Larger firms possess greater capabilities for digital transformation, and in certain sectors, the demand for digitalization is higher due to export requirements or competitive pressures.[2] However, differences in managerial practices account for as much variation in digital tool usage as firm size and sector combined (Figure 1.4). This capability gap manifests through informational limitations, behavioral biases, and organizational misalignments, directly impacting firms' productivity and competitiveness.[3]

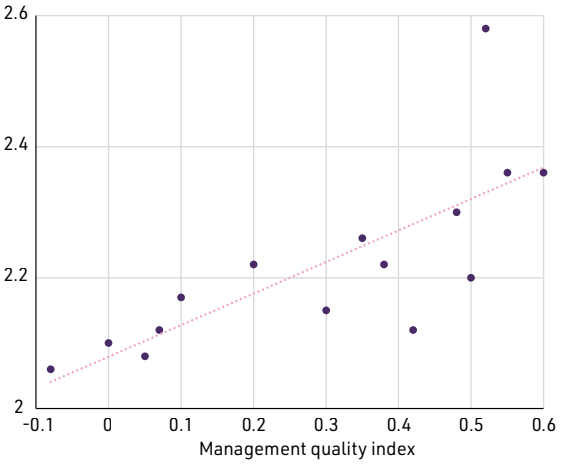
FIGURE 1.3 Enterprises using e-business applications by class size, EU, 2023



Source: Eurostat.

2. World Bank. (2023). *The Missing Element of Firm Digitalization.*
3. Cirera X., Comin D., and Cruz M. (2022). *Bridging the Technological Divide. Technology Adoption by Firms in Developing Countries.*

FIGURE 1.4 Correlation of use of technology and managerial practices in Bulgaria, Poland, and Slovakia



Source: World Bank team.

Current public support programs for MSME digitalization in the CEE region largely overlook non-technological barriers to digital transformation. An analysis of initiatives in Bulgaria, Poland, and Slovakia reveals that while MSMEs account for 98% of businesses and two-thirds of employment, most programs targeting these firms emphasize technological adoption and financial assistance.[4] In contrast, support mechanisms for larger firms, such as European Digital Innovation Hubs, include more comprehensive components aimed at building organizational and managerial capabilities. This difference in program design highlights a gap in addressing the complementary capabilities that smaller firms require for effective digital transformation.[5] This focus on technological aspects in MSME digitalization support programs is a potential reason why past initiatives have not succeeded in closing the digitalization gap between small firms and larger enterprises.

The Digitrans pilot aligns closely with the European Union's policy objectives of fostering digital and green transformation, addressing a critical knowledge gap in supporting MSME digitalization. While complementary capabilities are crucial for digitalization, there is limited empirical data on effective intervention models targeting these capabilities in small firms. The pilot implements a standardized support instrument across Bulgaria, Poland, and Slovakia, combining financial support with capability development components based on global best practices. This multi-country approach enables rigorous evaluation of intervention effectiveness across similar market contexts, aiming to develop a replicable model for advancing the European Union's twin transition goals.[6]

1.2 PROGRAM DESIGN

The pilot design was grounded in a systematic approach, combining in-depth analysis of MSME digitalization barriers in Bulgaria, Poland, and Slovakia with global evidence on effective support mechanisms. It leveraged key insights from analytical work identifying common challenges across these EU member states, while integrating considerations for country-specific contexts. The design prioritized developing an effective and scalable instrument

4. World Bank. (2023). *Gaps and Opportunities - Country-specific reports on Programs, Institutions and Barriers in Bulgaria, Poland, and Slovakia*.
5. *Ibidem*
6. European Commission. (2022). *Strategic Foresight Report. Twinning the Green and Digital Transitions in the New Geopolitical Context.*

tailored to small business needs, specifically targeting key non-technological and non-financial barriers to digitalization. Program partners were actively involved through knowledge exchange, collaborative design, and joint decision-making during structured project retreats (Figure 1.5). A prototype of the pilot instrument, tested with 25 firms in Sofia, Bulgaria, in second half of 2023, provided insights that shaped a standardized yet adaptable pilot instrument design for implementation across all three countries and validated costs and operational assumptions before the pilot phase.

FIGURE 1.5 Project Retreats for Pilot Design and Implementation

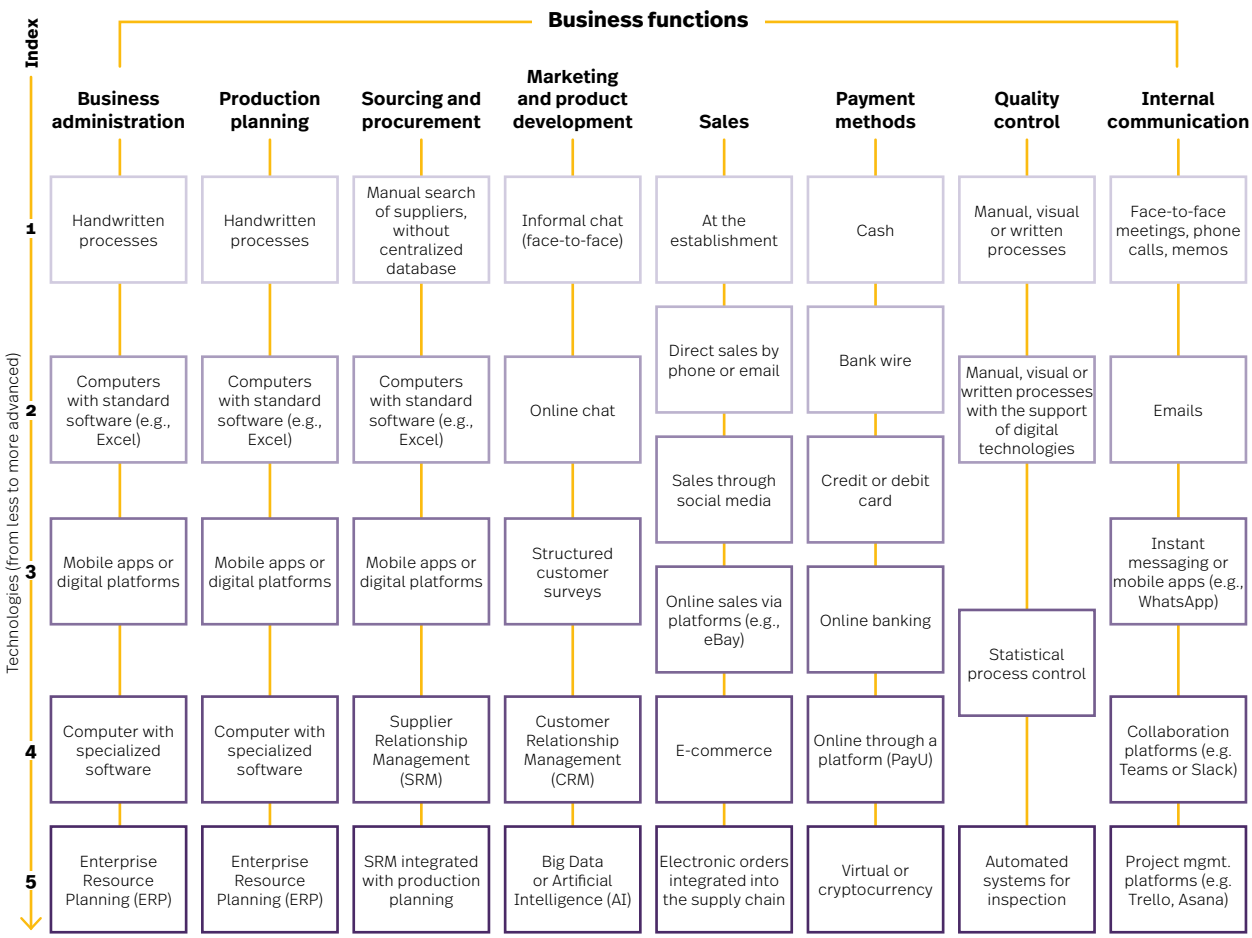
JUNE 2023 **FIRST PROJECT RETREAT: BRATISLAVA, SLOVAKIA**	• Objective: Use background research and needs analysis results to establish a framework and key assumptions for pilot design • Outcome: Set the stage for instrument development with agreed-upon design parameters and target group specification
MAY 2024 **SECOND PROJECT RETREAT: WARSAW, POLAND**	• Objective: Refine the pilot instrument design based on prototyping insights and early feedback • Outcome: Establish operational arrangements and ensure readiness for multi-country pilot rollout
MARCH 2025 **THIRD PROJECT RETREAT: SOFIA, BULGARIA**	• Objective: Discuss results of impact evaluation and lessons learned from pilot implementation. • Outcome: Learning workshops and discussions guide the future direction of planned and ongoing public programs.

Source: World Bank team.

The pilot focuses on firms with 5–35 employees across Bulgaria, Poland, and Slovakia, targeting those at a foundational level of digital sophistication and representing all sectors of the economy. This firm size segment exhibits the most pronounced digitalization gaps, and the presence of a minimum workforce is critical for the effective adoption and utilization of digital tools, thereby excluding firms with less than 5 workers. The sector-agnostic design emphasizes focus on general business functions that are universally applicable across industries, addressing critical gaps in horizontal digital capabilities (Figure 1.6). To ensure a stable operational baseline, eligibility is limited to firms that have been in operation for at least 12 months, increasing the likelihood of successful digital transformation. While recognizing the importance of sector-specific functions specific to individual industries, these were intentionally excluded from the scope to maintain a concentrated focus on foundational digital capabilities, which are evidenced as a gateway for digitalization of more specific business areas.[7]

7. World Bank. (2022). *Drivers of Productivity Growth in Poland: A Firm-Level Perspective on Technology Adoption and Firm Capabilities*.

FIGURE 1.6 General Business Functions framework



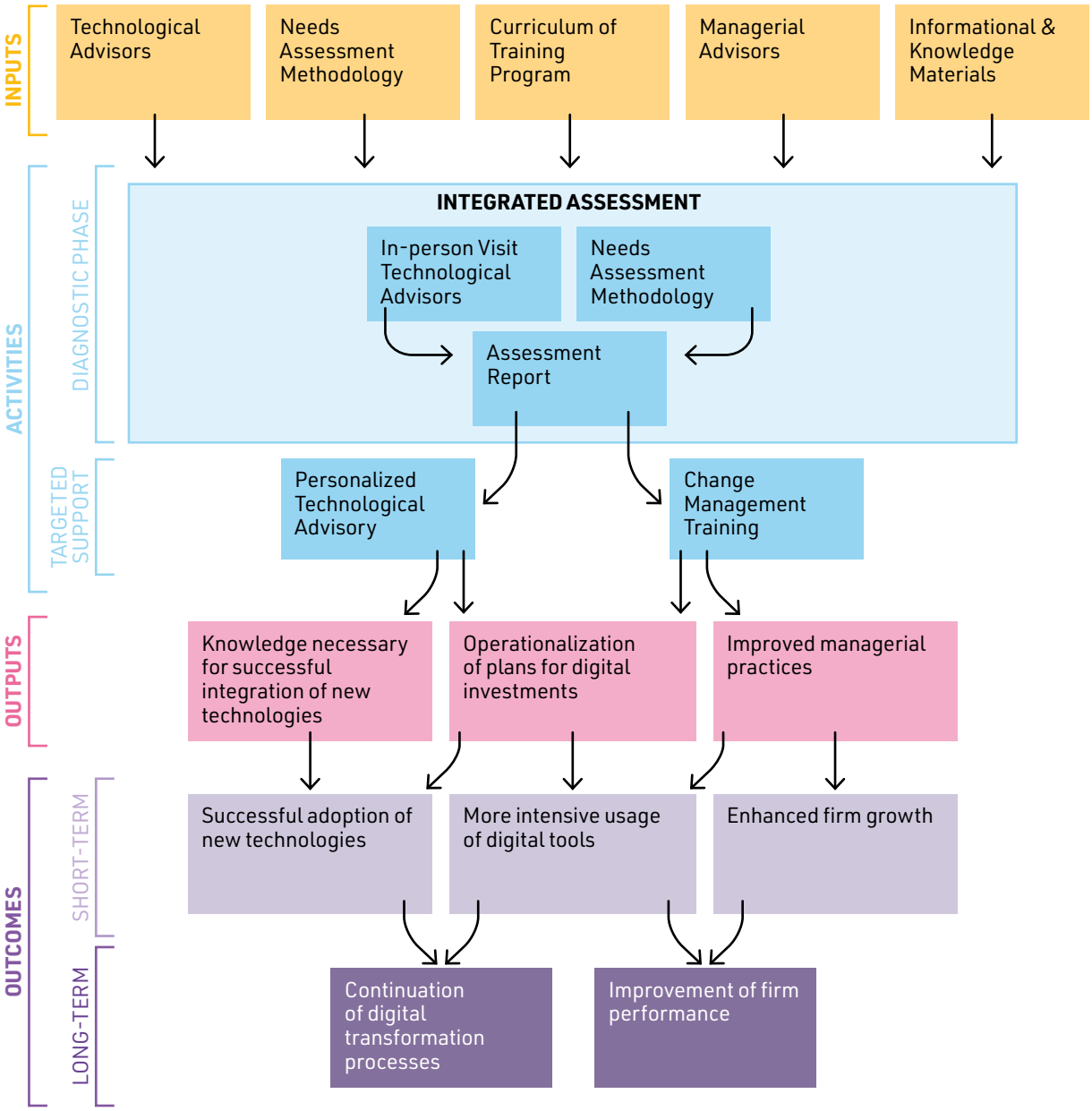
Source: World Bank team.

The pilot instrument comprises two components: one reflecting a traditional intervention focused on financial support and another introducing a novel approach to addressing gaps in complementary capabilities. The pilot applies identical support mechanisms across Bulgaria, Poland, and Slovakia, integrating Financial Subsidy with Technical Assistance to address both resource and capability constraints. The financial assistance component provides up to EUR 2,000 (or equivalent in local currency) per firm for the adoption of digital business tools. The technical assistance component offers standardized advisory services and training with a focus on strengthening internal capabilities. This dual-component design enables comparative analysis of intervention effectiveness, including an evaluation of the impact of financial support vis-à-vis its application combined with technical assistance to the comparison group of firms that receive neither type of support.

The pilot aims to accelerate digital transformation among MSMEs by addressing critical non-technological and financial barriers to technology adoption and usage. The technical assistance component is designed to tackle three core challenges: insufficient knowledge and information about appropriate technologies, financial constraints that hinder initial investments, and gaps in organizational capabilities necessary for effective technology

integration.[8] By targeting these foundational barriers, the program expects to drive increased levels of digitalization among participating firms (Figure 1.7). In the longer term, greater digital adoption is anticipated to translate into enhanced competitiveness and productivity, enabling firms to achieve productive growth and improved competitiveness in increasingly digitalized market environments.

FIGURE 1.7 Theory of Change of the pilot instrument



Source: World Bank team.

8. World Bank. (2023). - *Lessons for the EU Member States: Best practices for enhancing MSMEs Digitalization and Managerial Capacity Building*

The pilot design process encompassed four phases: research, prototyping, procurement, and implementation setup, spanning a total of two years. The preparation followed a systematic approach, moving from broad research and analysis to focused design and operational readiness (Figure 1.8). The initial research phase focused on analyzing both demand and supply factors affecting firm digitalization across Bulgaria, Poland, and Slovakia.[9] On the demand side, the team conducted surveys and qualitative interviews with MSMEs and ecosystem stakeholders to identify digitalization needs, constraints, and potential solutions for different firm segments. Supply-side analysis involved mapping existing public support instruments, private market offerings, and available technologies for target firms. The research combined analysis of secondary data with primary data collection through firm surveys and interviews with key digital ecosystem stakeholders.[10] Based on this research, the team specified target groups and designed intervention frameworks. The design was then tested through prototyping with 25 Bulgarian firms to assess feasibility and make necessary adjustments for specifying Terms of Reference and operational materials for the pilot implementation across three countries.

FIGURE 1.8 Timeline of design of the pilot instrument



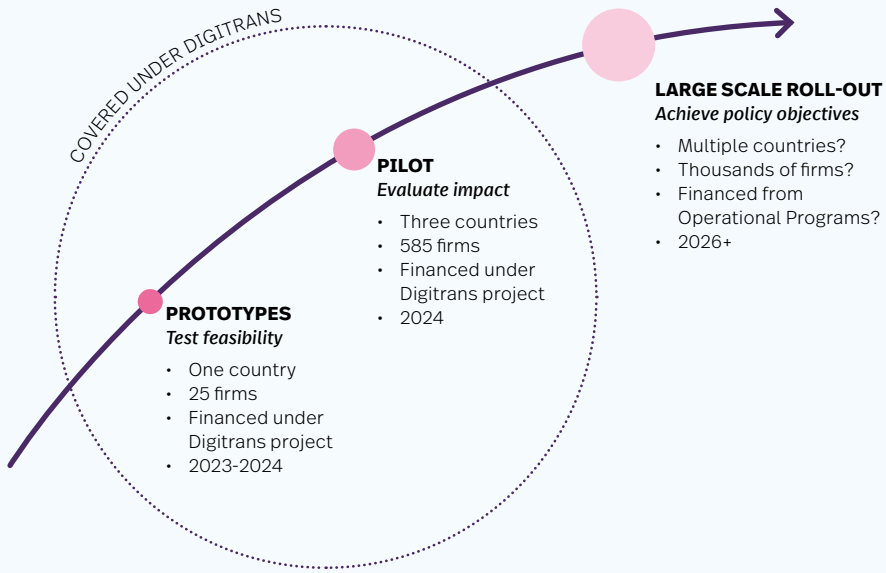
Source: World Bank team.

9. World Bank. (2024). *Gaps and Opportunities- State of Firms' Digitalization in Bulgaria, Poland, and Slovakia*.
10. These included digital associations, technology creators, technology integrators, business associations etc.

LESSON LEARNED

GAINS FROM PROTOTYPE TESTING

Prototyping was conducted in the Sofia region of Bulgaria from December 2023 to April 2024, involving 25 firms with 10–50 employees. The process mimicked all steps of the planned pilot instrument and tested five different variants of the support mechanism, each applied to groups of five firms. This approach provided insights into the feasibility and effectiveness of various design options, allowing for the refinement of the instrument and ensuring its readiness for broader rollout.



Prototyping revealed useful operational insights that significantly contributed to fine tuning and reshaped the implementation approach for the pilot instrument. The process uncovered unexpected benefits and pain points in the firm experience journey that were not visible in theoretical design.

- Initial digital materials had excessive technical language that was difficult to understand for some business owners; simplifying terminology increased training engagement in the pilot compared to prototypes
- Pre-designed timelines proved unrealistic for small business owners' availability; introducing flexible scheduling options prevented high dropout rates
- Standardized support materials required tiering by firm digital literacy level rather than by sector as originally planned

Prototype firms provided testimonials that proved essential for recruiting the pilot cohort, creating credibility that theoretical program descriptions couldn't achieve. Their feedback shaped communication materials that specifically addressed resistance points identified during prototype interactions.

Practical Application Components for Future Programs:

- **Pilot Design Refinement Tool**: A structured framework for evaluating target group definition and support component designs based on prototype feedback
- **Quick Application Process Template**: A streamlined 10-minute application form design with positive user experience metrics
- **Measurement Systems Setup Guide**: Guidelines for implementing baseline data collection protocols from program start
- **Service Delivery Materials Testing Protocol**: A methodology for validating content complexity and delivery methods with representative users
- **Stakeholder Feedback Integration Process**: A systematic approach to collecting and incorporating multi-stakeholder feedback before full-scale implementation
- **Program Credibility Building Kit**: Templates for generating and leveraging early success stories and testimonials in communication campaigns

The Financial Subsidy is designed to address market failures in digital tool adoption by reducing the upfront cost barriers that frequently deter MSMEs from investing in digital solutions. This component functions as a time-limited incentive, incentivizing firms to explore digital technologies that are often overlooked due to competing business priorities. The subsidy supports a broad range of digital tools aimed at enhancing general business functions, improving operational efficiency, and driving revenue and margin optimization. The design prioritizes efficiency through a streamlined disbursement process, ensuring timely and transparent allocation of funds (Box 1.1). This approach reflects a standard approach widely adopted in programs supporting MSME digitalization globally, focusing on mitigating financial constraints to accelerate digital adoption. Eligibility conditions for the purchases under the Financial Subsidy are listed in Annex 1.

BOX 1.1 Summary of Financial Subsidy Component

The implementation period for the Financial Subsidy component is structured as follows: **6 to 12 weeks** from the signing of the Beneficiary Agreement are allocated for pre-approval and purchases, with **weeks 13–14** designated for the review and reimbursement process. This 6-week initial period allows beneficiary firms to determine their planned purchases, while firms receiving Technical Assistance can simultaneously identify their Priority Recommendation and begin both the Change Management Training and Personalized Technology Advisory services before becoming eligible to submit pre-approval forms.

1. Pre-Approval Process

Beneficiary firms must initiate a pre-approval process before incurring eligible expenses. The process involves the following steps:

- **Submission and Review Timeline**: The pre-approval process should not exceed 5 working days from the time the MSME initiates the request.
- **Pre-Approval Documentation**: Firms must provide a completed form containing sufficient details to assess the eligibility of the proposed expenses. This documentation includes:

  a. For firms applying solely for the Financial Subsidy: specific details regarding the digital tool for which funding is sought.

  b. For firms applying for both the Financial Subsidy and Technical Assistance: an endorsement from the Technological Advisor confirming that the proposed expenses align with a priority recommendation.

2. Communication of Deadlines

The Consultant is responsible for ensuring beneficiary firms are aware of deadlines related to pre-approval requests.

- Firms must be notified of deadlines at least **10 working days** prior to the end of the eligible cost period if they have not initiated the pre-approval process.
- Timely reminders will ensure firms complete the necessary steps within the specified timeframe.

3. Purchase and Reimbursement Process

Once pre-approval is granted, firms can proceed with the purchase of approved items or services. The process includes:

- **Purchase Deadline**: Purchases must be made within the eligible cost period.
- **Submission of Reimbursement Request**: Firms must submit their reimbursement request, including proof of purchase and verification of alignment with pre-approval criteria, within 10 working days of completing the purchase.

4. Review and Reimbursement

The Consultant will review all reimbursement submissions to confirm they meet eligibility requirements.

- **Reimbursement Timeline**: Reimbursements will be processed **14 weeks** after the signing of the Beneficiary Agreement between the Consultant and the respective firm.
- Any discrepancies or additional documentation requirements will be addressed during this review period to ensure compliance.

Source: World Bank team.

The Technical Assistance component is designed to address gaps in complementary capabilities to digitalization through a structured set of training and advisory activities. This approach integrates diagnostic assessments, group training sessions, and individualized advisory support to guide participating MSMEs through all stages of preparation for broader and more effective use of digital tools (Table 1.1). The delivery model combines technological and managerial expertise at every stage, enabling firms to improve their technical capabilities while developing the organizational readiness required for sustained digital adoption. The process begins with a diagnostic phase that evaluates firms' digital needs and identifies barriers, providing a clear roadmap for transformation. This is followed by group training sessions that deliver practical knowledge and tools for implementing digital solutions. Finally, individualized advisory support offers tailored guidance to address specific challenges, ensuring that proposed solutions are actionable, practical, and aligned with resource constraints. Details of the Technical Assistance methodology are described in Annex 2.

TABLE 1.1 Summary of Technical Assistance Component

Week	Activity	Effort per firm– Consultant	Delivery person	Effort – Beneficiary firm
1	Onboarding webinar	0.2h per firm	Tech. and Mgmt. Advisor	2 hours
2	Integrated Assessment – Preparatory Call	3h (2 x 1.5h)	Tech. and Mgmt. Advisor	1.5 hour
2-3	Integrated Assessment – On-site visit	8 hours, in-person visit (2 x 4h)	Tech. and Mgmt. Advisor	5-8 hours
3-4	Integrated Assessment – Report preparation	4 hours (2 x 2h)	Tech. and Mgmt. Advisor	N/A
4	Integrated Assessment - Report Presentation Meeting	4 hours (2 x 2h)	Tech. and Mgmt. Advisor	2 hours
5-8	Group Change Management Training (4 sessions x 4 h)	2 hours	Mgmt. Advisor	16 hours
5-8	Personalized Technological Advisory	16 hours	Tech. Advisor	12 hours

Source: World Bank team.

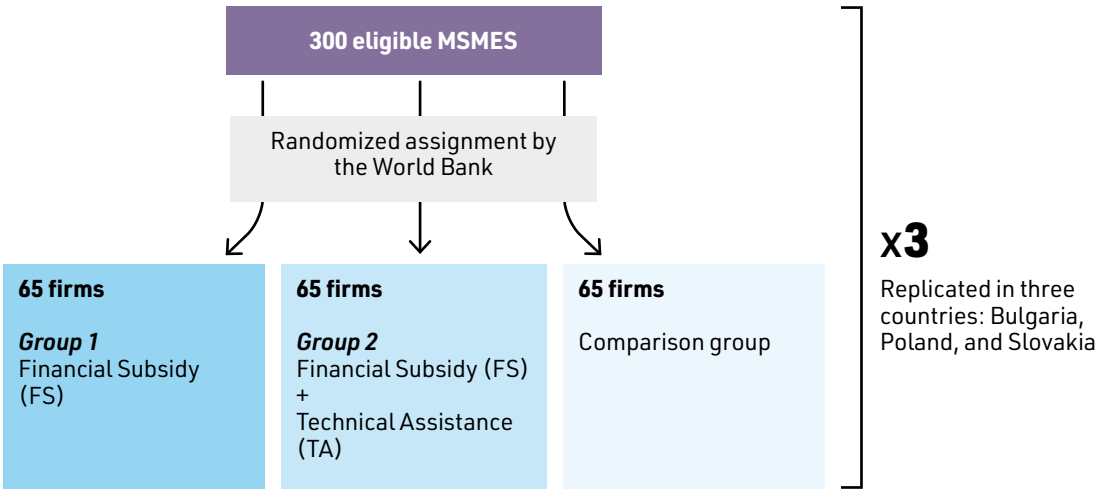
As a novel complement to traditional financial subsidy programs, the Technical Assistance component was introduced to fill capability gaps that often limit the effectiveness of financial support alone. By combining subsidies with structured training and advisory services, the program tests whether building firms' complementary capabilities can enhance the uptake and effective use of digital tools. This dual-track design enables a more comprehensive approach to supporting digitalization–addressing both financial constraints and capability barriers–and generates insights into how integrated support mechanisms can improve the effectiveness of public programs aimed at MSME digital transformation.

1.3 IMPLEMENTATION APPROACH

Financial subsidies and technical assistance were implemented as complementary interventions to assess the added value of capability-building support in driving digital adoption. Two distinct treatment arms–one offering only financial subsidies, and the other combining subsidies with technical assistance–enabled a clear comparison of outcomes. While financial support aimed to reduce the upfront cost of adopting digital tools, the technical assistance component addressed knowledge and skills gaps through targeted training and advisory services. This setup provided a practical framework to test whether financial incentives alone are sufficient, or whether additional capability-building is necessary to unlock deeper and more effective digital transformation.

The pilot evaluation design divided firms into three equal groups to isolate and measure distinct intervention impacts. Financial subsidies alone were planned to be provided to the first group of 65 firms, reflecting traditional digitalization voucher programs implemented across EU countries over the last decade. The second group received both financial subsidies and technical assistance - testing a novel approach that addresses firms' technological, financial, and complementary capability gaps (Figure 1.9). The third Comparison Group of 65 firms received none of the above support packages, enabling comparison with baseline outcomes. This three-group structure was replicated across three countries, bringing the total number of participating firms to 585.[11] The design enabled measurement of both isolated effects of financial subsidies by comparing Group 1 to Comparison Group, and additional impact of technical assistance by comparing Groups 1 and 2, while the Comparison Group established baseline outcomes for assessing overall program effectiveness.

FIGURE 1.9 Beneficiary groups of the pilot instrument



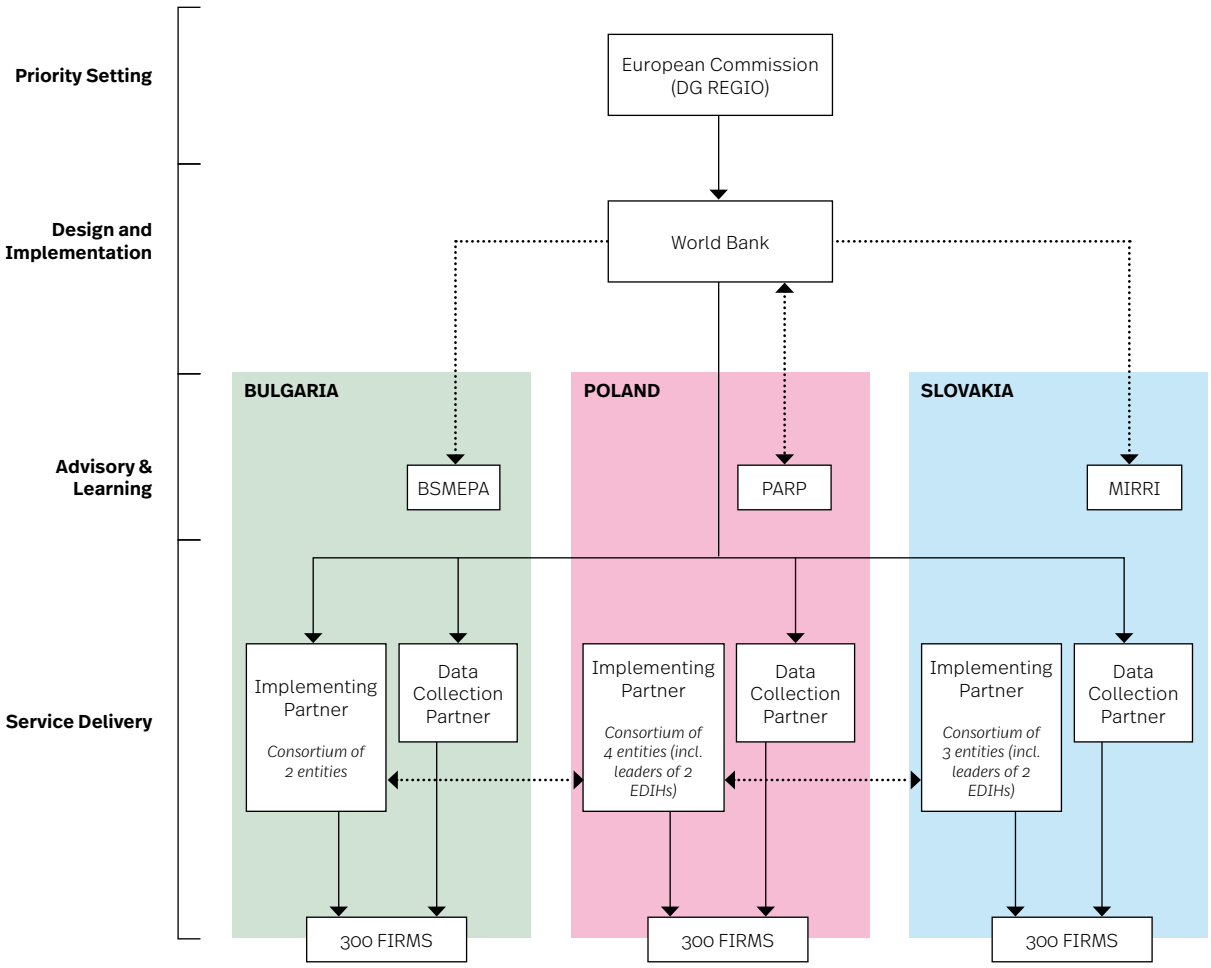
Source: World Bank team.

The equal allocation of firms ensured statistical equivalence between groups at baseline, enabling rigorous impact evaluation. The assignment by the World Bank research team gave each eligible firm and probability of being placed in any group, with firms informed about all possible support packages during application but learning their specific assignment only after completing the application process. This implementation approach ensured that both observable and unobservable firm characteristics were balanced across groups at the start of the program. By establishing equivalent groups at baseline, any subsequent differences in digitalization or performance could be attributed to the pilot interventions rather than pre-existing firm characteristics. The process has maintained transparency while preventing selection bias, as firms applied without knowing which specific support package they would receive.

The pilot was implemented with separation between program delivery, data collection, and impact evaluation. Three distinct institutions were involved in each country - Implementing Partners, Data Collection Partners, and the World Bank research team – with identical institutional setup replicated across all participating countries (Figure 1.10). The Implementing Partners - private firms and NGOs, including those managing European Digital Innovation Hubs in Poland and Slovakia - administered financial subsidies and coordinated technical assistance delivery according to the Implementation Manual, maintaining separation from impact measurement activities. Data Collection Partners, established market research firms, conducted baseline and follow-up surveys independent of group assignment decisions, ensuring unbiased measurement of outcomes. The research team designed the pilot instrument, evaluation methodology, conducted group assignments, and analyzed results without direct firm interactions. This separation of roles enabled focused execution of responsibilities while preventing conflicts of interest and maintaining evaluation integrity.

11. Ultimately, the number of firms assigned to the pilot instrument groups was slightly increased in order to maximize the utilization of resources procured under the pilot, which resulted in the approach outlined in Chapter 4.

FIGURE 1.10 Institutions involved in pilot implementation



Source: World Bank team.

Following the prototyping phase, the World Bank team competitively procured both implementing and data collection partners in each of the three participating countries. The selection process prioritized partners with proven experience in delivering firm-level support programs and conducting high-quality survey research. Implementing Partners – including private firms, NGOs, and institutions affiliated with European Digital Innovation Hubs – were chosen based on their operational capacity, sector expertise, and ability to deliver both financial and advisory services in line with the Implementation Manual. Data Collection Partners were established research firms with a track record of executing business surveys and meeting rigorous data quality standards. Each procurement followed a two-stage process, starting with a public call for Expressions of Interest (EoIs), followed by the shortlisting of selected applicants (Table 1.2). Between the two stages, interested applicants were able to submit clarification questions, which were addressed by the World Bank team and published publicly to ensure transparency and equal access to information. Further details on the requirements and evaluation criteria for both Implementing and Data Collection Partners are provided in Annex 3.

TABLE 1.2 Summary of Procurement Outcomes by Country and Partner Type

	Bulgaria	Poland	Slovakia
Implementing Partners			
# of submitted EoIs	11	13	8
# of shortlisted offers	7	7	5
Data Collection Partners			
# of submitted EoIs	3	5	4
# of shortlisted offers	1	2	2

Source: World Bank team.

After selection, all partners engaged in a collaborative preparation phase, which involved refining implementation protocols, localizing training materials, and jointly testing delivery processes to ensure readiness for rollout. Co-design workshops were held with partners across countries to align on core implementation elements, leverage each partner's strengths, and exchange best practices. These workshops served to streamline operational preparations while accelerating the rollout timeline.

2

OUTREACH AND PARTICIPANT SELECTION

The Digitrans outreach and selection process balanced three key requirements: making the program easy and accessible for applicant firms to minimize administrative burden, generating demand for a novel support program, and ensuring alignment with EU regulations for replicability. External communication emphasized the program's focus on enhancing firm productivity through access to digital tools and expert guidance. The experimental design–including random assignment and the presence of a Comparison Group–was communicated transparently in participation materials. This approach aimed to mirror real-world conditions in which firms opt into support programs based on their growth potential rather than research participation. The outreach strategy was informed by evidence on effective communication with small firms, combining multi-channel delivery with messaging that highlighted tangible benefits and fostered trust through clarity and consistency.

2.1 OUTREACH STRATEGY

The outreach campaign aimed to generate interest from 300 eligible firms in each of the three pilot countries between May and July 2024. Achieving this target required overcoming specific challenges posed by the implementation approach. Notably, firms were offered to apply without knowing in advance which support package they would receive, as group assignment occurred only after application–marking a departure from typical EU-funded programs. The strategy needed to present the program in an attractive and credible way while clearly communicating the possibility of receiving three packages of support, which was essential to preserve evaluation integrity. The outreach covered all regions in Bulgaria and Slovakia, and three selected regions in Poland (Mazowieckie, Małopolskie, and Warmińsko-Mazurskie), representing approximately 40% of Polish enterprises. Additional goals included securing one-third participation from the manufacturing and construction sectors and reaching firms without prior experience in public support programs. Outreach activities were designed and implemented by the country-level Implementing Partners, with technical guidance from the World Bank drawing on international good practices and cross-country learning.

FIGURE 2.1 Timeline of Pilot Recruitment per country

	Slovakia	Bulgaria	Poland
Recruitment launch	**May 17th**	**June 6th**	**May 22nd**
Cohort 1	YES	NO	NO
# of firms	150	–	–
TA delivery	June-July	–	–
Cohort 2	YES	YES	YES
# of firms	150	300	300
TA delivery	September-October	September-October	October-November

Source: World Bank team.

The outreach strategy leveraged multiple communication channels to generate demand for the new program within a compressed timeframe. Global evidence on effective outreach for public support programs targeting small firms highlighted the value of combining direct emails, social media campaigns, business association networks, and targeted events. Performance metrics were used to monitor each channel's effectiveness in reaching priority firms, allowing for real-time adjustments and cost-efficiency analysis. This multi-channel approach reflected best practices in omni-channel communication while enabling a comparative assessment of outreach methods. The World Bank team defined baseline communication standards in the Terms of Reference for Implementing Partners to ensure consistency across countries. Partners then tailored the intensity and use of channels based on local market conditions and their operational experience, while maintaining alignment with core program messages and outreach objectives.

The outreach messaging emphasized the program's value proposition for small businesses and its procedural simplicity—helping it stand out in a landscape crowded with other public support instruments. Communications focused on three core messages: business improvement through tailored support, a streamlined application process requiring minimal documentation, and expert guidance throughout participation. Messaging highlighted specific examples of potential benefits, such as support for technology adoption or management training, while maintaining transparency about the randomized assignment of support packages after application. Marketing materials also emphasized the program's flexibility to address individual firm needs and its grounding in international best practices. Each country's project website conveyed consistent core messages, with visual elements adapted to local contexts (Figure 2.2). Implementation Partners further reinforced program credibility by offering real-time support to potential applicants, responding to inquiries, and assisting with the application process. Description of the pilot published on the websites is listed in Annex 4.

FIGURE 2.2 Screenshots of the websites per country

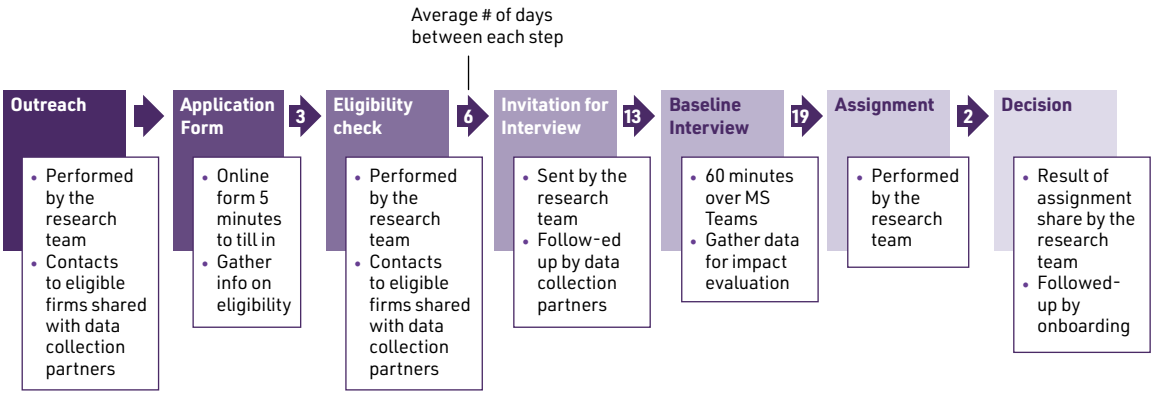


Source: World Bank team.

2.2 OUTREACH DELIVERY

The outreach process established clear timelines and minimized firms' effort while meeting impact evaluation requirements. The outreach process was streamlined, requiring up to 90 minutes from firms to complete the application procedures while ensuring that the average time between applying and receiving information was less than two months (Figure 2.3). The process was designed to minimize effort from applicants while satisfying all requirements for the rigorous impact evaluation. It began with a media launch and email campaigns to initiate outreach, which continued through multiple channels to create program awareness. Interested firms could complete a five-minute application form on the website. Eligible firms were automatically screened, and their contact information was shared with the Data Collection Partner, who invited them for interviews. After interviews were completed, the World Bank team allocated firms and communicated the application results to them via personalized e-mails.

FIGURE 2.3 Process of recruitment and origins with average time in days between steps



Source: World Bank team.

The outreach activities were supported both by project partners and business ecosystem stakeholders to amplify reach. National SME agencies and ministries provided official patronage and used their communication channels, enhancing program credibility as a legitimate public support instrument. Additional partners, including chambers of commerce and industry associations, shared program information through their networks and hosted online and in-person information sessions. Regional chambers proved particularly effective in reaching manufacturing firms, while startup hubs helped access firms outside traditional support networks. This multi-stakeholder approach, combining government endorsement with established business networks, strengthened the program's positioning as a trustworthy support instrument.

Multi-channel outreach with repeated exposure through targeted delivery drove program applications most effectively. Analysis showed multiple exposure channels driving applications - 56% of applicants encountered program information through at least two sources, confirmed by qualitative findings that repeated exposure built program credibility. Direct email campaigns achieved 6% application conversion, making them the most cost-effective outreach channel, while social media reached 7,000-19,000 firms per country at 3% engagement rate, ranking second in cost-effectiveness. Business associations provided crucial third-party validation through in-kind support, reaching 120-210 firms through co-organized information meetings. High-intensity approaches like telephone campaigns and in-person visits achieved highest conversion rates of 10-20%, but at the highest cost per application.

LESSON LEARNED

STRATEGIES FOR SMARTER, FASTER OUTREACH

The pilot implementation demonstrated key insights about recruitment approaches during challenging periods. A data-driven approach measuring channel performance allowed for strategic resource allocation based on actual, daily-tracked conversion metrics.

Messaging that addressed specific business pain points ("spending too much time on paperwork?" rather than "digital transformation opportunity") generated higher response rates than technical program descriptions. Geographic analysis showed different engagement patterns between primary and secondary cities, with some secondary locations showing stronger response rates than initially anticipated.

- Message testing identified optimal length (under 100 words) and timing (Tuesday/Wednesday mornings) correlating with higher open and response rates
- Materials featuring testimonials from previous participants achieved approximately 5 times the engagement of standard program descriptions
- Partnerships with business organizations provided additional credibility and reach to target audiences

The summer recruitment period required adapting the contact strategy to account for vacation schedules. Multiple touchpoints across channels and extended response hours outside standard business times showed improved completion rates compared to traditional single-channel approaches.

Communication using concrete business terminology rather than policy language correlated with higher application completion rates and more positive feedback from participating firms.

Practical Application Components for Future Programs:

- **Channel Performance Dashboard**: A real-time tracking system for monitoring outreach effectiveness across multiple communication channels
- **Message Testing Framework**: A methodology for rapidly testing and optimizing outreach messaging based on open rates, response rates, and conversion metrics
- **Geographical Targeting Matrix**: A data-driven approach to allocating outreach resources based on regional response patterns and business demographics
- **Multi-Channel Contact Strategy**: A coordinated approach to reaching potential participants through complementary channels with optimized timing and frequency
- **Business-Centric Communication Templates**: A library of proven messaging frameworks that resonate with business owners by addressing concrete pain points and benefits

2.3 APPLICATION RESULTS

Applications were primarily received during periods of active outreach, with significant surges corresponding to the intensity of outreach activities. Following each campaign launch, there was a sharp increase in the cumulative number of applications across all countries. In contrast, periods without active outreach–such as weekends–showed minimal application activity, indicating that without proactive engagement, applications were not spontaneously generated (Figure 2.4). Over 90% of total applications were submitted during less than 20% of the days when outreach efforts were strongest, while more than half of the recruitment days registered fewer than three applications in each country (Figure 2.5). In the absence of active outreach, application rates were significantly low. Continuous engagement is necessary to drive application submissions, as organic or delayed responses were minimal.

Ongoing improvements in messaging and fine-tuning of outreach strategies led to better targeting, resulting in a higher share of eligible applications over time. Initially, the broad outreach approach resulted in a lower ratio of eligible to total applications. By focusing on specific industries and regions and improving communication about eligibility, targeting became more effective, leading to a steady rise in the ratio of eligible applications during the recruitment period (Figure 2.6). However, the conversion rate from website visits to eligible applications remained relatively stable over time, suggesting that attracting the appropriate audience to the website was more important than on-site messaging changes (Figure 2.7). In Bulgaria and Slovakia, approximately 20–30 website visits were needed to obtain one eligible application, while in Poland, about 50 visits were required. These findings indicate that while targeted outreach increased the proportion of eligible applicants, the efficiency of converting website visitors to eligible applications depended more on the quality of the audience reached than on adjustments to website content.

FIGURE 2.4 Cumulative Number of Applications by Day of Outreach Duration per country

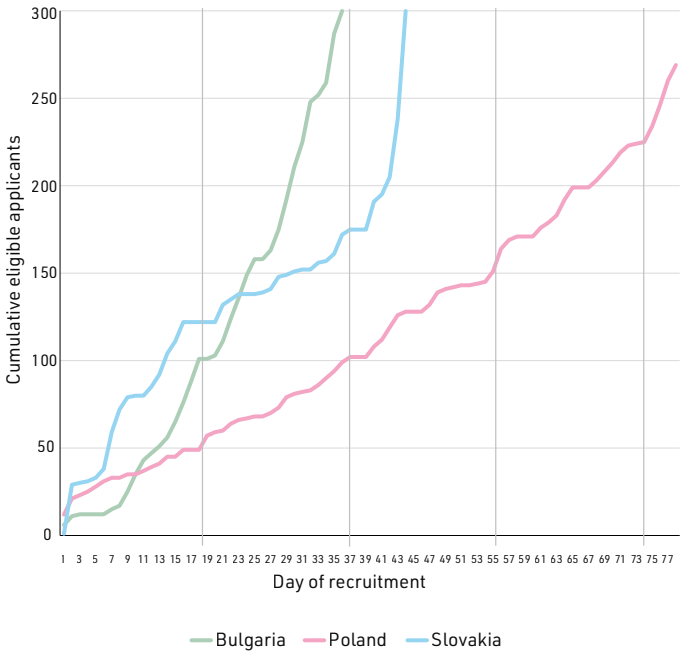
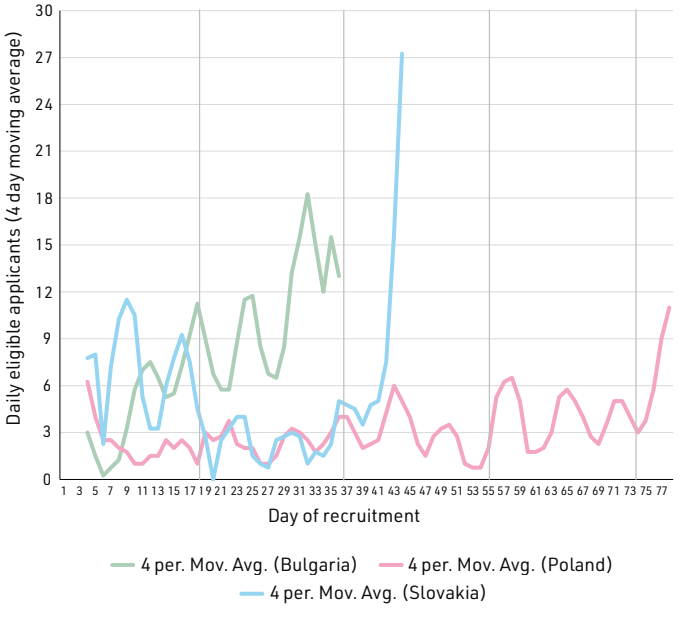


FIGURE 2.5 Daily Number of Applications (4-day moving average) per country



Source: World Bank team.

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FIGURE 2.6 Weekly Share of Eligible Applications in Total Applications per country

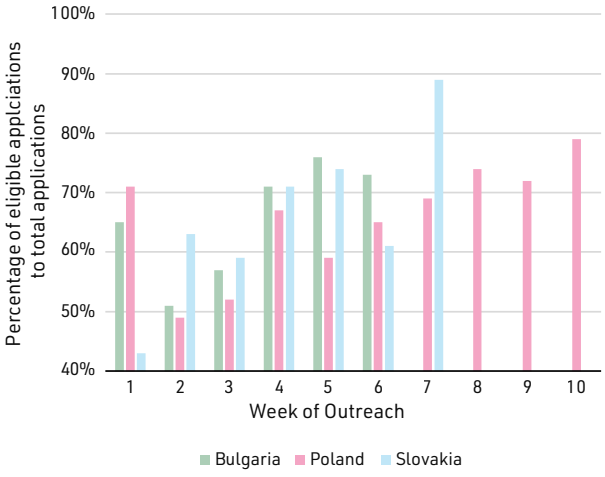
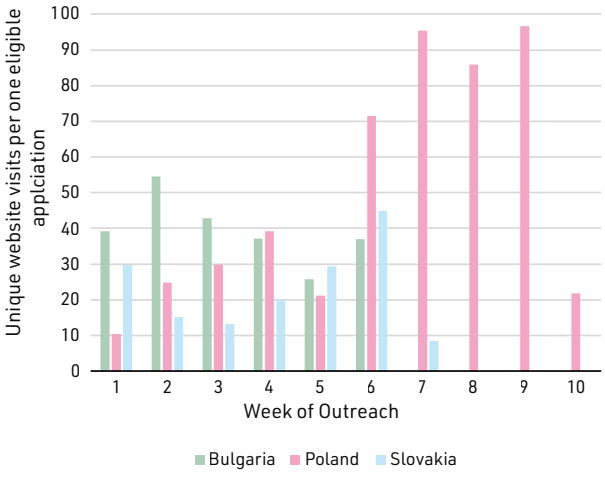


FIGURE 2.7 Weekly Ratio of Website Visits per Eligible Application per country

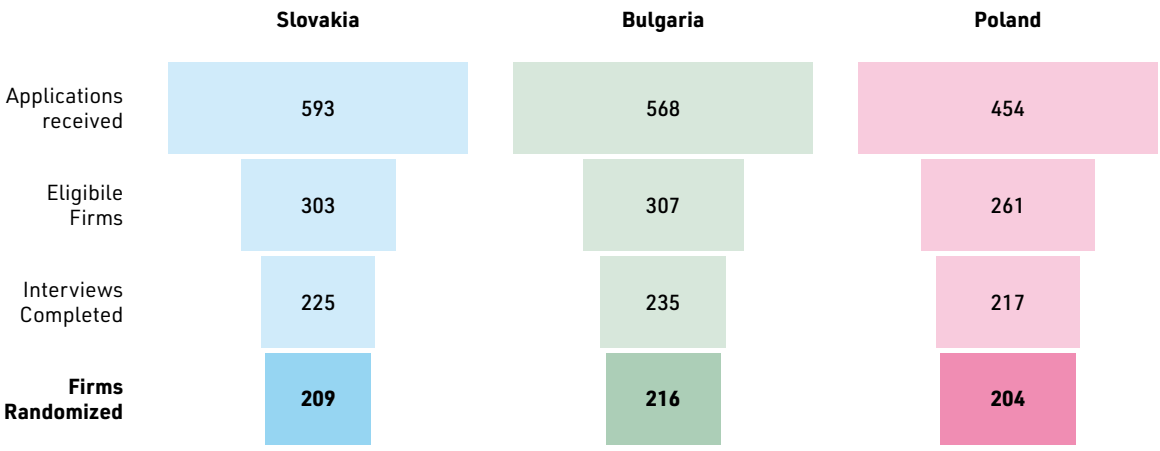


Source: World Bank team.

Recruitment efforts generated close to or above 300 eligible applications in less than two months in each country. In total, over 1,500 applications were submitted, with approximately 60% meeting the eligibility criteria. The number of firms ultimately randomized in each country exceeded initial projections – reflecting both the high volume of eligible applicants and implementation efficiencies identified during the preparation phase with Implementing Partners, which enabled support for a larger number of firms. The expanded sample size also helped safeguard statistical power in anticipation of potential attrition (Figure 2.8). This adjustment was informed by monitoring information from past training and advisory programs aimed at building firm capabilities, where up to one-third of selected firms did not complete the intended intensity of participation, particularly in digitalization-focused initiatives for MSMEs.[12]

12. World Bank. (2020). *Poland: Return on Investment of Public Support to SMEs and Innovation*.

FIGURE 2.8 Recruitment results per country



Source: World Bank team.

LESSON LEARNED

ENGAGING FIRMS THROUGH LOW ENTRY BARRIERS AND TRANSPARENT PROCESSES

The pilot recruitment process yielded valuable insights about effectively engaging firms in support programs. Several key principles emerged that significantly impacted participation rates and applicant satisfaction:

- **Minimize Entry Costs and Ensure Transparency**:
  - Firms are willing to apply to programs even if they are uncertain about the benefits, as long as the cost of entry is low, and the process is transparent.
- **Concentrate Resources on Active Outreach Management**:
  - Allocating more resources to actively manage outreach efforts is more effective than spreading them thinly.
    - This approach reduces the duration of recruitment.
    - It allows for professional communication with more firms from the target group.
    - Firms are more satisfied when the process doesn't take too long.
- **Adapt Strategies Based on Performance**:
  - Update the website, messaging, and outreach strategies or channels when existing methods are not effective.
    - Continuous adaptation ensures outreach remains effective and responsive to firm needs.

- **Avoid Unnecessary Activities**:
  - Focus on actions that generate applications and enhance program trustworthiness.
    - Avoid activities that do not contribute to these goals, such as:
    - Media coverage that doesn't reach the target audience.
    - Engaging stakeholders not directly connected to the firms (e.g., attending events with accountants instead of business owners).

- **Prioritize Direct Contact with Firm Owners or Key Personnel**:
  - Direct engagement with firm owners or key decision-makers is crucial.
  - This ensures clear communication and increases the likelihood of participation.

Practical Application Components for Future Programs:

- **Application Process Optimization Tool**: A framework for evaluating and streamlining application procedures to minimize entry barriers

- **Outreach Resource Allocation Model**: A methodology for concentrating resources on high-yield outreach activities rather than dispersing efforts

- **Performance-Based Adaptation Protocol**: A system for continuously monitoring outreach effectiveness and making rapid adjustments based on real-time data

- **Direct Engagement Framework**: Guidelines for establishing and maintaining direct communication with business owners and key decision-makers

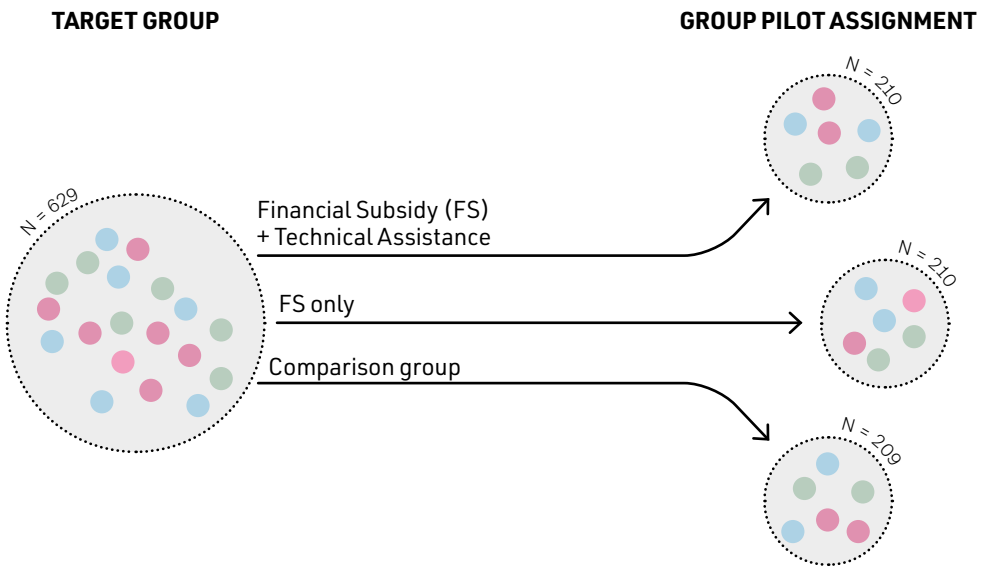
- **Outreach Activity Assessment Matrix**: A tool for evaluating potential outreach activities based on their likelihood of generating qualified applications

2.4 BENEFICIARY ALLOCATION

Beneficiary firms were allocated into three groups, with each group comprising approximately 200 firms in total—formed by assigning around 65 to 72 firms per group in each of the three countries. In each country, approximately 200 firms were randomized across the three groups, ensuring balanced geographical representation and comparability. The three groups were defined as follows: (i) firms receiving only benchmarking reports, (ii) firms receiving financial subsidies in addition to benchmarking reports, and (iii) firms receiving both financial subsidies and technical assistance (Figure 2.9). The equal distribution of firms across groups and countries allows for robust comparison between different types of support and controlling for country-specific effects in the analysis.

13. Beneficiary assignments employed RCT-aligned procedures for the SME digitalization support program, implemented by the Romania North-Eastern Development Agency under the ERDF-cofinanced Operational Program in 2023, in full compliance with national and European regulations governing EU funding.

FIGURE 2.9 Pilot groups allocation



Source: World Bank team.

Firms were allocated to intervention groups using a randomization process in which all eligible firms that completed the baseline interview had an equal probability of being assigned to any of the three groups. Randomization was conducted separately within strata defined by country and firm characteristics to ensure balanced representation. In each country, six strata were created by combining two firm size categories (micro and small) with three levels of digital sophistication. Within each stratum, firms had equal chances of assignment to any group. This stratified randomization approach ensured comparability across groups, improved balance in key characteristics, and increased the statistical power and precision of the impact estimates. The process followed established statistical best practices, with full documentation retained to support transparency, auditability, and future replication. The allocation method was also designed to comply with national and EU regulations governing MSME support programs co-financed through EU Operational Programs, supporting its potential for broader application in other publicly funded interventions.[43]

Allocation resulted in balanced groups across a broad range of firm characteristics, creating a solid foundation for rigorous impact evaluation. Firms across groups show similar levels in key outcomes of interest - digital adoption and use, digital practices and knowledge, managerial practices, and firm performance measures. Balance extends to business characteristics such as size, sector, business age, revenue levels, and use of public support, as well as business owner characteristics including gender, education, business experience, and socio-behavioral factors like risk aversion and locus of control. The absence of systematic differences between groups at baseline ensures that subsequent changes can be reliably attributed to the support received rather than pre-existing characteristics. This validates the study's ability to generate credible evidence about effectiveness of different types of support in enhancing firm digitalization.

3

PARTICIPANT CHARACTERISTICS

The pilot engaged a diverse group of small and medium-sized enterprises selected through an open application process. Baseline data collected from these firms–through application forms and structured surveys–provided insights into their sectoral composition, size distribution, and digital readiness prior to intervention. Participating firms represented a range of industries, with a concentration in services and manufacturing, and were broadly comparable across treatment and comparison groups. Key baseline indicators, such as digital tool usage, managerial practices, and perceived barriers to digitalization, were used to establish pre-intervention equivalence and inform subsequent analysis.

3.1 DATA COLLECTION METHODOLOGY

The primary method of data collection for the Digitrans Pilot is a structured 60-minutes survey administered via video call by trained enumerators. The survey focuses on firm digitalization and related business practices. Its design builds on insights from the 2023 Digitrans Survey and previous digitalization policy initiatives. Each randomized firm participates in multiple survey rounds, beginning with a baseline and followed by a round approximately six months later. This panel structure enables a randomized controlled trial (RCT) analysis, allowing the research team to attribute observed changes in outcomes to the different support services provided through Digitrans. While the core structure of the survey remains consistent across rounds, certain modules–such as those on stable firm characteristics (e.g., internet connectivity)–are excluded or streamlined in the follow-up to avoid redundancy. The instrument is designed to support ongoing assessment of the short-, medium-, and potentially long-term effects of the pilot.

Baseline surveys were conducted between June and September 2024 across the three participating countries. Among firms eligible and included in the randomization, 209 were surveyed in Slovakia between June 4 and July 19; 216 in Bulgaria between July 29 and August 9; and 204 in Poland between August 9 and September 20. The staggered timing of survey implementation across countries is only due to differences in implementation windows across country partners, and is not reflective of a research strategy. In Slovakia, a longer survey window was required due to the cohort-based delivery of support under the pilot instrument. In Poland, the later timeline resulted from a longer recruitment window. The World Bank Digitrans team directly coordinated survey implementation in Slovakia, which helped ensure consistent data collection practices and allowed for the transfer of lessons learned to data collection partners in Bulgaria and Poland.

The data collection process was designed to be uniform across countries. Enumerators in each location were trained jointly by the local data collection partner and the World Bank Digitrans team. Training included an overview of the program's objectives and design, a detailed walkthrough of the questionnaire, participation in test interviews, and a short assessment to confirm readiness and suitability for the digitalization-focused survey content. Data collection was conducted through three independent contracts with consulting firms, one in each country. To maintain quality of the data collection process and satisfy applicants' requirements, survey coordination and delivery were managed by local consultants from the research team who were independent from the respective implementation partners. Contact details for these local coordinators were shared with all survey participants, enabling firms to raise questions or resolve issues independently of the data collection partners.

Video-call interviews conducted by trained enumerators were selected as the primary data collection method to combine high data quality with scalability and cost-efficiency. This approach enabled rich, interactive engagement with respondents–closely mirroring the quality of face-to-face interviews–while allowing for centralized oversight and efficient implementation across multiple countries. The baseline survey lasted an average of 62 minutes per firm, reflecting both the detailed nature of the questionnaire and the ability of respondents to remain engaged throughout. Video interviews ensured consistent delivery of questions, real-time clarification of any doubts, and strong respondent focus. Compared to self-administered surveys, which were considered but ultimately not used, this method provided significantly more reliable and comparable data across firms and contexts.

BOX 3.1 Questionnaire Sections

The questionnaire is divided into seven main sections, with all the sections designed to help us examine and explain changes in digitalization for participating firms. These sections are:

1. Firm Characteristics: This section comprises basic information about the firm, and double checks the information provided by firms in the application form.

Key Outcome Sections

2. Technology Adoption and Use: This is the largest section of the survey, and seeks to cover the digital methods adopted and used by firms across the eight GBFs. This section follows a standard framework developed in the World Bank, called the Firm Adoption of Technology (FAT) framework. The core adoption and usage questions are used to create the "GBF Index" which serve as measures of digitalization, and our primary outcome indicator for impact analysis. The GBF Index is divided into the extensive and intensive margin, which each range from 1 (least sophisticated) to 5 (most sophisticated). The extensive margin represents the average technological sophistication of the most advanced methods for each GBF, while the intensive margin represents the average sophistication of the most used methods for each GBF.

3. Firm Performance: This section inquires as to firms' business performance metrics, such as revenues, profits, assets/investments value, participation in trade, and business linkages. While some of these may not see change in the short term due to interventions, it is important to collect them at baseline for future analysis.

Secondary Outcome Sections

4. Digital Practices: The Digital Practices Index assesses how firms approach software management, updates, data security, and their decision-making processes for adopting new tools. This section also included a series of questions regarding the respondent firm's preparatory actions taken during their most recent acquisition of a digital tool. This index is a weighted average of three subindices, those being the:

i. Systems Subindex (30%), which covers the practices of firms regarding updating tools, backing up data, and training staff in functionalities.

ii. Integration Subindex (30%), which covers the use and scope of professional integration services, as well as the implementation into everyday business processes, including data migration, frequency of use, and feedback mechanisms.

iii. Information Subindex (40%), which covers the number of factors a firm assessed during their most recent adoption of a digital tool, as well as the methods they used for assessing those factors.

The Digital Practices Index and its Subindices are normalized to scales ranging from 0 to 100.

5. Managerial Practices: The Managerial Practices Index investigates broader organizational behaviors, such as employee performance monitoring, business planning, and change management strategies. This index is a weighted average of five subindices, those being the:

i. Monitoring Subindex (25%), which covers the presence, collection, and usage of performance indicators.

ii. People Subindex (25%), which covers the usage of employee performance incentives, and actions surrounding promotions and dismissals.

iii. Change Management Subindex (25%), which covers customer feedback mechanisms, ideation strategies, as well as change implementation and iteration.

iv. Planning Subindex (12.5%), which covers the usage and scope of business targets.

v. Risk Management Subindex (12.5%), which covers the scope of risk identification and mitigation throughout business strategies and projects.

The Managerial Practices Index and its Subindices are normalized to scales ranging from 0 to 100.

Heterogeneity Analysis

6. Financial Preferences: Firms are asked to play two hypothetical games, which help us to determine their market digital tool preferences, as well as their discount rate for money-now vs. money-later.

7. Business Perceptions: We draw from two established psychometric, Likert-scale-based tests to provide firms with about two dozen of questions on the respondents' attitudes and self-assessed behaviors.

The questionnaire frequently utilizes multiple-choice or objective numerical format questions. This design helps to promote comparability of results between respondents, especially for the Technology Adoption and Use section, in which most of the included questions have been validated and implemented in various other country contexts through the World Bank. Open-ended free-response questions were only used in places where it would not be feasible to provide suggested answers, such as when asking respondents for the names of specific digital tools they use. However, some multiple-choice questions also had an option for a respondent to respond with "Other," and then further specify an answer. The structure and content of the questionnaire, including all survey sections, are detailed in Annex 5.

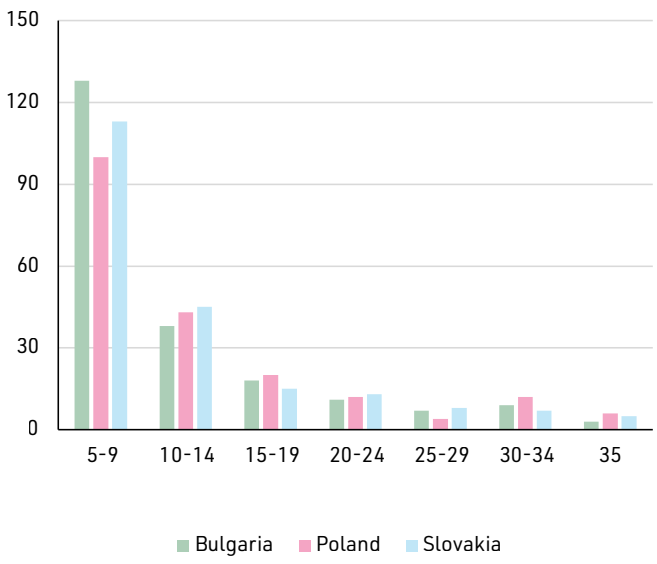
3.2 PARTICIPATING FIRMS' DEMOGRAPHICS

The baseline survey data provides insight into the characteristics of firms participating in the Digitrans Pilot and how they compare to the broader population of firms. This subsection examines how pilot firms differ from the wider market by comparing them to a subset of respondents from the 2023 Digitrans Survey and 2022 Firm-level Technology Adoption Survey carried out in Poland.[14] To ensure comparability, the analysis focuses only on firms with 5 to 35 employees, excluding smaller and larger firms that could distort average values and bias the comparison.

Baseline Business Characteristics

The distribution of firm sizes in the Digitrans Pilot closely reflects that of the broader economy. In both the Pilot and the prior diagnostic surveys, 56% of eligible firms are classified as micro-enterprises (fewer than 10 employees), while the remaining 44% are small enterprises (10–49 employees). In the Pilot sample, this corresponds to 340 micro firms and 284 small firms. The employment distribution is similarly aligned, with a clear concentration in the 5–9 employee range and a sharp decline in the number of firms as size increases (Figure 3.1). For instance, fewer than 10% of firms in the Pilot report having more than 20 employees, consistent with broader patterns in the economy. Overall, the Pilot sample is broadly representative of the size distribution of eligible firms in the target economies.

FIGURE 3.1 Distribution of Pilot Firm Employment

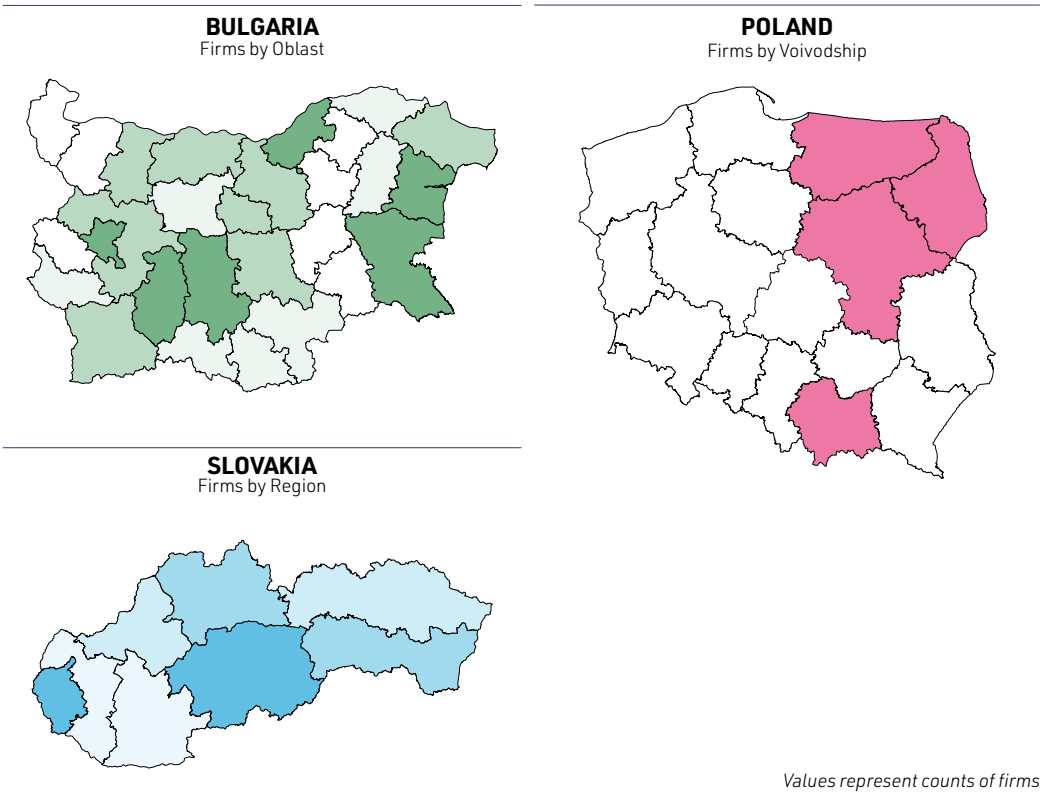


Source: World Bank team.

14. Detailed analysis of the results of these surveys could be find in World Bank. (2023). Gaps and Opportunities - Country-specific reports on Programs, Institutions and Barriers in Bulgaria, Poland, and Slovakia.

While Pilot firms are distributed across their respective countries, there is geographic clustering in the capital regions of Bulgaria and Slovakia, whereas firms in Poland are more evenly spread across the three eligible regions. In Poland, this distribution was guided by operational considerations to facilitate delivery of the technical assistance. In Bulgaria and Slovakia, both Implementing Partners were based in the capital cities and drew on existing professional networks during recruitment, contributing to the observed concentration (Figure 3.2). As a result, the geographic distribution of Pilot firms may not be fully representative of the national firm population. Geographic data collected during the baseline will allow control for location effects in the impact analysis. This is particularly relevant because the prior diagnostic surveys indicated that firms based in capital regions tend to be more digitally advanced – especially in areas such as marketing and payment systems.

FIGURE 3.2 Geographical distribution of Pilot Firms

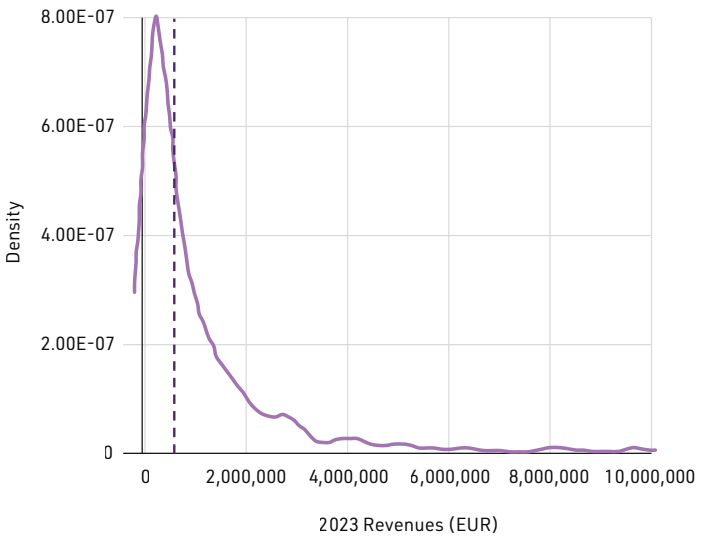


Source: World Bank team.

Two-thirds of Pilot firms report annual revenues between EUR 300,000 and EUR 600,000. The distribution is concentrated around this range, with a median revenue of EUR 510,000 (Figure 3.3). The distribution closely mirrors that of the general population of micro and small firms with 5 to 35 employees. Revenue levels are relatively consistent across countries, and similar patterns are observed within each national sample–indicating that outreach efforts attracted comparable segments of eligible firms in all three contexts. Asset values show a similar concentration, with more than 75% of firms reporting holdings below EUR 1 million. Together, these indicators point to a sample composed largely of firms operating with limited financial scale, that constitute a large group of firms with low digital sophistication across the European Union.

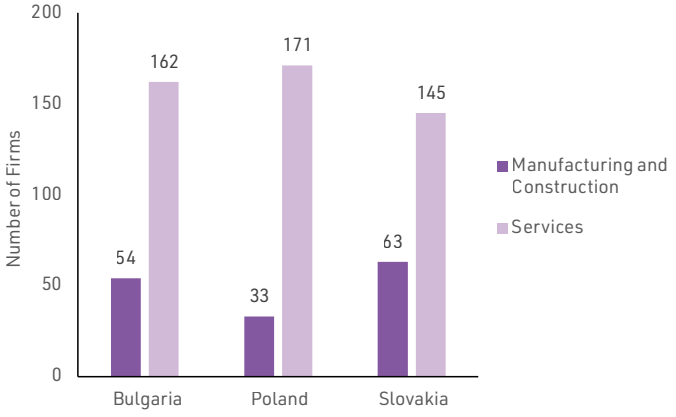
Most Pilot firms operate in the services sector, with around one-quarter coming from manufacturing or construction. Overall, 24% of participating firms are in manufacturing or construction, while 76% are in services. Bulgaria has the highest share of manufacturing and construction firms at 30%, followed by Slovakia at 25%, and Poland at 16% (Figure 3.4). Although these shares are slightly below the initial target, the overall sample composition supports meaningful analysis across sectors. Since the primary impact evaluation will be conducted on the full cross-country sample, this distribution allows for controlling sectoral differences in outcomes. Importantly, no significant differences in revenue, profit, or productivity levels are observed across sectors – when comparing the pilot baseline data to either of the diagnostic surveys.

FIGURE 3.3 Distribution of Pilot Firms' Revenue



Source: World Bank team.

FIGURE 3.4 Pilot Firm Sectors, by Country



Source: World Bank team.

3.3 BASELINE FIRM CAPABILITIES

The digital capabilities of Pilot firms are assessed using the Firm-level Adoption of Technology (FAT) framework, which evaluates digital sophistication across eight general business functions. For each function, firms receive two scores on a scale from 1 to 5: an adoption score, reflecting the most advanced digital tool available within the firm for that function (also referred to as the extensive margin), and a usage score, indicating the level of sophistication of the tool used most frequently (the intensive margin). A firm's adoption index is calculated as the average of its eight adoption scores, while the usage index is the average of the eight usage scores.[15] Together, these indicators offer a structured view of both the breadth of digital tools adopted and the depth of their integration into daily operations.

15. For more information on the FAT framework and its approach to measuring levels of digitalization, see World Bank (2022), *Bridging the Technological Divide: Technology Adoption by Firms in Developing Countries*.

Baseline data indicate that the digital sophistication of Pilot firms is in line with the wider population of micro and small firms in the participating countries. The median adoption index is 2.95, suggesting that most firms have basic digital tools, such as Excel or equivalent software, available across business functions, with more advanced tools present in only one or two areas (Figure 3.5). The median usage index is lower, at 2.23, highlighting a gap between what firms have access to and what they actively use in day-to-day operations (Figure 3.6). In practice, most firms continue to rely on analogue methods such as handwritten notes, paper records, and manual coordination for routine business tasks. This pattern points to a digitalization profile where initial tools may be in place, but digital integration into core workflows remains limited.

FIGURE 3.5 Extensive Margin per Country

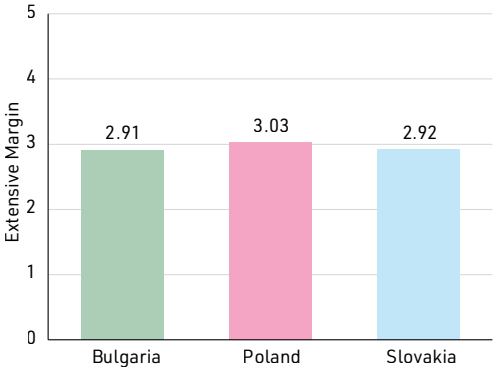
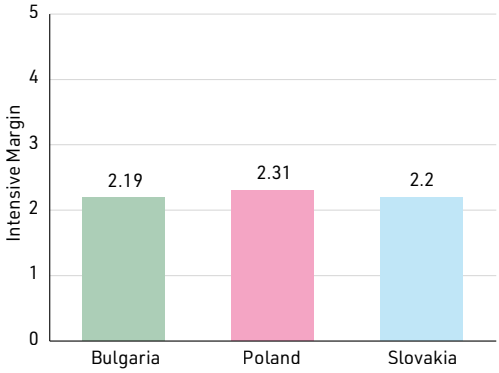


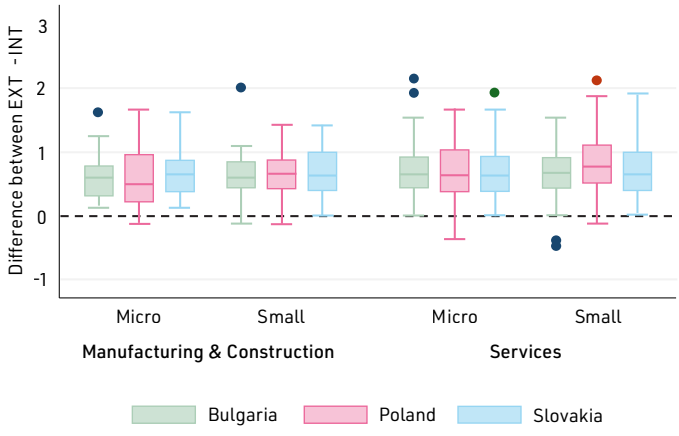
FIGURE 3.6 Intensive Margin per Country



Source: World Bank team.

Firms do not use the most advanced digital tools they have adopted. Across countries, sectors, and firm sizes, adoption consistently outpaces usage, indicating that firms often possess more advanced solutions than are reflected in their routine operations. This gap is evident among both micro and small firms and does not vary between manufacturing and services (Figure 3.7). Its persistence suggests that the barrier to effective digitalization lies less in access to technology and more in the internal capacity to integrate and operationalize it. These findings underscore the relevance of capability-building support as a complement to financial incentives, particularly for firms that have already acquired but not yet embedded digital tools into daily workflows. A discrepancy of approximately 0.5 to 1 point between adoption and usage is consistent with findings from diagnostic surveys conducted both in the pilot countries and internationally.

FIGURE 3.7 Differential between GBF Margins per Country, by Size and Sector



Source: World Bank team.

Digital sophistication of pilot firms is comparable across the three pilot countries. Adoption scores (extensive margin) are similar in Bulgaria and Slovakia, with slightly higher values observed in Poland. However, no significant differences are found in usage scores (intensive margin), suggesting that Polish firms, despite having more advanced tools on average, do not use them more frequently than their peers. This supports the decision to analyze firms from all three countries as a single sample for impact evaluation, given their overall similarity in baseline digital readiness. It also reinforces the rationale for complementary support: higher availability of tools alone does not lead to greater digital integration without targeted assistance.

Among Pilot firms, small enterprises in the services sector show higher digital sophistication than micro firms. On average, small service firms score 0.15 points higher on the adoption index and 0.11 points higher on the usage index, with both differences statistically significant at the 0.1% level (Figures 3.8 and 3.9). A difference of 0.15 on the index corresponds approximately to a one-level increase in adoption or usage within a single general business function. In manufacturing and construction, no significant difference is observed between micro and small firms. Similar patterns were identified in diagnostic studies conducted in the pilot countries, where small service firms tended to demonstrate higher digital sophistication in general business functions. In contrast, manufacturing firms showed greater digitalization in sector-specific processes – areas not covered by the Pilot's measurement framework, which focuses on general business functions.[16]

FIGURE 3.8 Extensive Margin per Country, by Size and Sector

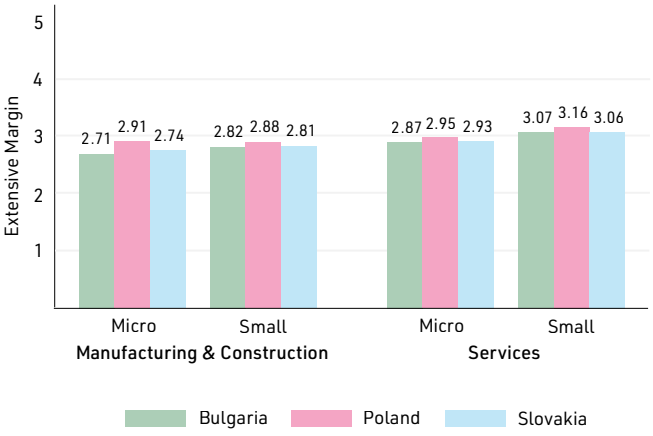
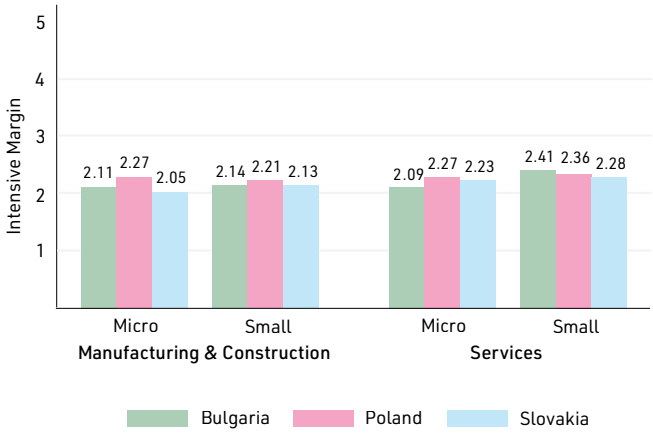


FIGURE 3.9 Intensive Margin per Country, by Size and Sector



Source: World Bank team.

16. Sector-specific business functions in the FAT framework refer to digital processes that are central to a firm's core production activities and vary by industry. Unlike general business functions–which cover areas like sales, payments, or internal communication–sector-specific functions reflect how technology is used in production itself. In food processing, for example, this includes the level of automation and digital control in functions such as ingredient testing, blending, anti-bacterial treatment, packaging, and storage. These functions are not assessed in the Digitrans Pilot, which focuses exclusively on general business functions applicable across sectors.

Among Pilot firms, the highest levels of digital sophistication are observed in business administration and payments, while marketing, sales, and quality control rank lowest. High adoption in administration reflects regulatory requirements that compel firms to use digital systems for tracking and reporting. In the case of payments, widespread access to payment terminals and instant transfer systems has led to nearly universal uptake, with 94% of firms adopting the highest level of digitalization in this area. In contrast, lower digital maturity in marketing, sales, and quality control reflects their less immediate operational urgency, making them more likely to be overlooked. This pattern mirrors the broader firm population, where similar gaps are observed between compliance-driven and discretionary areas of digital adoption.[17]

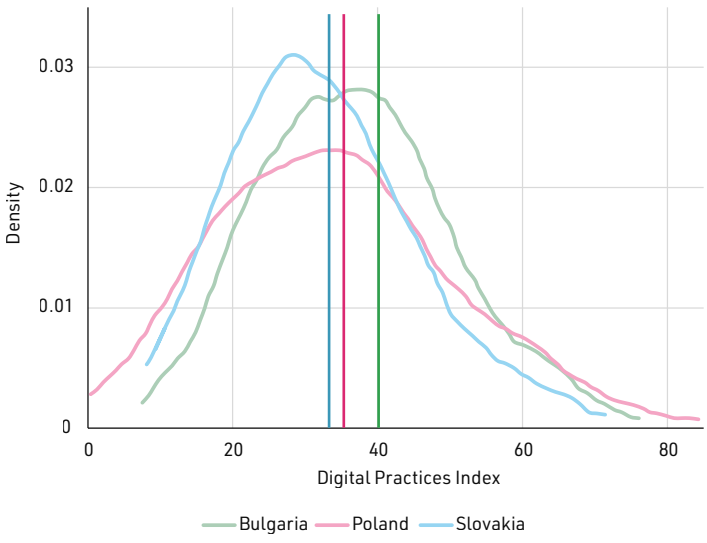
Secondary Outcomes

In addition to core digital adoption and usage, the Digitrans Pilot also tracks secondary outcomes related to how firms manage digital processes, adapt to change, and build internal knowledge around digitalization. To capture these dimensions, the baseline survey includes dedicated sections that assess specific practices and capabilities. Responses are scored to construct two composite measures: a digital practices index and a managerial capability index.

The Digital Practices Index captures how firms approach software management, data security, and digital integration. It combines three weighted subindices: Systems (30%), which covers processes for software updates, data backups, and staff training; Integration (30%), which reflects the use of integration services and the extent to which digital tools are embedded in daily operations; and Information (40%), which measures the breadth of factors firms consider when adopting new digital tools. Together, these components provide a structured view of how firms govern and sustain their digital tools.

Baseline data indicate meaningful differences in digital practices across the three Pilot countries. Firms in Bulgaria report the highest median score on the Digital Practices Index, followed by Poland, while firms in Slovakia score slightly lower. In addition to having lower scores on average, the distribution in Slovakia is more compressed, with fewer firms exhibiting strong digital management practices and a concentration of scores near the lower end of the scale (Figure 3.10).

FIGURE 3.10 Digital Practices Index, Distributions per Country



Source: World Bank team.

17. A detailed discussion of these patterns and underlying drivers can be found in World Bank (2024), Gaps and Opportunities: State of Firms' Digitalization in Bulgaria, Poland, and Slovakia.

There are significant correlations between a firm's Digital Practices Index result and both its size (understood as employment) and sector group. We find that small, as opposed to micro, firms display a 5-percentage point higher DP Index score, significant at the 0.1% level. It is possible that the larger employment levels of firms necessitate greater efforts to implement digitalization changes. Alternatively, the availability of extra staff capacity could enable more formal processes to be undertaken when implementing new digital tools.

On the sectoral level, we find that services firms score 3.5 percentage points higher on the DP Index than firms in manufacturing and construction, significant at the 5% level. This suggests that there is less observed digital formality in the manufacturing and construction sectors, which could potentially be alleviated through complementary support.

The Managerial Practices Index captures the extent to which firms apply structured approaches to performance monitoring, strategic planning, and organizational change. It is composed of five weighted subindices: the Monitoring subindex (25%), which assesses the presence, collection, and use of performance indicators; the People subindex (25%), which covers employee performance incentives, promotions, and dismissal practices; the Change Management subindex (25%), which evaluates customer feedback mechanisms, ideation strategies, and change implementation; the Planning subindex (12.5%), which reflects the scope and structure of business targets; and the Risk Management subindex (12.5%), which measures the extent of risk identification and mitigation in business strategies and projects. Together, these components provide a structured view of a firm's capacity to plan, adapt, and manage organizational change in a data-informed manner.

The distribution of Managerial Practices Index scores shows notable variation across the three pilot countries. Median scores are highest in Bulgaria (53%), followed by Poland (44%) and Slovakia (43%), with the overall sample median at 47%. These country differences are statistically significant at the 0.1% level, indicating meaningful variation in the application of structured management practices. While the general distribution shapes remain broadly comparable, firms in Slovakia exhibit a slightly narrower spread, with scores more concentrated around the median (Figure 3.13).

FIGURE 3.11 DP Index per Country, by Size

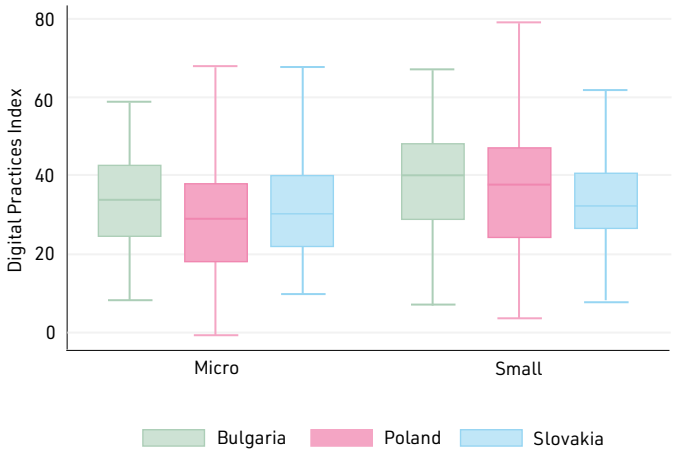
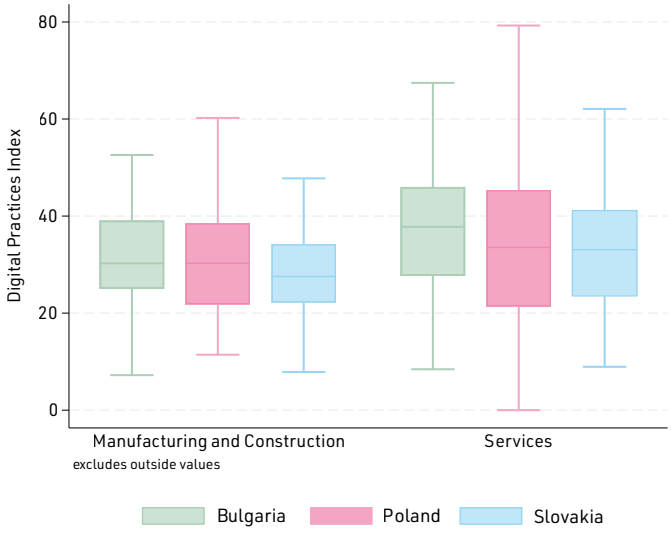
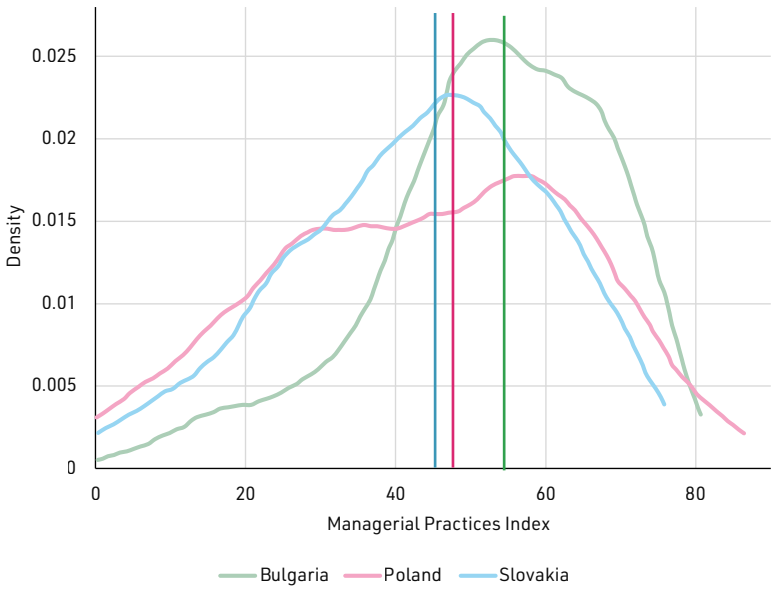


FIGURE 3.12 DP Index per Country, by Sector



Source: World Bank team.

FIGURE 3.13 Managerial Practices Index, Distributions per Country



Source: World Bank team.

A significant relationship is observed between firm size, measured by employment, and the Managerial Practices Index. Small firms score 6.6 percentage points higher than micro firms, significant at the 0.1% level. This may reflect the greater organizational complexity and resource availability of small firms, which can support more structured management practices. No significant differences were detected between sectors, suggesting that managerial structure is relatively consistent across services, manufacturing, and construction firms in the sample.

4

PROGRAM IMPLEMENTATION

The implementation of the pilot followed a structured model anchored in three core components: benchmarking, financial support, and technical assistance. Benchmarking reports were generated using data from baseline interviews, offering firms personalized information on their digital maturity relative to peers. Financial subsidies were disbursed through a simplified process designed to facilitate the acquisition of digital tools with minimal delays and transparency. Technical assistance was delivered through a combination of on-site engagements, group workshops, digital readiness assessments, and personalized advisory support. Real-time feedback mechanisms and regular monitoring enabled adaptive management, ensuring the program remained responsive to firm needs while safeguarding data quality and compliance.

Program implementation in Slovakia followed a two-cohort structure, while Bulgaria and Poland operated with single cohorts. Slovakia's implementation split 60-70 firms equally between Cohort 1 (July-August) and Cohort 2 (September-November), enabled by early initiation of firm recruitment. In Bulgaria and Poland all firms were supported from September to December (Figure 4.1). This phased approach in Slovakia allowed testing implementation procedures, gathering operational feedback, and marginally refining program materials before expanding support to remaining countries in fall. The iterative implementation structure generated practical insights that improved program delivery across all participating countries.

FIGURE 4.1 Pilot implementation timeline per country

	MONTH											
	May-24	June-24	July-24	Aug-24	Sept-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Closing of activities Mar-25	End of Digitrans Apr-25
Recruitment	SK	SK	SK									
		BG	BG	BG								
	PL	PL	PL	PL	PL							
Data collection	SK C1	SK C1	SK C2	SK C2			SK C1	SK C1		SK C2	SK C2	
			BG	BG								
				PL	PL						PL	PL
Support delivery		SK C1	SK C1	SK C1	SK C2	SK C2	SK C2					
					BG	BG	BG					
						PL	PL	PL				

Slovakia | Bulgaria | Poland

Source: World Bank team.

LESSON LEARNED

BENEFITS OF COHORTS-BASED IMPLEMENTATION

The phased implementation approach in Slovakia, with two distinct cohorts, yielded significant operational benefits that enhanced program effectiveness across all countries. Key advantages observed include:

- **Refined Implementation Materials and Procedures**
  - Refinements to instrument design from practical experience increased program effectiveness while reducing time spent on basic administrative tasks, creating more space for value-adding interactions.
    - Group webinars replaced repetitive individual explanations
    - Enhanced implementation guides and meeting scenarios
    - Better aligned training components for cohesive learning experience

- **Improved Advisory Service Quality**
  - Advisors and trainers developed deeper understanding of common challenges and effective solutions in program implementation. This accumulated expertise translated into more targeted, comprehensive support for subsequent firms while maintaining planned timelines.
    - Advisors applied practical insights from Cohort 1
    - More targeted solutions for common implementation challenges
    - Expanded scope of advisory support while maintaining timelines

- **Streamlined Administrative Processes**
  - Streamlined administrative processes based on actual operational needs rather than theoretical assumptions. These refinements maintained comprehensive program oversight while reducing time burden on both implementing team and participating firms.
    - Streamlined documentation based on actual firm needs
    - Simplified monitoring while maintaining data quality
    - Reduced processing requirements for firms and administrators

Practical Application Components for Future Programs:

- **Phased Implementation Playbook**: A structured approach to program rollout with built-in refinement points between cohorts
- **Advisory Knowledge Management System**: A framework for capturing, organizing, and transferring insights between advisor teams and program phases
- **Administrative Process Optimization Template:** A methodology for evaluating and streamlining administrative requirements based on early implementation feedback
- **Cross-Cohort Learning Framework**: Guidelines for systematic identification and application of lessons from early participants to improve experience for later cohorts
- **Continuous Improvement Tracking Tool**: A standardized system for monitoring operational efficiency metrics across program phases

4.1 BENCHMARKING REPORT

Benchmarking reports provided firms with detailed comparative analysis of their digital maturity against comparable peers. The reports, delivered as 32-slide automated presentations, utilized data from application forms and baseline interview to create individualized profiles of strengths and gaps. Each report mapped the firm's adoption of digital tools across eight General Business Functions and AI applications (Table 4.1), comparing them to similar enterprises within their country, sector (professional services, business services, or manufacturing and construction), and size category (micro or small). The comparative analysis drew from multiple data sources, including data on other Digitrans beneficiaries and findings from broader digitalization surveys conducted during the project's diagnostic phase, covering over 300 firms in Bulgaria and Slovakia, as well as the 2022 Firm-level Technology Adoption Survey of 1,500 Polish firms.[18] This benchmarking approach aimed to help firms identify their digitalization strengths and areas for improvement, while demonstrating the tangible value of their participation in the assessment process.

TABLE 4.1 Structure of Benchmarking report

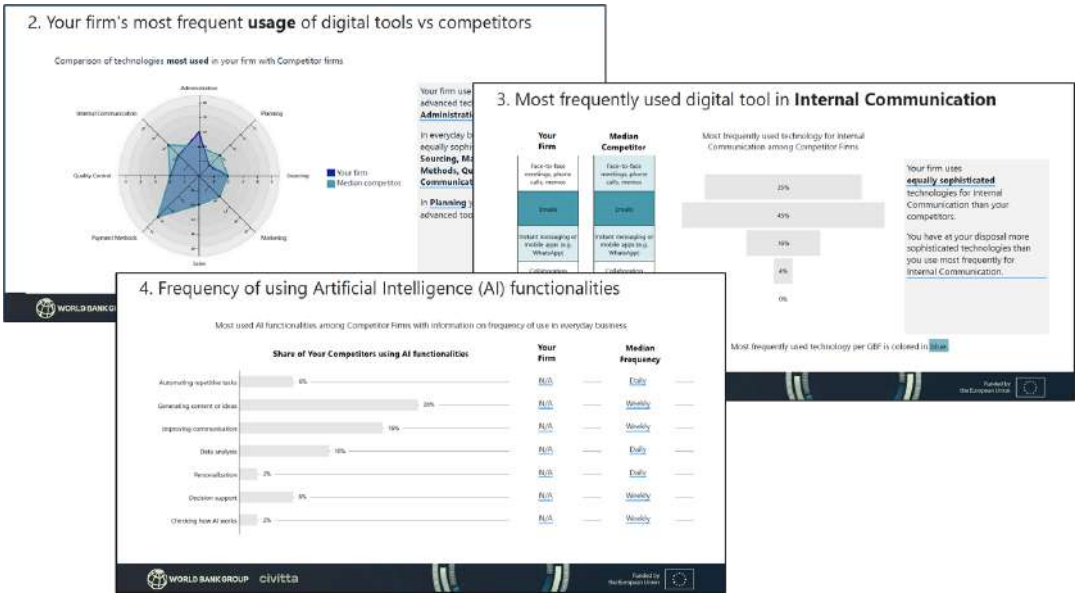
Section	Content Summary
Introduction *(3 slides)*	Overview of the report's purpose: benchmarking your firm's digitalization compared to competitors, based on data from interviews and surveys across Slovakia, Bulgaria, and Poland.
Digitalization Index *(4 slides)*	Comparative analysis of your firm's digital sophistication using general business functions (GBFs) and technologies. Measures adoption and usage of digital tools.
Digitalization per General Business Function *(16 slides)*	Detailed insights into technology adoption and use across eight GBFs: Administration, Planning, Sourcing, Marketing, Sales, Payment Methods, Quality Control, and Internal Communication.
Use of Artificial Intelligence *(4 slides)*	Assessment of AI tool adoption and usage. Includes competitor comparison, highlighting functionalities like task automation, data analysis, and personalization.
Next Round of Benchmarking *(1 slide)*	Invitation for participation in the second benchmarking round to track progress and gain updated insights.

Source: World Bank team.

The benchmarking report demonstrated to firms the immediate value of data collected during interviews and application process and was used for facilitation of service delivery. Information from the benchmarking reports served as foundational data for advisors and trainers delivering Technical Assistance, enabling informed initial interactions with participating firms (Figure 4.2). This data-driven approach allowed advisors to demonstrate expertise and preparedness from the first meeting, fostering trust and highlighting the value of firms' participation in the assessment process. The data management system implemented strict confidentiality measures, processing peer comparison data only in aggregate form.[19] Access to individual firm information was restricted through a controlled distribution system, ensuring that only designated advisors working directly with specific firms could view their detailed reports, in accordance with data protection regulations and non-disclosure agreements between firms and advisors.

18. World Bank. (2023). *Gaps and Opportunities - Country-specific reports on Programs, Institutions and Barriers in Bulgaria, Poland, and Slovakia*.
19. Each firm was benchmarked against a peer group, selected from 18 possible combinations defined by country (Bulgaria, Poland, or Slovakia), sector group (consumer services, business services, or manufacturing & construction), and firm size (micro or small).

FIGURE 4.2 Excerpts from the Benchmarking Report



Source: World Bank team.

The second benchmarking report will be provided to all participating firms following their endline interview, delivering an analysis of their digitalization progress over the six-month implementation period against comparable competitors. This report maintains the initial benchmarking approach while adding a temporal dimension, enabling firms to assess their advancement in digital adoption and identify both areas of significant progress and remaining opportunities for improvement. These structured benchmarking reports demonstrate how administrative data, collected primarily for program application and evaluation purposes, can be leveraged to enhance program delivery and generate additional value for participants. This approach presents potential applications beyond the current program scope, with benchmarking reports potentially serving as evidence-based justification for targeted digitalization support in future initiatives, helping to streamline program design and resource allocation.

4.2 FINANCIAL SUBSIDY

Financial subsidy take-up was equal across treatment groups, with approximately 90% of firms ultimately receiving support. However, the timing of pre-approval submissions differed. Financial Subsidy-only firms showed two peaks—at program launch and near the deadline—reflecting pre-existing investment intentions. Firms receiving combined support concentrated submissions closer to the deadline, typically after working with advisors to refine digital tool selections. Around 5–10% of pre-approved firms did not proceed with financial subsidy claims, despite completing initial documentation. The approval process also varied by group: Technical Advisors reviewed tool alignment for firms receiving Technical Assistance, while Financial Management Specialists oversaw approvals for Financial Subsidy-only participants to ensure neutrality. Pre-approval reviews were completed within 2–3 days, and subsidy disbursements were issued within 5–10 working days.

LESSON LEARNED

FIRMS REQUIRE PROACTIVE REMINDERS EVEN IN SIMPLE PROGRAMS

The pilot implementation revealed that even with streamlined application procedures and reduced administrative burdens, firms still required substantial guidance to complete applications successfully. Key insights include:

- **Provide Active Support Despite Program Simplification**:
  - Even with simplified procedures, firms require consistent guidance to complete applications without relying on external consultants.
  - Business owners need to develop new skills and attitudes to manage the process independently.

- **Prioritize Direct Communication Channels**:
  - Phone calls and direct engagement prove more effective than written manuals or PDF guidelines.
  - Personal contact enables immediate clarification of questions and maintains momentum.

- **Demonstrate Clear Requirements Through Examples**:
  - Share completed document samples to illustrate exact expectations.
  - Provide unambiguous templates for all required documentation.

- **Maintain Regular Follow-up Systems**:
  - Implement systematic deadline reminders.
  - Track application progress to identify potential bottlenecks early.

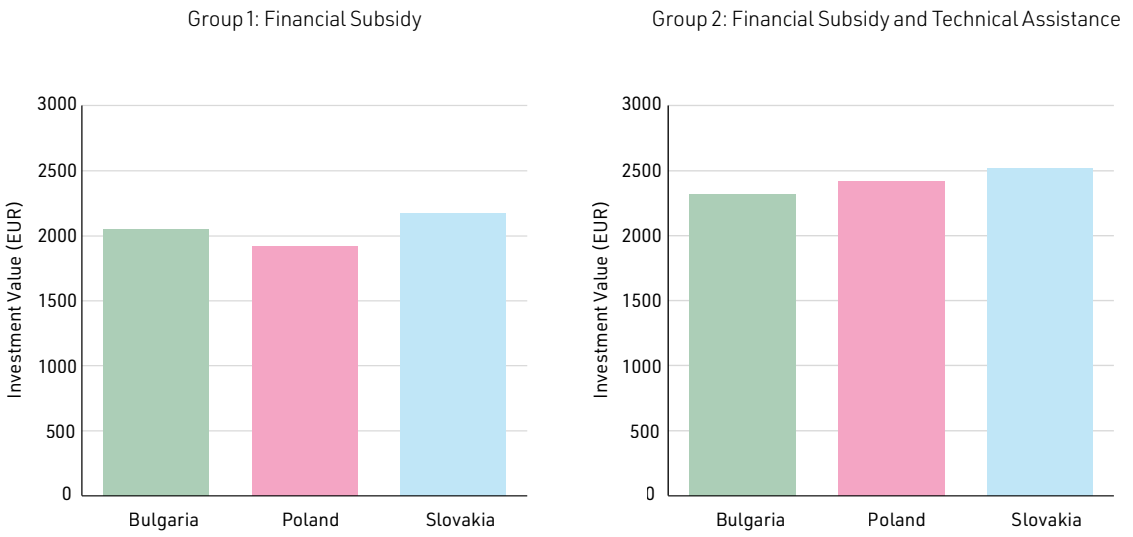
- **Allow Justified Flexibility in Implementation**:
  - Accommodate reasonable deadline extensions when firms face structural constraints:
    - Digital tool trial periods affecting payment schedules
    - Extended vendor negotiations
    - Internal change management processes requiring additional time

Practical Application Components for Future Programs:

- **Application Progress Tracking System**: A digital dashboard for monitoring firm application status with automated alert triggers for follow-up actions
- **Documentation Example Library**: A repository of properly completed application samples for different firm types and funding categories
- **Flexible Deadline Management Framework**: Guidelines for assessing and responding to extension requests with standardized criteria for approval
- **Proactive Communication Sequence**: A template schedule of reminder messages via multiple channels (email, SMS, phone) with escalation protocols
- **Self-Service Application Support Portal**: A structured online resource combining FAQs, video tutorials, and live chat options for immediate assistance

Firms receiving Technical Assistance invested more than those receiving only the Financial Subsidy, contributing additional out-of-pocket funds beyond the program limit. On average, Technical Assistance participants spent approximately 30% more than the EUR 2,000 net subsidy value, covering the additional EUR 500–700 from their own resources. In contrast, most Financial Subsidy-only firms limited their purchases to the reimbursement ceiling (Figure 4.3). Out-of-pocket spending was reported by 42% of firms in the Technical Assistance group, compared to 17% in the Financial Subsidy-only group. Technical Assistance recipients also purchased fewer distinct tools but selected solutions with more user licenses and longer subscription periods, indicating a focus on deeper implementation.

FIGURE 4.3 Average Value and Distribution of Financial Subsidy investments by Group per country



Source: World Bank team.

Technical Assistance recipients prioritized fewer but more advanced and higher-value digital solutions, reflecting a focus on deeper and more strategic implementation. Firms in both treatment groups invested in digital tools targeting similar business functions, with over 70% of tools supporting Administration, Planning, or Marketing activities (Figure 4.4). However, firms receiving Technical Assistance selected more advanced solutions on average, with higher levels of functionality and more frequent inclusion of customization services. Cloud-based SaaS tools accounted for 80% of acquisitions among Technical Assistance recipients, compared to fewer than 50% among Financial Subsidy-only firms (Figure 4.5). Technical Assistance participants also purchased fewer tools overall but selected subscriptions that were over 50% longer in duration and had higher per-user or per-license values (Figure 4.6).

FIGURE 4.4 Distribution of Tools Purchased from Financial Services by GBF

Group 1: Financial Subsidy

T1	Admin	Planning	Sourcing	Marketing	Sales	Payment Methods	Quality Control	Internal Comm	
1	0%	0%	0%	0%	0%	0%	0%	0%	**0%**
2	4%	0%	0%	0%	0%	0%	2%	4%	**10%**
3	0%	0%	2%	4%	2%	0%	0%	2%	**10%**
4	13%	10%	2%	13%	0%	0%	0%	8%	**46%**
5	10%	6%	0%	6%	12%	0%		2%	**35%**
6					0%	0%			**0%**
	27%	**15%**	**4%**	**23%**	**13%**	**0%**	**2%**	**15%**	

Average technological sophistication of technologies purchased **4.04**

Group 2: Financial Subsidy and Technical Assistance

T1	Admin	Planning	Sourcing	Marketing	Sales	Payment Methods	Quality Control	Internal Comm	
1	0%	0%	0%	0%	0%	0%	0%	0%	**0%**
2	3%	3%	0%	0%	0%	0%	0%	5%	**11%**
3	0%	0%	0%	0%	0%	0%	2%	2%	**3%**
4	8%	3%	2%	14%	2%	2%	0%	5%	**34%**
5	12%	9%	3%	11%	8%	0%		8%	**51%**
6					2%	0%			**2%**
	23%	**15%**	**5%**	**25%**	**11%**	**2%**	**2%**	**18%**	

Average technological sophistication of technologies purchased **4.39**

Source: World Bank team.

The initial digital sophistication level correlated with the complexity of purchased tools, though this relationship was weaker among Technical Assistance recipients. Among Financial Subsidy-only firms, higher initial digitalization corresponded with more sophisticated tool purchases, resulting in distinct sophistication levels between more and less digitalized firms. Technical Assistance recipients showed smaller differences in tool sophistication between firms of varying initial digital maturity, with less digitalized firms selecting tools at comparable complexity levels to their more digitalized counterparts. This pattern was particularly evident in Administration, where recipients of Technical Assistance almost universally invested in digital tools at 4 or 5 level of sophistication, compared to beneficiaries of Financial Subsidy than on average opted for tools at level 2 of sophistication.

4.3 TECHNICAL ASSISTANCE

The Technical Assistance component reached high completion rates, with three quarters of beneficiary firms across countries receiving certification. Firms qualified for certification by attending at least 80% of planned Technical Assistance interactions, which included Integrated Assessment, Priority Recommendations, and Personalized Technological Advisory sessions. Over 95% of eligible firms completed the Integrated Assessment phase and received Priority Recommendations, making them eligible for Financial Subsidies. The firms' engagement varied by activity type - while most participated in 2 or more Personalized Technological Advisory sessions, the Group Change Management training saw lower participation rates of 50-70% per session. The evidence shows strong overall engagement in individual advisory activities, with lower participation for group sessions, suggesting firms prioritized personalized support over group training formats.

Project participation was driven by incentives rather than contractual penalties to encourage engagement in Technical Assistance activities. While all firms committed to participate in all project activities by signing Beneficiary

FIGURE 4.5 Division of cloud / desktop of purchased tools per group

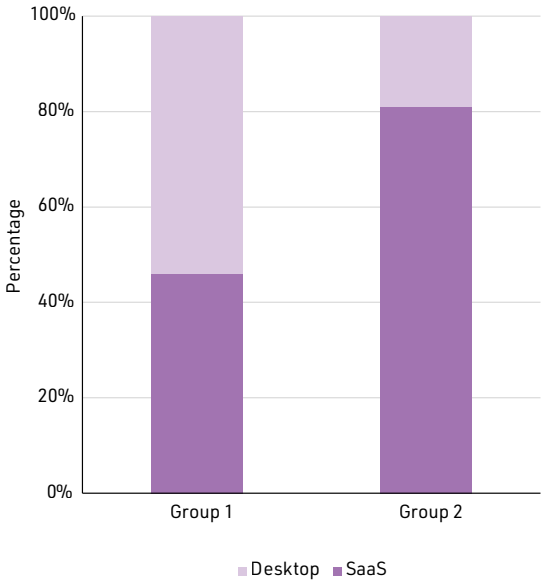
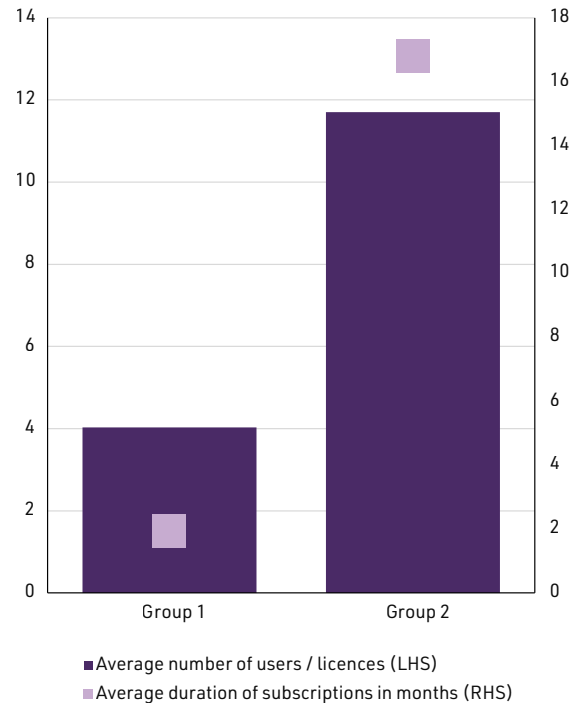


FIGURE 4.6 Number of users and duration of licenses per group



Source: World Bank team.

Agreements, enforcement strategically linked completion of the Integrated Assessment to Priority Recommendations, which were required to access Financial Subsidies. This approach effectively ensured high participation in initial assessment phases. On top of that the project team employed scheduling strategies to drive participation: Personalized Technological Advisory achieved success through flexible, individually scheduled sessions, while Group Training Management experienced lower attendance despite early communication of fixed dates and personalized phone reminders. The data suggests that flexible scheduling combined with financial incentives generated higher participation rates than rigid group schedules, even when well-communicated in advance.

4.3.1 INTEGRATED ASSESSMENT

Priority recommendations in the Integrated Assessment consisted of 3-5 items per firm, concentrating on general business functions, particularly sales and marketing operations which mirrored the areas where Financial Subsidies were most frequently utilized. Marketing recommendations dominated in Bulgaria, while planning and internal communication recommendations were more common in Poland and Slovakia. One-third of firms received priority recommendations related to other general business functions, with administration being the most frequent areas for improvement identified. Other business functions rarely appeared in the priority recommendations. The organizational components of these recommendations primarily focused on establishing processes from scratch, as most firms lacked existing procedures and clear allocation of responsibilities in these general business areas. The technological component of recommendations focused on maximizing usage of existing digital tools and adopting function-specific software solutions where needed, with emphasis on integration between all tools to ensure their effective utilization.

Priority Recommendations remained largely unchanged throughout the program implementation, with individual firms requesting modifications based on evolving knowledge. The initial recommendations, developed through the Integrated Assessment, served as a stable roadmap for firms' technological improvements. Fewer than one in ten firms requested changes or expressed dissatisfaction with their Priority Recommendations during the Assessment phase. Changes occurred when firms gained deeper insights during training sessions and Personalized Technological Advisory, revealing previously unidentified opportunities or constraints. While firms gained additional insights through training sessions and Personalized Technological Advisory, these learnings typically refined implementation approaches rather than altered core recommendations.

4.3.2 GROUP CHANGE MANAGEMENT TRAINING

Training participation showed patterns of engagement, with two thirds of beneficiary firms attending more than half of the available sessions. Of these actively participating firms, 83% developed complete implementation plans for Priority Recommendations that included all curriculum-mandated elements such as timeline, resource allocation, and success metrics. Analysis of attendance records revealed that initial participation strongly predicted continued engagement - firms present at the first two sessions maintained consistent attendance throughout the program, while those absent at the start did not re-engage later. Between training sessions, participating firms generally completed their assigned tasks, often utilizing support from technical advisors during Personalized Advisory Sessions to refine their work. However, the quality of outputs varied significantly, reflecting differences in time and attention firms dedicated to these assignments - ranging from detailed analyses with concrete action steps to basic outlines that met minimum requirements.

LESSON LEARNED

REAL-TIME BENEFICIARY FEEDBACK ENABLES PROGRAM ADAPTATIONS DURING IMPLEMENTATION

The pilot implementation demonstrated significant benefits from establishing continuous feedback mechanisms and allowing for real-time program adaptations. Creating structured channels for beneficiary input throughout the program cycle resulted in meaningful improvements to service delivery:

- **Establish Continuous Feedback Mechanisms:**
  - Provide beneficiary firms with access to feedback forms from program allocation.
  - Enable comments on any program aspect at any time during implementation.
  - Create clear channels for suggesting improvements or reporting issues.

- **Implement Rapid Response to Feedback**:
  - Review feedback regularly with research teams and implementing partners.
  - Make program adjustments during implementation rather than waiting for final evaluation.
  - Document and track changes to measure their effectiveness.

- **Examples of Successful Program Adaptations**:
  - Introduction of group information sessions: Added between Cohorts 1 and 2 in Slovakia. Reduced time spent by advisors explaining program basics. Increased participation rates and development impact.
  - Enhanced Training Accessibility: Added online catch-up sessions for missed in-person trainings. Maintained priority of in-person attendance while providing flexibility.
  - Expanded Advisory Support: Added managerial consulting to address organizational bottlenecks.
  - Broadened technical advisory scope to cover multiple recommendations: Deepened technology integration support based on firm needs, when work on Priority Recommendation was commenced.

Practical Application Components for Future Programs:

- **Continuous Feedback System**: A multi-channel mechanism for collecting, categorizing, and prioritizing beneficiary input throughout program implementation
- **Rapid Adaptation Protocol**: A framework for quickly assessing feedback and implementing appropriate program adjustments
- **Program Flexibility Matrix**: A decision-making tool for determining which program elements can be modified in response to feedback without compromising core objectives
- **Adaptation Impact Tracking Tool**: A methodology for measuring the effects of mid-course program changes on participation, satisfaction, and outcomes
- **Cross-functional Adaptation Team Structure**: Guidelines for assembling and operating teams responsible for implementing and monitoring program adjustments

Training participation rates remained consistent across different delivery formats, including online sessions, in-person training in capital cities, and regional locations. Even firms that confirmed attendance sometimes missed sessions, with two main reasons reported: urgent business priorities that emerged after confirmation and lack of backup personnel to cover for the designated participant's regular duties. To maintain engagement levels, each country implemented specific make-up strategies for firms that missed scheduled sessions, with over 50% of absent firms utilizing these catch-up options. The approaches varied by country - Bulgaria organized additional online sessions with trainers, Poland provided recordings of training sessions, and Slovakia offered individual consultations with management trainers to help firms complete their change implementation plans. These flexible attendance solutions helped firms stay on track with program requirements despite occasional absences from primary training sessions.

4.3.3 PERSONALIZED TECHNOLOGICAL ADVISORY

Over 90% of eligible firms completed at least two advisory sessions, with technical staffing emerging as a key differentiator in reaching advisory objectives. Firms with in-house technical personnel achieved advisory objectives in half the time, requiring an average of two meetings compared to four meetings for firms without technical staff. This technical expertise also influenced preparation levels – most firms with technical staff consistently identified their preferred investment tools from the Priority Recommendation before the first advisory session, while firms without technical personnel rarely made this determination independently. Firms with higher participation in Change Management were also more engaged in the Technological Advisory Training engagement correlated with higher program utilization overall, as firms attending more training sessions averaged 3.4 advisory sessions and 9.5 advisor hours, compared to 2.7 sessions and 6.8 hours for firms with lower training participation.

The Beneficiary Handbook served as a centralized, cloud-based document repository for all project activities and materials. This individualized document shared with beneficiary firms over the cloud drive, structured as a business presentation, consolidated advisory outputs, training materials, and firms' own work products throughout the program implementation cycle. The Handbook's structure followed the project's progression - starting with process overview, then expanding with assessment reports, training materials, and technological advisory documentation as these activities were completed. Each firm's Handbook incorporated inputs from multiple project experts, including at least two advisors and one trainer, along with supporting annexes and documentation. This approach provided firms with a comprehensive record of their project journey, enabling them to track progress during implementation and reference materials for future organizational changes or technology adoption initiatives.

4.4 FEEDBACK FROM IMPLEMENTATION

Digitrans program participant data shows consistent results across all three implementing countries. Firms rated program support delivery above 8.0 on a 10-point scale in each country, with two-thirds of individual interactions receiving satisfaction scores above 85% and the remaining third above 70%. The program components delivered specific outcomes: 95% of firms received relevant recommendations from Integrated Assessments, 85% completed Priority Recommendation preparation during Personalized Technological Assessment, and 70% completed change implementation plans over the course of Group Change Management Training. The application process, particularly the endline interview step which firms reported

as unnecessary and mundane, was identified as the main area requiring improvement. Nine out of ten participating firms reported recommending the program to other business owners.

Implementation partners reported on the feasibility of delivering planned support in allocated time frames, noting high relevance of the structured process for beneficiary firms. Companies participated actively in both online and in-person consultations, meeting deadlines and openly sharing information about their processes during site visits. Advisors identified distinct support needs based on firm size - companies with 20 or more employees typically sought specific digitalization solutions, while smaller firms required fundamental business support in sales and business model development. However, the combination of technical and management advisory proved effective across all firm sizes, with design of the support program having enough flexibility to adapt to needs of heterogenous firms from the target group. Implementation partners emphasized that streamlined administration enabled firms to focus on implementation, while the coordinated timing of different support activities maintained consistent engagement throughout the program.

LESSON LEARNED

ENGAGING IMPLEMENTING PROVIDERS INTO FEEDBACK LOOPS AND LEARNING

Implementing partners identified several operational challenges during the program's execution and offered valuable suggestions for optimization. Their feedback, collected systematically throughout implementation, highlighted key areas for improvement:

- **Consultant Preparation and Time Allocation**
  - Service providers initially lacked adequate preparation and coordination time allocation
    - Introduce standardized 1-hour preparation block for consultants to review client data and align approach
    - Implement direct consultant-client scheduling system
    - Establish clear travel compensation framework based on distance/time metrics

- **Client Preparation Process**
  - Pre-engagement client preparation required streamlining
    - Launch standardized webinars for all participating companies before individual engagements
    - Provide benchmark reports prior to first consultations
    - Create visual process guides (infographics) outlining program stages and financial allocations

- **Support Scope Expansion**
  - Companies required broader business support beyond digital transformation
    - Expand program scope to include optional modules: change management, sales, business models, communication

        - Implement parallel management and technical consulting throughout the program duration
        - Allow flexibility in allocating consulting hours between technical and management support based on company needs

- **Post-Implementation Support**
    - Post-implementation support emerged as critical success factor
        - Structure extended support phase combining technical setup and process adjustment
        - Establish follow-up consulting sessions for companies implementing multiple tools
        - Create framework for measuring benefits after one year of implementation

- **Administrative Approach**
    - Program administration approach impacted participation
        - Maintain minimal administrative requirements during implementation
        - Create flexible meeting formats combining online and in-person sessions
        - Allow companies to adjust training focus based on specific needs

Practical Application Components for Future Programs:

- **Consultant Preparation Protocol**: A standardized process for consultant preparation including templates for client profile reviews and inter-consultant coordination
- **Client Engagement Roadmap**: A visual framework outlining program stages, expectations, and milestones for participating firms
- **Modular Support Menu**: A configurable set of support modules allowing firms to customize their assistance package based on specific needs
- **Extended Support Framework**: A structured approach to post-implementation support with clear objectives, methodologies, and success metrics
- **Administrative Minimization Toolkit**: Guidelines for reducing administrative burden while maintaining necessary oversight and documentation

5

IMPACT EVALUATION

This chapter presents findings from the Digitrans pilot program, analyzing how different support schemes impacted small business digitalization and managerial capabilities. Based on a rigorous randomized controlled trial (RCT) methodology with over 600 participating MSMEs across the three countries, the impact evaluation compares the effectiveness of providing (i) Financial Subsidy alone and (ii) Financial Subsidy combined with Technical Assistance against a comparison group receiving only benchmarking reports. The analysis examines multiple outcomes, including digital tools adoption, usage patterns, managerial practices, and investment behaviors, and discussed how these findings could inform future policies aiming to enhance MSME competitiveness.[20] Details on the estimation strategies, outcome types, and regression specifications are provided in the Box 5.1 below.

BOX 5.1 Estimating Pilot Effects

The evaluation follows a structured data collection process: firms completed an application form, participated in a baseline survey, were randomly assigned to one of three groups (Financial Subsidy only, Financial Subsidy + Technical Assistance, or comparison group), received their assigned interventions, and then completed a follow-up survey approximately 3 months post-intervention.

The impact evaluation draws on experimental data from a RCT conducted across Bulgaria, Poland, and Slovakia. To estimate the effects of different support interventions on firm outcomes, the impact evaluation includes baseline measures collected during the application form and a baseline survey on all groups, as well as a follow-up measure collected through a survey on all groups 2-3 months post interventions.

The analysis follows a consistent regression framework:

- **For outcomes measured at both baseline and endline** treatment effects are estimated using an **ANCOVA regression estimation**. This approach improves statistical precision by controlling for initial outcome levels.
  - The specification includes controls for:
    - Baseline value of the outcome variable
    - Stratification (strata) fixed effects[21]
- **For outcomes measured only at endline**, effects are estimated using a linear regression model that includes stratification (strata) fixed effects, but exclude baseline values.
- **Balance tests at baseline and for those at** are analyzed using descriptive statistics and mean difference testing to assess data quality and randomization validity.
- Across all outcomes, results are presented as **treatment effects** for the Financial Subsidy (FS) and Financial Subsidy + Technical Assistance (FS+TA) groups relative to the comparison group.

This estimation strategy ensures comparability across outcomes and enhances the validity of the causal interpretation of treatment effects.

Source: Digitrans team.

The remainder of this chapter presents the empirical findings structured around key thematic areas: program participation and attrition, digital tool adoption and usage, firm capabilities and practices, and heterogeneity of effects across different types of firms and owners. Each section highlights where the interventions led to statistically and economically meaningful changes, drawing on the methodological approach outlined above.

20. McKenzie, D. (2020). *Small Business Training to Improve Management Practices in Developing Countries: Reassessing the Evidence*. World Bank Research Observer.
21. Stratification was based on country (Bulgaria, Poland, Slovakia), sector group (services vs. manufacturing & construction), and baseline digitalization level (low, medium, high). These strata fixed effects are included as controls in all regression models.

5.1 PROGRAM PARTICIPATION AND ATTRITION ANALYSIS

This section examines survey participation patterns and their implications for the validity of impact analysis results. Through systematic analysis of response rates, selection patterns, and balance tests, we examine if there are systematic differences at baseline between responding firms during follow-up surveys.

5.1.1 SURVEY RESPONSE RATES

High response rates and balanced experimental groups at baseline provide a methodologically robust foundation for impact analysis. The baseline survey achieved 100% response rate as completion was a prerequisite for program participation, while the endline survey conducted 2-3 months post-intervention achieved response rates of nearly 90% among firms that received either Financial Subsidy or both Financial Subsidy and Technical Assistance, compared to approximately two-thirds participation rate among comparison group firms (Table 1). This pattern was consistent across all three countries, with treated firms responding at rates approximately 20-30 percentage points higher than comparison group firms (Figure 1). The response rate remained similar between FS and FS+TA recipients, indicating that any support beyond the benchmarking report equally motivated firms to complete the endline interview. Slovakia achieved the highest overall response rate at 83%, followed by Bulgaria at 81% and Poland at 67%. While differential attrition occurred between treatment and comparison groups, the maintained balance in baseline characteristics across respondents (as shown in the following section) ensures valid counterfactual analysis of intervention impacts.[22]

TABLE 1 Survey Response Rates

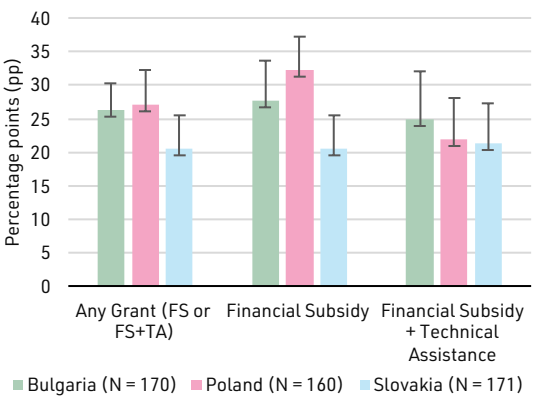
Per group, per country

	Overall	FS	FS+TA	Comparison
All countries	81%	90%	86%	63%
	501	*189*	*180*	*132*
Bulgaria	81%	89%	86%	61%
	170	*64*	*62*	*44*
Poland	78%	93%	82%	61%
	160	*63*	*56*	*41*
Slovakia	83%	89%	89%	68%
	171	*62*	*62*	*47*

Source: Digitrans team.

FIGURE 1 Treatment Effects on Response Rate

Per group, per country



Note: Treatment effects on survey response rate are shown in percentage points, per group and per country.

22. Gertler et al. (2016). *Impact Evaluation in Practice*, 2nd Ed., World Bank.

5.1.2 SELECTION ANALYSIS

Respondents and non-respondents exhibited statistically equivalent baseline characteristics, suggesting the limited risk of systematic selection patterns biasing results. Analysis comparing baseline characteristics of respondent and non-respondent firms revealed minimal differences across multiple measured dimensions. In Slovakia, respondents differed from non-respondents only in the Digital Practices Systems Sub-Index, with this difference being small and statistically significant only at the 10% level. With only 3 out of 36 comparisons showing statistically significant differences between respondents and non-respondents, the analysis confirms minimal selection bias. This ensures that differential response rates do not compromise the validity of impact estimates (Figure 2).

FIGURE 2 Baseline Characteristics Comparison: Respondents vs. Non-Respondents

Per country

Balanced
*p<0.1
**p<0.05
***p<0.01

Baseline Characteristic	Bulgaria	Poland	Slovakia
Size (employment)	0.36	-1.36	-1.89
	0.781	*0.196*	*0.149*
Share of manuf. Firms	-0.01	-0.01	-0.1
Stratification variable	*0.849*	*0.784*	*0.784*
Revenue log EUR	0.47	0.21	-0.27
	0.447	*0.23*	*0.363*
Profit margin pct.	-3.71**	0.12	0.01
	0.045	*0.196*	*0.688*
Total Assets log EUR	0.47	-0.26	-0.19
	0.262	*0.469*	*0.697*
Labor productivity EUR	37.02	63.16	24.72
	0.305	*0.717*	*0.578*
Firm Age years	1.86	1.35	-0.95
	0.208	*0.242*	*0.54*
GBF Intensive Index	0.03	-0.74	-0.03
Stratification variable	*0.941*	*0.747*	*0.974*
GBF Extensive Index	-0.05	-0.01	-0.04
	0.572	*0.873*	*0.7*
Digital Practices Index	0.4	-1.17	3.25
	0.878	*0.47*	*0.211*
Systems Sub-Index	5.4	5.83	7.68*
	0.219	*0.128*	*0.094*
Information Sub-Index	3.61	-1.49	2.03
	0.255	*0.708*	*0.452*

Note: Table shows differences in means between respondents and non-respondents for each baseline characteristic.

Green cells indicate balanced characteristics with no statistically significant differences.
Out of 36 total comparisons, 3 (8.3%) show statistically significant differences.

Source: Digitrans team.

5.1.3 BALANCE TESTS AMONG RESPONDENTS

The research sample maintained balance across respondents from treatment groups despite differential response rates, further strengthening the evaluation's internal validity. Statistical tests across dozens of baseline variables, including firm size, sector, age, revenue, digital maturity indices, profit margins, labor productivity, and owner characteristics, revealed no systematic differences between firms that responded to the endline survey across the three experimental groups. The maintained balance was evident both at the pooled level across all countries and within individual country samples, with imbalance limited to just one characteristic in the pooled analysis and one each in Bulgaria and Poland, while Slovakia maintained complete balance between groups (Figure 3). With only 3 of the 108 total comparisons showing any statistically significant differences, the overall balance achieved high standards for experimental validity. This balanced composition is critical for valid causal inference, as it increases the chance that any observed differences in outcomes can be attributed to the program interventions rather than to pre-existing firm characteristics.

FIGURE 3 Balance of Baseline Characteristics Across Treatment Groups

Per group, per country

Balanced

*p<0.1

**p<0.05

***p<0.01

Note: Table shows balance of baseline characteristics across treatment groups. "Pooled" columns show differences with p-values for all countries combined. Country columns use color coding with stars to indicate statistical significance. Green cells indicate balanced characteristics (p ≥ 0.1).

C = Control group, FS = Financial Subsidy only, FS+TA = Financial Subsidy + Technical Assistance.

Out of 108 total comparisons (12 variables × 3 pairwise comparisons × 3 countries), only 3 (2.8%) show statistically significant differences.

Baseline Characteristic	Pooled (All countries)			Bulgaria			Poland			Slovakia		
	C vs FS	C vs FS+TA	FS vs FS+TA	C vs FS	C vs FS+TA	FS vs FS+TA	C vs FS	C vs FS+TA	FS vs FS+TA	C vs FS	C vs FS+TA	FS vs FS+TA
Size (employment)	0.86	1.12	0.26									
	(0.80)	*(0.83)*	*(0.79)*									
Share of manuf. Firms	-0.03	-0.02	0.01									
Stratification variable	*(0.05)*	*(0.05)*	*(0.04)*									
Revenue log EUR	-0.07	-0.21	-0.15									
	(0.19)	*(0.21)*	*(0.20)*									
Profit margin pct.	0.01	0.05	0.04									
	(0.02)	*(0.05)*	*(0.05)*									
Total Assets log EUR	-0.16	-0.02	0.14									
	(0.26)	*(0.25)*	*(0.25)*									
Labor productivity EUR	416.1	878.65	924.75									
	(558.93)	*(848.28)*	*(940.91)*									
Firm Age years	0.16	0.46	0.31									
	(0.97)	*(1.01)*	*(0.90)*									
GBF Intensive Index	0.03	0.06	0.03									
Stratification variable	*(0.05)*	*(0.05)*	*(0.05)*									
GBF Extensive Index	0.06	0.07	0.01									
	(0.06)	*(0.06)*	*(0.06)*									
Digital Practices Index	-0.18	0.15	0.33									
	(1.70)	*(1.79)*	*(1.66)*									
Systems Sub-Index	-3.56	0.37	3.93									
	(3.03)	*(3.11)*	*(2.76)*									
Information Sub-Index	1.18	-0.57	-1.75									
	(2.22)	*(2.26)*	*(2.17)*									

Source: Digitrans team.

5.2 IMPACT ON DIGITAL ADOPTION AND USAGE

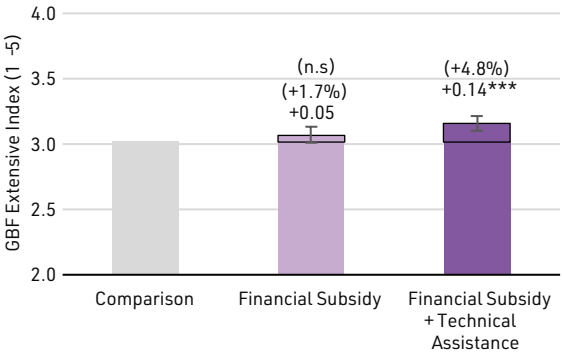
This section explores how the two sets of interventions impacted the firms' adoption of technologies, and the extent to which these technologies were integrated into usage in business operations. By examining both the extensive margin (adoption of new tools) and intensive margin (usage of adopted tools), we provide a comprehensive assessment of how financial and technical support influence digitalization outcomes across different business functions and firm types.

5.2.1 TECHNOLOGY ADOPTION

The combination of Financial Subsidy and Technical Assistance yielded modest but statistically significant improvements in digital technology adoption, while Financial Subsidy alone produced minimal and non-significant effects. In the combination of all countries, firms receiving the combined package of FS + TA had a 4.8% increase on average in the Technology Adoption GBF Extensive Index (+0.14 points, p<0.01) compared to the comparison group (Figure 4). Country-specific analysis reveals this effect was driven primarily by firms in Slovakia (+0.17 points, p<0.10) and Poland (+0.13 points, p<0.05), with Bulgarian firms showing no significant improvement (Figure 5). The impact of the combined FS+TA support, while appearing modest, represents adopting on average one more advanced tool in one business function–equivalent to half the difference between adoption scores in the least digitalized quarter of firms and the median firms in Poland. This suggests that even small improvements in adoption metrics can represent meaningful technological advancement in specific business functions.

FIGURE 4 Impact on Digital Technology Adoption Index

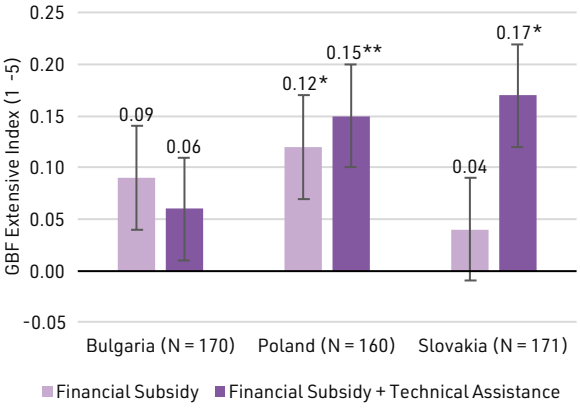
Per group, all countries



*Note: * p < 0.10; N = 432; "n.s." = not statistically significant.*

FIGURE 5 Technology Adoption Impact

Per group, per country



*Note: * p < 0.10, ** p < 0.05; N = 432.*

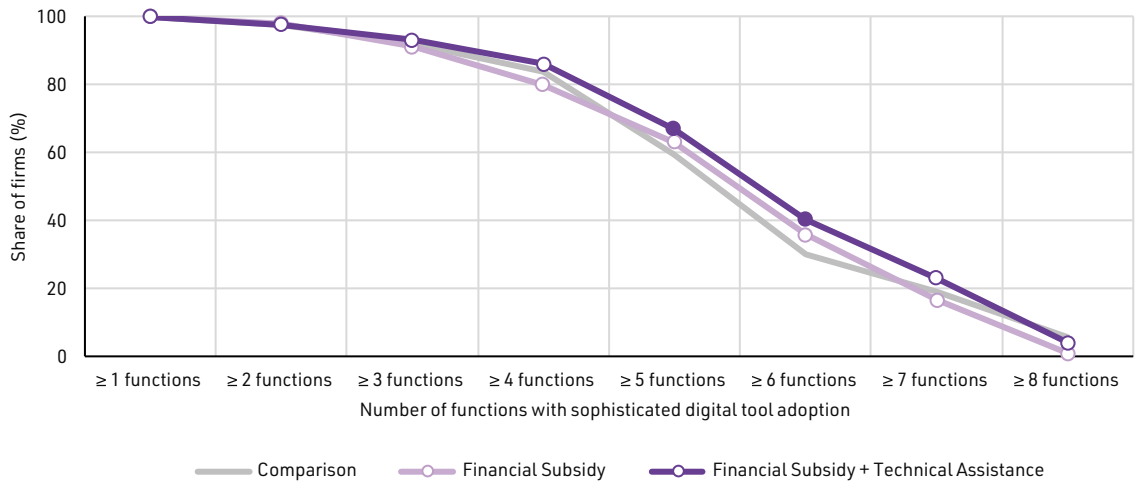
Source: Digitrans team.

Firms in all groups, including those receiving only benchmarking support, increased their technology adoption during the six-eight month evaluation period. The comparison group demonstrated improvements in their technology adoption scores between baseline and endline measurements, reflecting the strong market-driven digitalization trend affecting all MSMEs in the pilot countries. The lack of statistically significant differences between the Financial Subsidy only group and the comparison group suggest that financial support alone may have primarily substituted investments firms would have made with their own resources anyway, essentially crowding out private investment rather than generating additional adoption. In contrast, the statistically significant improvements observed in the FS+TA group indicate that complementary support enabled firms to adopt more sophisticated technologies beyond natural trends.

The interventions produced measurable differences in the breadth of sophisticated digital tool adoption across business functions. While many firms in all groups adopted sophisticated technologies in at least one or two business functions, larger divergences across the three groups appeared at higher levels of usage of sophisticated technologies (Figure 6). The firms receiving FS+TA support were 10 percentage points more likely than those in the comparison group to adopt sophisticated digital technologies in three or more general business functions. The greatest differences emerged at the five to six-function thresholds, with the FS+TA group showing a 15-20 percentage point advantage over the comparison group (70% vs. 50% at five functions and 45% vs. 25% at six functions). These patterns indicate that Technical Assistance provided a meaningful advantage in supporting broader digitalization, enabling firms to extend sophisticated tool use across multiple areas of operation–an effect not observed, statistically, among firms that received only the financial subsidy.

FIGURE 6 Adoption of Sophisticated Digital Tools Across Business Functions

Per group, all countries



Note: N = 384. Markers indicate statistical significance: full marker = significant; empty marker = not significant.

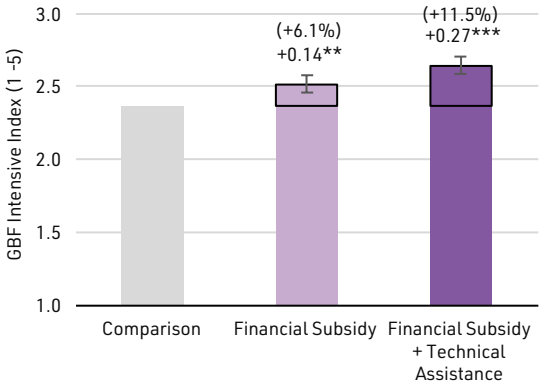
Source: Digitrans team.

5.2.2 TECHNOLOGY USAGE

The combined Financial Subsidy and Technical Assistance enhanced the usage intensity of digital tools, generating impacts over two times larger than the Financial Subsidy alone. The digital usage index, measuring the sophistication of most frequently used digital tools, increased by 0.27 points (+11.5%, p<0.01) relative to the comparison group among those receiving the combined Technical Assistance and Financial Subsidy, compared to 0.14 points (+6.1%, p<0.10) for the Financial Subsidy-only firms (Figure 7). This pattern was consistent across countries, with particularly strong effects in Slovakia (+0.29 points, p<0.01) and Poland (+0.21 points, p<0.05) (Figure 8). Qualitative insights from pilot implementation suggest that the beneficiaries of the combined FS+TA not only started to use newly acquired tools more frequently, but they have also increased usage of the most advanced tools they had adopted prior to participation in the pilot instrument but kept underutilized. This differential impact on usage suggests capability-building components were effective in helping firms operationalize their digital investments beyond simply acquiring new technologies.

FIGURE 7 Impact on Digital Technology Usage Index

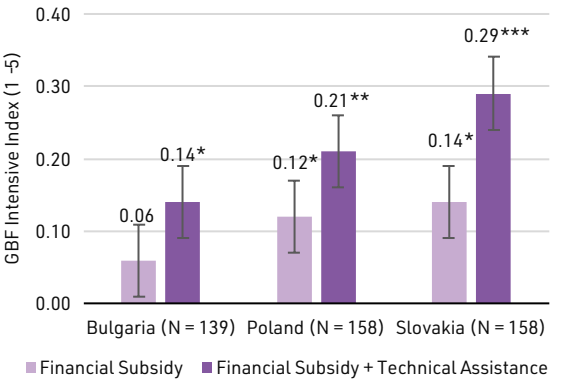
Per group, all countries



*Note: * p < 0.10; *** p < 0.01; N = 395.*

FIGURE 8 Technology Usage Impact

Per group, per country



*Note: * p < 0.10; ** p < 0.05; *** p < 0.01; N = 384.*

Source: Digitrans team.

Increase in digital tool usage over time were modest for firms without complementary support, contrasting with the more robust background trend in technology adoption. While comparison group firms showed substantial improvements in technology adoption during the evaluation period, their usage intensity increased minimally without intervention. This suggests that market pressures may drive initial technology acquisition, but effective implementation requires overcoming additional barriers, which is consistent with the result of the survey data analysis of MSME digitalization in Bulgaria, Poland, and Slovakia preceding the design of the pilot instrument.[23] Financial Subsidy alone generated limited usage improvements beyond the

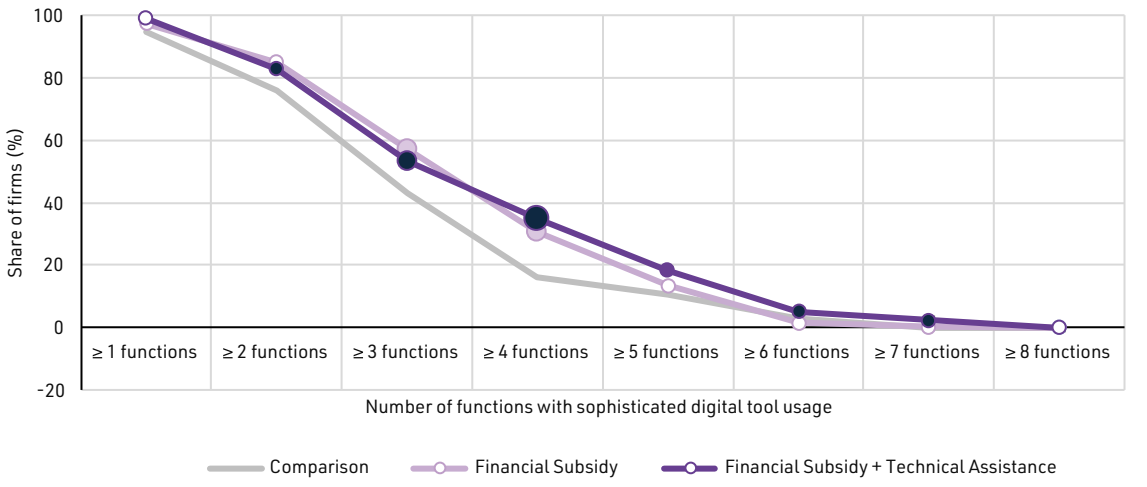
23. World Bank. (2024). *Gaps and Opportunities - Country-specific reports on Programs, Institutions and Barriers.*

counterfactual trend, while adding Technical Assistance more than doubled the impact. This contrasts with adoption findings, suggesting that capability constraints may represent a more persistent obstacle to usage than to initial technology acquisition.

The pilot instrument significantly increased the breadth of sophisticated digital tool usage across business functions. On average across all three countries, approximately 70% of FS+TA firms use advanced digital technologies in three or more business functions compared to 55% of FS-only firms and just 40% of comparison group firms (Figure 9). This divergence becomes more pronounced at higher thresholds, with about 45% of FS+TA firms using sophisticated technologies in four or more functions versus 35% of FS-only firms and only 15-20% of comparison firms. The advantage of complementary support persists through six or more functions, while the FS-only advantage over the control group diminishes beyond four functions. This pattern demonstrates how technical assistance can be a strong instrument to promote digitalization across multiple business areas, enabling firms to integrate sophisticated tools more broadly throughout their operations rather than in isolated functions.

FIGURE 9 Usage of Sophisticated Digital Tools Across Business Functions

Per group, all countries



Note: N = 356. Markers indicate statistical significance: full marker = significant; empty marker = not significant.

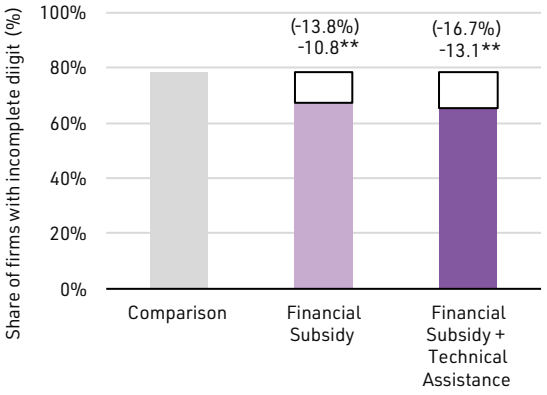
Source: Digitrans team.

Both variants of the pilot instrument reduced the prevalence of incomplete digitalization to a similar degree. Incomplete digitalization is defined as a situation where the firm is not most frequently using the most sophisticated digital tool they have adopted for a given business function, essentially leaving advanced capabilities untapped. The share of firms with incomplete digitalization decreased by 11.5% in the FS+TA group and by 13.4% in the FS-only group compared to the comparison group (Figure 10). This improvement was most pronounced in Bulgaria (-16% for FS and -17% for FS+TA), with smaller but still significant reductions in Poland and Slovakia (Figure 11). However, the similar performance of both intervention types

should be interpreted carefully. FS-only firms adopted fewer new sophisticated tools overall, making it relatively easier for them to fully utilize their smaller digital portfolio. In contrast, FS+TA firms adopted a higher number of new tools, which extended the potential for incomplete digitalization.

FIGURE 10 Incomplete Digitalization Impact

Per group, all countries

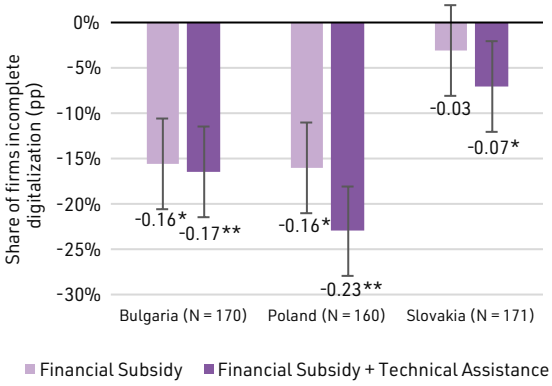


*Note: * p < 0.10; *** p < 0.01; N = 432.*

Source: Digitrans team.

FIGURE 11 Reduction in Incomplete Digitalization

Per group, per country

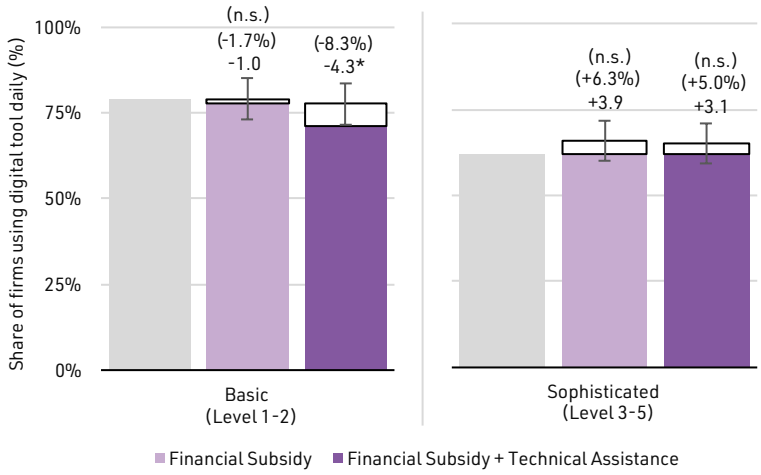


*Note: * p < 0.10; N = 432.*

Neither intervention variant significantly impacted the frequency of tool usage or broadened the number of users within participating firms. The data shows no statistically significant differences across groups in the share of firms using daily either basic or sophisticated tools (Figure 12). While the share of firms in the pilot interventions groups using basic digital tools decreased and conversely there was an increase relative to the control of group in the share of firms using sophisticated tools, these differences are small and not significant. Neither intervention variant significantly impacted the distribution of digital tool users across organizational levels. CEO/owner usage remained consistently high across all groups, while usage patterns among senior staff, junior staff, and external collaborators showed only minimal variations between intervention and comparison groups. Although the FS+TA group demonstrated slightly higher usage rates at some organizational levels, these differences were not statistically significant (Table 2). Qualitative insights suggest that while implementing narrower organizational changes in firms–focusing primarily on modifying how existing users utilize digital tools–is manageable in the short term, triggering and implementing broader organizational changes in digital usage patterns across organizations requires more time and intensity than the three-month intervention period provided.

FIGURE 12 Frequency of Digital Tool Usage by Sophistication Level

Per group, all countries



Note: N = 429. Differences are not statistically significant.

Source: Digitrans team.

TABLE 2 Digital Tool Users by Organizational Role

Per group, all countries

	Basic (Level 1-2)			Sophisticated (Level 3-5)		
	Comparison	Financial Subsidy	Financial Subsidy + Technical Assistance	Comparison	Financial Subsidy	Financial Subsidy + Technical Assistance
Owner	96.8	95.2	95.0	89.0	90.0	92.0
	(0.9)	*(1.1)*	*(1.1)*	*(1.6)*	*(1.5)*	*(1.4)*
CEO	30.5	37.0	29.2	36.8	36.5	43.8
	(2.3)	*(2.4)*	*(2.3)*	*(2.4)*	*(2.4)*	*(2.5)*
Director or Lead staff	15.5	4.0	3.2	4.8	12.5	8.0
	(1.1)	*(1.0)*	*(0.9)*	*(1.1)*	*(1.3)*	*(1.4)*
Senior staff	29.0	32.2	38.8	31.5	28.8	42.0
	(2.3)	*(2.3)*	*(2.4)*	*(2.3)*	*(2.3)*	*(2.5)*
Junior staff	14.8	30.2	32.8	23.8	34.5	38.0
	(1.8)	*(2.3)*	*(2.3)*	*(2.1)*	*(2.3)*	*(2.4)*
External collaborators	0.0	0.0	0.0	16.5	16.2	19.5
	(0.0)	*(0.0)*	*(0.0)*	*(1.9)*	*(1.8)*	*(2.0)*

Share of firms where a given group of employees are using tool <10% 10-29% 30-59% 60%+

Note: Numbers in the table show the percentage of firms where each organizational role uses digital tools at end-line; N = 391

Source: Digitrans team.

5.2.3 BUSINESS FUNCTION-SPECIFIC EFFECTS

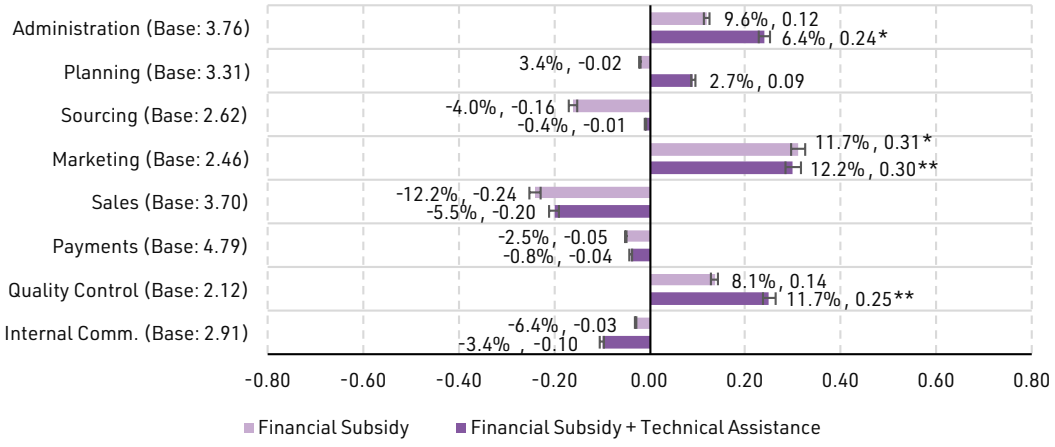
Following from the general results, we explore in this section how the impacts of Financial Subsidy alone versus combined Financial Subsidy and Technical Assistance differed across the eight general business functions analyzed. By studying function-specific effects, we identify which business operations are most responsive to different support types and where complementary assistance generates the greatest marginal value, insights that can help optimize future digitalization support targeting and intensity.

Technology Adoption

At the level of individual general business functions, the combined support drove adoption in 3 out of 8 GBFs, while financial subsidy alone improved adoption in one function. The combined FS+TA intervention produced statistically significant improvements in Administration, Marketing, and Quality Control, with effects equivalent to half of the firms in this group adopting a higher level of digital sophistication in these business functions (Figure 13). Financial Subsidy alone generated a significant improvement in Marketing, of a similar magnitude to the combined support. These patterns indicate that while financial resources alone can drive meaningful improvements, combining with technical assistance provides additional value. Both interventions showed negative effects in the adoption of a Sales digital tool, though this result is not statistically significant. Payments, which started with the highest baseline sophistication level, saw minimal changes across all groups.

FIGURE 13 Technology Adoption Impact by Business Function

Per group, all countries



*Note: * p < 0.10; *** p < 0.01; N = 387.*

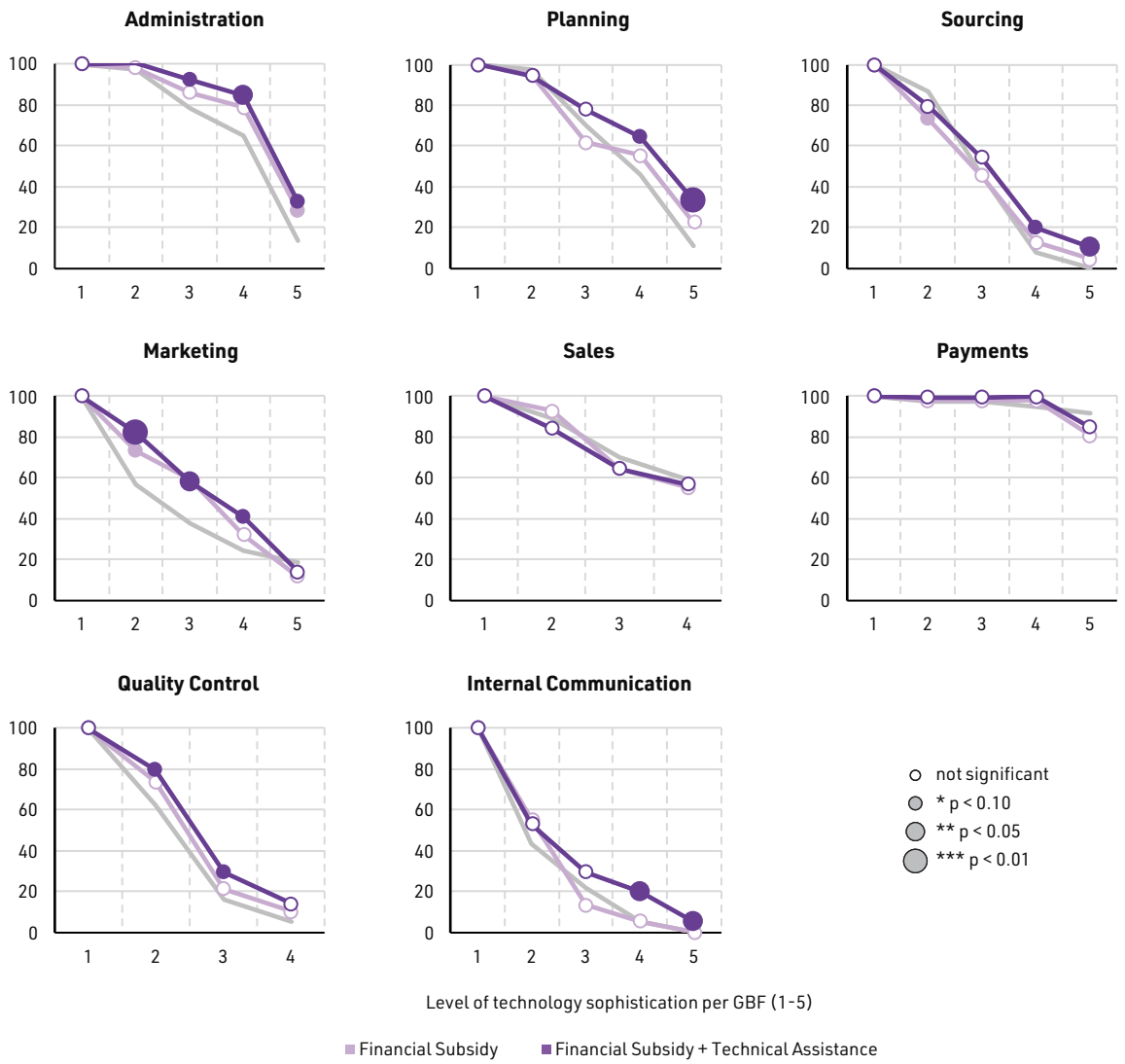
Source: Digitrans team.

The combined FS+TA increased the share of firms adopting sophisticated digital tools in all business functions except Sales and Payments. Firms receiving complementary support achieved higher adoption rates of Level 3+ technologies in six out of eight business functions compared to other groups (Figure 14). The highest change was observed in Planning,

where 40% of FS+TA recipients adopted ERP class systems versus 20% of FS-only firms and 15% of control group firms. Marketing showed similar patterns, with FS+TA firms maintaining significantly higher adoption of technologies across all sophistication levels. These patterns indicate that technical assistance enabled firms to adopt more sophisticated technologies across the entire spectrum, stimulating transitions from level 2 to level 3 technologies as well as from level 4 to level 5. This demonstrates the intervention's relevance for firms at different points in their digitalization journey.

FIGURE 14 Adoption of Advanced Technologies by Business Function

Per group, all countries



Note: N = 397

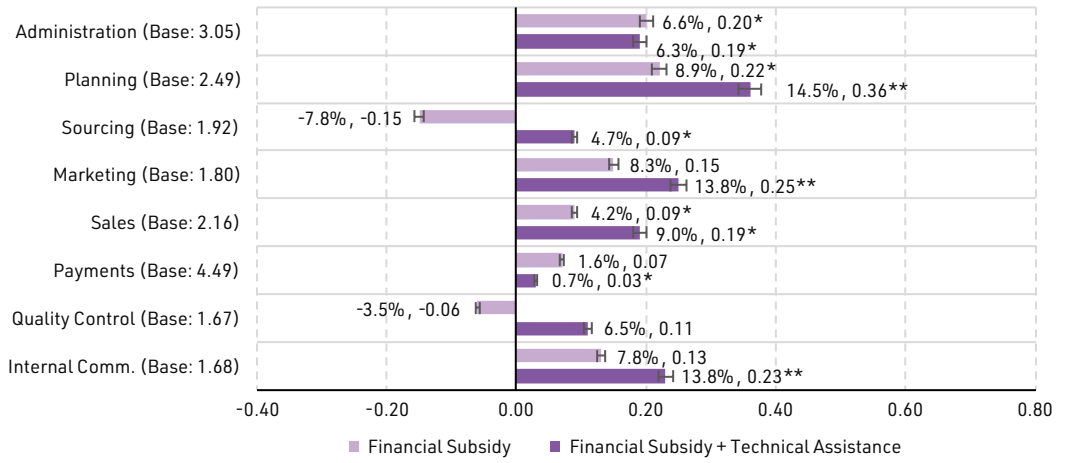
Source: Digitrans team.

Technology Usage

The Financial Subsidy coupled with Technical Assistance improved the sophistication level of technologies most frequently used across six business functions, while the Financial Subsidy alone improved tools used in only two functions. FS+TA firms showed improvements in Planning, Marketing, Sales, and three other functions, with the largest gains in Planning (Figure 15). The scale of improvement in Planning corresponded to roughly half of complementary support recipients advancing by one level in sophisticated tools, with smaller but meaningful improvements in other functions. FS-only firms demonstrated significant improvement in usage only in Administration and Planning, with gains comparable to the FS+TA group in these particular functions. These results suggest that complementary capability-building is essential for maximizing technology usage across diverse business operations, especially those not driven by external regulations and requiring changes to established workflows and staff competencies.

FIGURE 15 Technology Usage Impact by Business Function

Per group, all countries



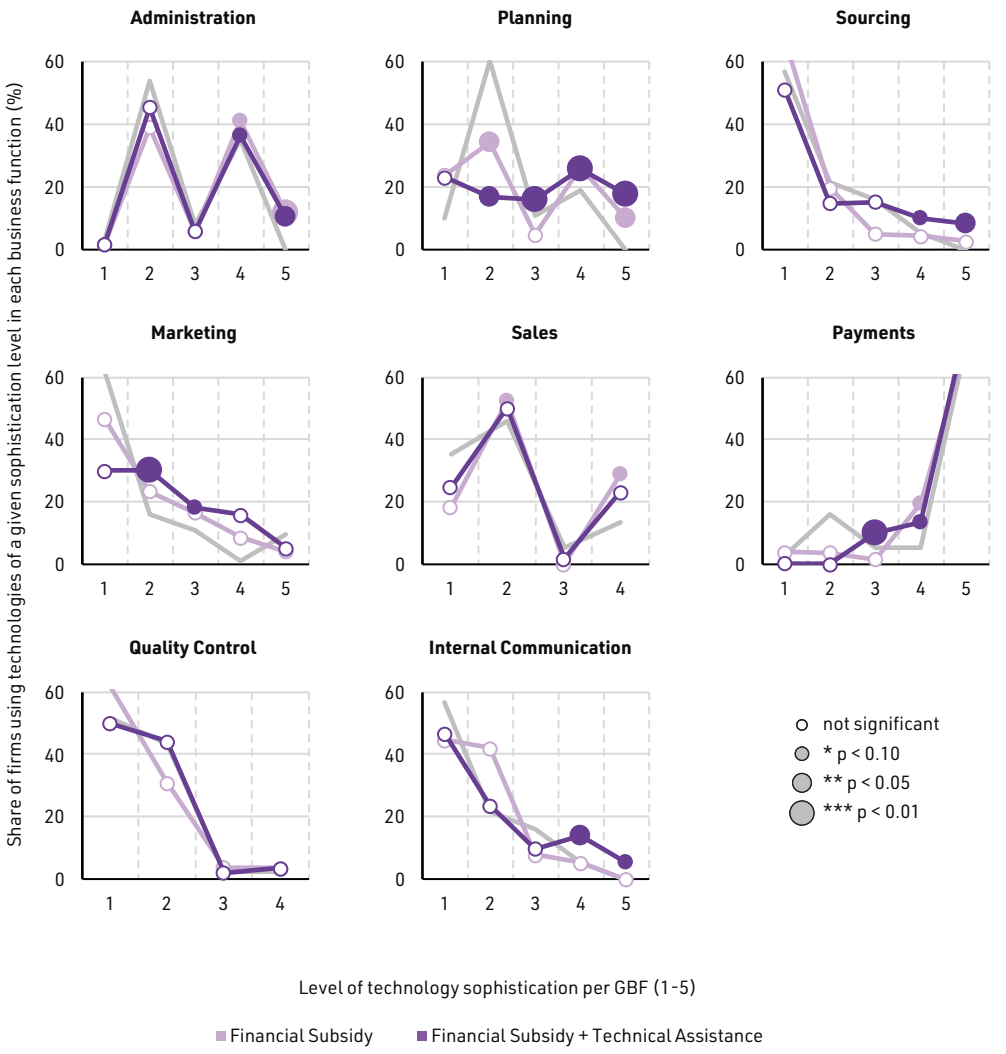
*Note: * p < 0.10; *** p < 0.01; N = 378*

Source: Digitrans team.

The combined support shifted the distribution of technologies most frequently used by firms across business functions. In Planning, approximately one-third of the FS+TA firms–the largest segment within this group–regularly used sophisticated digital tools such as dedicated planning platforms and WorkOS systems, compared to predominant reliance on Excel-type tools in both the FS-only and comparison groups (Figure 16). Similar shifts occurred in Marketing, where FS+TA firms most frequently used Excel to gather customer information and develop products, while the other groups primarily relied on pen and paper methods. While Financial Subsidy alone sometimes decreased the share of firms using basic tools and increased adoption of more advanced options, complementary support more consistently triggered broader shifts toward sophisticated digital tool usage across multiple business functions.

FIGURE 16 Usage of Advanced Technologies by Business Function

Per group, all countries



Note: N = 378

Source: Digitrans team.

The firms receiving the combined FS+TA increased the ability to utilize advanced features of sophisticated digital systems. Among firms that adopted complex enterprise systems, those receiving complementary support were significantly more likely to implement advanced functionalities. In ERP implementations, FS+TA recipients more frequently used sophisticated capabilities like business administration automation (56% vs. 42%) and real-time decision-making (36% vs. 24%) compared to FS-only firms (Figure 17). The pattern was even more pronounced for CRM systems, where 82% of FS+TA firms effectively managed client communications compared to 67% of FS-only firms and just 41% of comparison group firms. These differences demonstrate that technical support helps firms move beyond basic implementation to leverage sophisticated capabilities that generate greater business value.

FIGURE 17 Implementation of Sophisticated Digital System Functionalities

Per group, all countries

		Comparison	Financial Subsidy	Financial Subsidy + Technical Assistance
ERP (N= 278)	Automate business administration tasks	39.7	42.3	**56.2****
		(6.5)	*(8.1)*	*(7.9)*
	Standardize workflows, responsibilities and processes	28.4	34.6	30.6
		(6.3)	*(7.6)*	*(7.5)*
	Generate demand of financial forecasts	23.6	19.0	23.4
		(5.8)	*(-6.4)*	*(6.8)*
	Make resources (staffing, inputs, etc.) decisions using real-time data	27.3	23.8	**36.4****
		(6.1)	*(7.9)*	*(7.3)*
	Any of the above	75.9	73.0	67.1*
		(5.9)	*(7.3)*	*(7.5)*
CRM (N= 111)	Manage communication with clients (leads, after-sale contact, scheduling, payments)	41.2	**67.4****	**82.2*****
		(12.3)	*(13.3)*	*(12.7)*
	Automate sales and clients reports	62.5	40.1*	66.4
		(12.5)	*(14.8)*	*(13.9)*
	Data analysis to identify trends, or predict future occurences	58.8	**29.4****	54.5
		(12.3)	*(15.6)*	*(13.4)*
	Personalize customer experience or marketing campaigns based on data analytics	39.0	**11.2****	41.1
		(13.1)	*(12.7)*	*(14.3)*
	Any of the above	80.0	72.4	**68.4***
		(10.7)	*(10.5)*	*(9.0)*
EPM (N= 79)	Owner	58.8	**65.2***	**74.6****
		(12.3)	*(19.0)*	*(18.1)*
	CEO	52.9	32.7	44.6
		(12.5)	*(19.2)*	*(18.4)*
	Director or Lead staff	46.8	29.9	41.1
		(11.3)	*(16.9)*	*(16.1)*
	Senior staff	23.5	13.7	25.3
		(10.6)	*(13.1)*	*(15.9)*
	External collaborators	94.1	84.2	87.4
		(5.9)	*(14.3)*	*(12.0)*

Share of firms using given functionality: <10% | 10-19% | 20-29% | 30-39% | 40-49% | 50%+

Note: Cells and corresponding colors show the share of firms reporting use of advanced functionalities within adopted ERP and CRM.
*** indicates statistical significance at the 1% level relative to the comparison group.*
** indicates significance at the 10% level.*
Differences are calculated in reference to the control (comparison) group within each functionality.

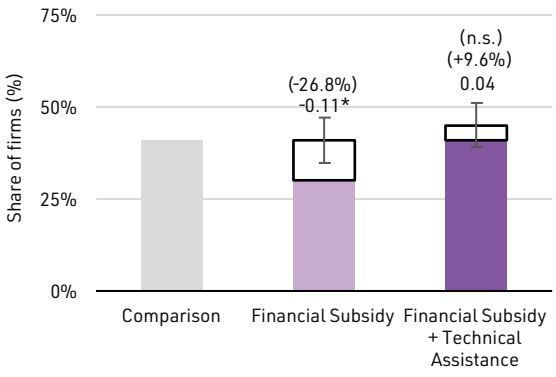
Source: Digitrans team.

5.2.4 ARTIFICIAL INTELLIGENCE

The pilot instruments did not increase the adoption of AI tools but increased its usage. Financial Subsidy recipients showed around one quarter decrease in AI adoption (-11 percentage points), while FS+TA firms showed much smaller 9% decrease (-4 percentage point), though only the effect in FS group was statistically significant (Figure 18). On the country level, results were also mixed, with no statistically significant change in any of the countries (Figure 19). Between baseline and endline measurements, AI adoption increased slightly across all groups, including the comparison group, suggesting that firms are experimenting with AI technologies regardless of support due to the current widespread market momentum around these technologies. This background trend indicates that a more focused intervention specifically targeting AI capabilities might be necessary to influence adoption rates beyond the natural market-driven experimentation already occurring.

FIGURE 18 AI Tool Adoption Impact

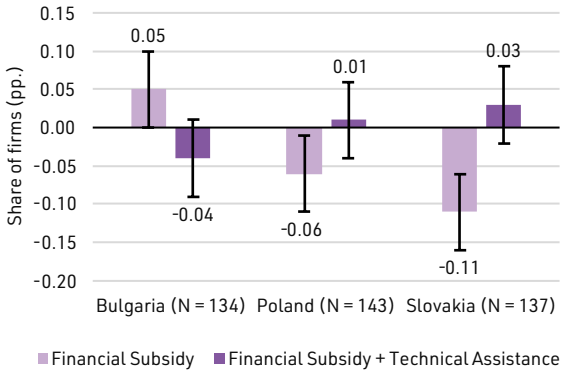
Per group, all countries



*Note: * p < 0.10; N = 418; n.s. = not significant*

FIGURE 19 AI Adoption Impact

Per group, per country



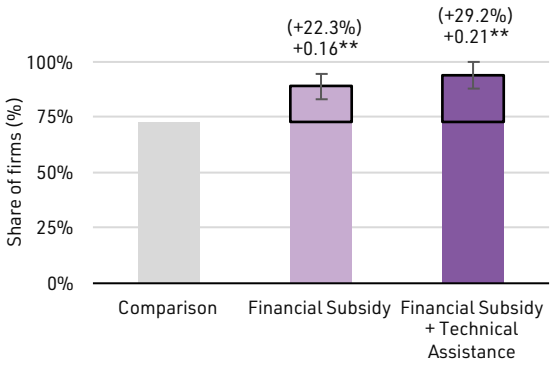
*Note: * p < 0.10; n.s. = not significant*

Source: Digitrans team.

Complementary technical assistance support paired with financial support resulted in higher improvements in AI usage intensity compared to Financial Subsidy alone. FS+TA firms demonstrated a 38.7% increase in AI usage (+26 percentage points, p<0.05), while Financial Subsidy recipients showed a smaller but still significant 22.1% increase (+15 percentage points, p<0.10) compared to the comparison group (Figure 20). The frequency distribution of AI usage reveals further differences, with FS+TA firms using AI tools more regularly than other groups (Figure 21). Approximately 45% of FS+TA firms used AI daily compared to 40% of FS-only firms and 30% of comparison firms. Less than 5% of FS+TA firms reported "rarely" using AI, while 15% of FS-only firms and 30% of comparison firms fell into this low-usage category. The elimination of occasional users in the FS+TA group indicates that the combined technical assistance with financial subsidy helps firms integrate AI tools into regular business operations rather than treating them as experimental technologies.

FIGURE 20 AI Tool Usage Impact

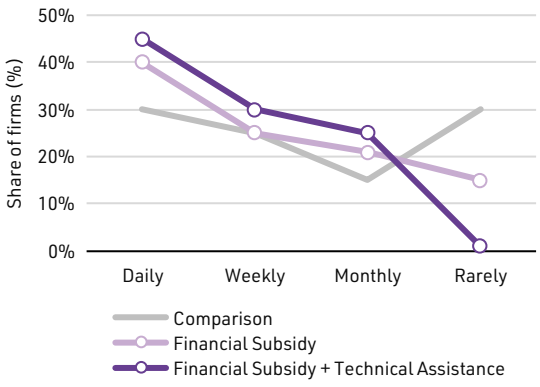
Per group, all countries



*Note: * p < 0.10; N = 84; n.s. = not significant*

FIGURE 21 AI Usage Frequency

Per group, per country



*Note: * p < 0.10; N = 84; n.s. = not significant*

Source: Digitrans team.

The combined approach shifted AI usage toward analytical and personalization applications rather than basic tasks. FS+TA firms were more likely to use AI for data analysis (75%) compared to FS-only (47%) and comparison firms (41%) (Table 3). Similarly, personalization applications were more common in FS+TA firms (31%) than in other groups (24% and 12%). Conversely, FS+TA firms were less likely to use AI for experimentally "checking how AI features work" (25%) compared to comparison firms (59%), suggesting more mature implementation. Basic applications like content generation remained high across all groups (83-92%), though firms from all groups showed relatively high usage of AI for automating repetitive tasks (53% vs. 39-41% in intervention groups).

TABLE 3 AI Functionality Usage by Application Type

Per group, all countries

	Comparison	Financial Subsidy	Financial Subsidy + Technical Assistance
Automating repetitive tasks	53%	39%	41%
	0.514	*0.499*	*0.500*
Generating content or ideas	92%	87%	83%
	0.243	*0.344*	*0.403*
Improving communication	76%	57%	75%
	0.437	*0.507*	*0.447*
Data analysis	41%	47%	**75%****
	0.507	*0.491*	*0.447*
Personalization	12%	24%	**31%***
	0.332	*0.470*	*0.479*
Decision support	35%	39%	38%
	0.493	*0.499*	*0.500*
Checking how AI features work	59%	37%	**25%****
	0.507	*0.470*	*0.447*

Share of firms where a given group of employees are using tool: <10% | 10-19% | 20-29% | 30-39% | 40-49% | 50%+

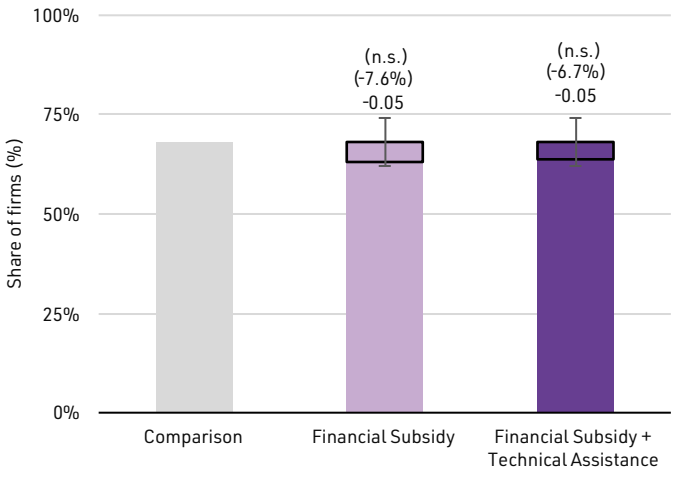
*Note: * p < 0.10; N = 84; n.s. = not significant*
Table reports the percentage of firms using AI for each application type by treatment group.
*** indicates statistical significance at the 1% level relative to the comparison group.*
** indicates significance at the 5% level; * at the 10% level.*
Comparisons reflect differences between FS+TA and FS-only groups relative to the control group across functionalities.

Source: Digitrans team.

The interventions had no significant impact on AI adoption intentions among firms not currently using these technologies. Among non-AI users, approximately 65-70% across all groups expressed intent to adopt AI tools within the next 12 months, with no statistically significant differences between the Financial Subsidy, combined Financial Subsidy and Technical Assistance, and comparison groups (Figure 22). This pattern held across different types of AI applications, from basic content generation to more sophisticated analytical tools. The lack of differentiation in adoption intentions suggests that general digitalization support may not specifically address the unique barriers to AI implementation, such as data readiness, algorithmic understanding, or AI-specific integration challenges. The uniformly high stated intentions across all groups, including the comparison group, likely reflect the broader market momentum around AI technologies rather than intervention-specific effects.

FIGURE 22 Intent to Adopt AI Among Non-Users

Per group, all countries



*Note: * p < 0.10; N = 467; n.s. = not significant*

Source: Digitrans team.

5.3 IMPACT ON FIRM CAPABILITIES AND PRACTICES

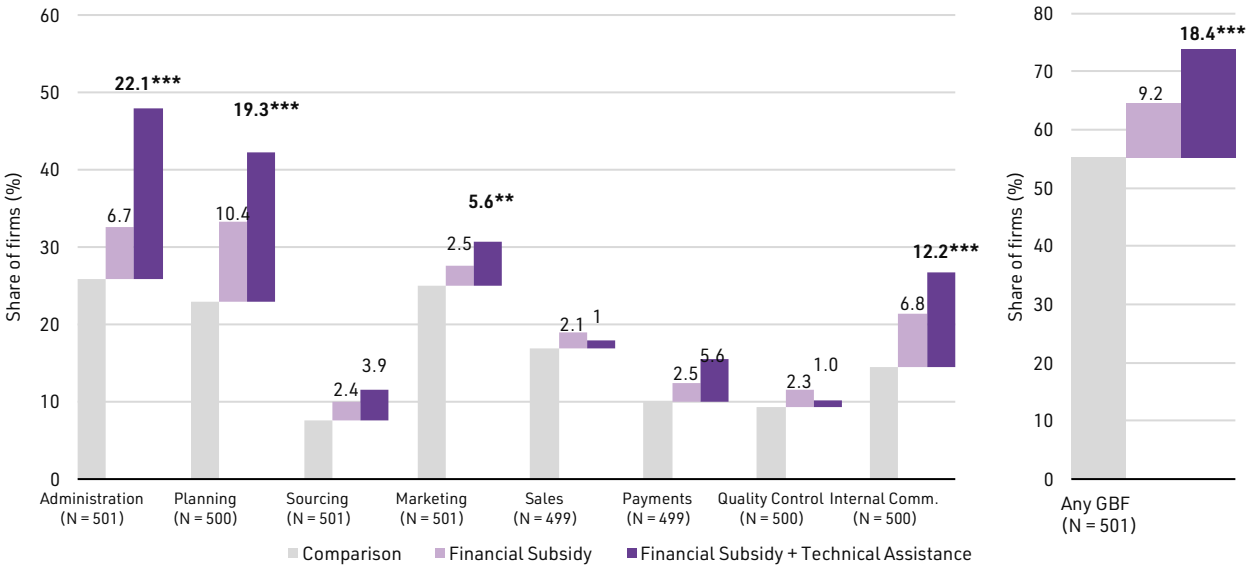
This section examines how the support packages influenced firms' organizational structures, managerial decisions, and broader digital knowledge–factors that determine whether technology investments deliver sustainable performance improvements.

5.3.1 MANAGERIAL PRACTICES

The combined Financial Subsidy and Technical Assistance significantly increased organizational changes across multiple business functions, while the Financial Subsidy alone had no significant impact. When asked at endline about organizational changes implemented since baseline, FS+TA firms were 18.6 percentage points more likely to report changes in Administration and 20.9 percentage points more likely to report changes in Planning functions compared to the comparison group (Figure 23). Marketing also saw substantial organizational restructuring, with the FS+TA group showing a 9.8 percentage point increase compared to the comparison group, while the FS-only group showed only a 4.2 percentage point increase. While more firms in the FS-only group on average introduced changes in all business functions, these differences were not statistically significant in any case. Overall, 70% of FS+TA firms implemented organizational changes in at least one business function, compared to about 50% in the comparison group and a statistically insignificant increase in the FS-only group. These patterns indicate that complementary support is essential for triggering the organizational adjustments needed to fully leverage digital tools in everyday business.

FIGURE 23 Organizational Change Implementation by Business Function

Per group, all countries



*Note: * p < 0.10; N = 467; n.s. = not significant*

Source: Digitrans team.

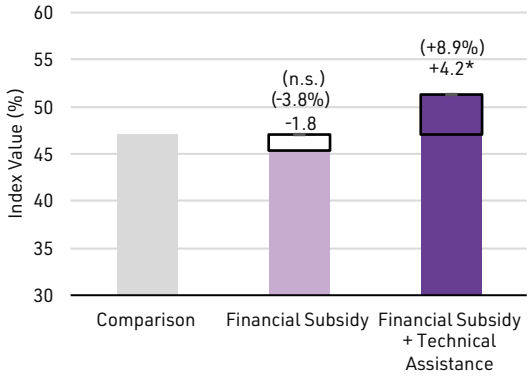
The combined Financial Subsidy and Technical Assistance improved managerial practices, while the Financial Subsidy alone had no effect. FS+TA firms demonstrated a 9.3% improvement in managerial practices (+4.31 percentage points, p<0.05) compared to a non-significant 2.5% decrease (-1.15 percentage points) among FS-only firms (Figure 24). The change was driven by the improvements in change practices (+12.9 percentage points, p<0.05) and planning practices (+6.0 percentage points), while monitoring and people management showed minimal or negative changes across both intervention groups (Figure 25). The improvement in change management among firms receiving complementary results from the intervention implementation, as the group managerial training within the Technical Assistance component specifically aimed to focus on change in practices in the context of digital upgrading in the firm. The training equipped managers with structured approaches to implement organizational changes and introduce digital tools, explaining the stronger organizational adjustments observed in the FS+TA group.[24]

5.3.2 DIGITAL PRACTICES AND KNOWLEDGE

Both interventions improved firms' digital knowledge, with the combined Financial Subsidy and Technical Assistance producing twice as large gains relative to the Financial Subsidy alone. FS+TA firms demonstrated a 25.6% increase in digital knowledge scores (+11.4 percentage points, p<0.01) compared to a 13.5% increase (+6.03 percentage points, p<0.01) among FS-only firms (Figure 26). The improvement patterns across knowledge sub-indices reveal important differences in intervention impacts. Both groups showed similar decreases in basic information knowledge (-5.9 and -5.2 percentage points), but FS+TA firms demonstrated much stronger gains in systems knowledge (+6.0 percentage points versus +1.2 percentage points) and maintenance knowledge (+12.9 percentage points versus +0.4 percentage points) (Figure 27). These results suggest that while both interventions enhanced overall digital knowledge, the combined Financial Subsidy and Technical Assistance specifically improved understanding of more complex, implementation-related aspects rather than basic concepts. This focused knowledge enhancement likely contributed to the more effective technology usage observed in the FS+TA group.

FIGURE 24 Impact on Managerial Practices Index

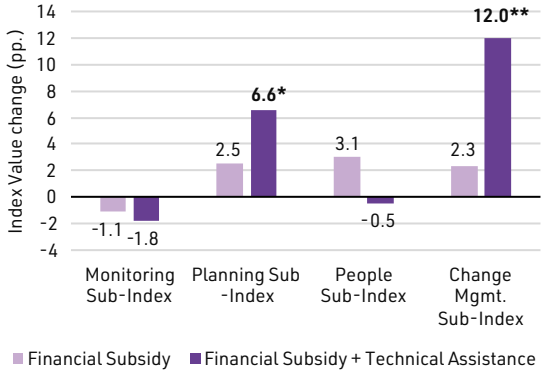
Per group, all countries



*Note: * p < 0.10; N = 342; "n.s." = not statistically significant.*

FIGURE 25 Impact on Managerial Practices Index Components

Per group, per country



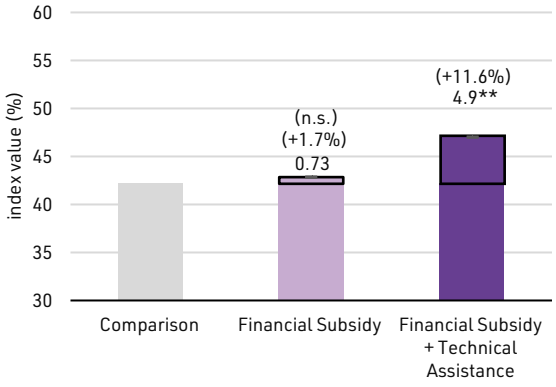
*Note: * p < 0.10; N = 342; "n.s." = not statistically significant.*

Source: Digitrans team.

24. Bloom et al. (2013). *Does Management Matter? Evidence from India*. Quarterly Journal of Economics

FIGURE 26 Impact on Digital Practices Index

Per group, all countries

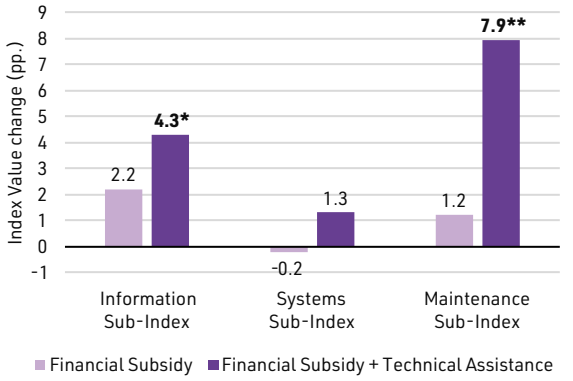


*Note: * p < 0.10; N = 382.*

Source: Digitrans team.

FIGURE 27 Impact on Digital Practices Index Components

Per group, per country

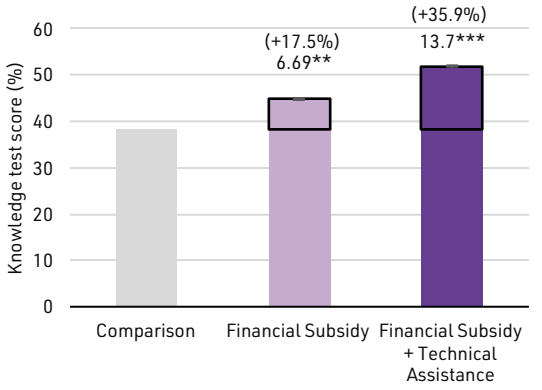


*Note: * p < 0.10; N = 382*

The combined Financial Subsidy and Technical Assistance improved firm owners' and key managers digital knowledge, while the Financial Subsidy alone had minimal effect. Digital knowledge was measured through a 10-question test at endline that assessed investment decisions, cybersecurity practices, and implementation aspects of digital business tools.[25] FS+TA firms demonstrated an 18.2% increase in digital knowledge scores (+7.6 percentage points, p<0.05) compared to a non-significant 2.3% increase (+0.98 percentage points) among FS-only firms (Figure 28). While the complementary support did not include formal training on digital technologies, relevant knowledge was transferred through ongoing interactions with technical advisors. This knowledge differential helps explain the greater adoption and usage intensity observed among FS+TA firms, as understanding how technologies connect to business processes is critical for effective implementation beyond basic adoption.

FIGURE 28 Impact on Digital Knowledge Test Scores

Per group, all countries



*Note: * p < 0.10; N = 501; n.s. = not significant*

Source: Digitrans team.

25. The digital knowledge test consisted of 10 questions: 8 multiple-choice questions with 4 options each and 2 yes/no questions. The multiple-choice questions covered: (1) primary benefit of digital marketing tools; (2) technology designed for collecting and managing customer information; (3) how firms use data analytics in business operations; (4) technology that integrates various aspects of the firm (finance, HR, sourcing) into one system; (5) technologies that do not allow real-time inventory tracking; (6) most important step before introducing new business software; (7) most effective way to determine if a new business process is working; and (8) best way to involve staff when implementing changes. The yes/no questions addressed whether on-premise technologies are more secure than cloud-based applications and whether subscription models require larger upfront investment than traditional software purchases. All questions included a 'don't know' option.".

5.4 HETEROGENEITY ANALYSIS

The impact of interventions varied substantially across different types of firms and owners. Firm and owner characteristics significantly moderated program effects, with patterns of heterogeneity varying across outcome dimensions. The most consistent moderating factors were firm size, initial digitalization level, location, owner's self-assessment of digital skills, and locus of control (Table 4). These systematic differences in treatment effects help identify which types of businesses benefited most from different support packages and provide empirical evidence on the distribution of impacts across the business population.

TABLE 4 Treatment Effect Heterogeneity by Firm and Owner Characteristics

Per group, all countries

	Digital Tool Adoption and Usage			Firm Capabilities				
	Technology Adoption	Technology Usage	AI Usage	Organizational Changes	Managerial Practices	Digital Knowledge	Digital Practices	Notes
Firm Characteristics								
Size								Smaller firms saw larger impact than larger firms.
Initial Level of Digitalization								Firms with low digital sophistication experienced greater impact than more advanced firms.
Location								Firms in capital cities had larger gains than those outside capitals.
Country								Firms in Poland and Slovakia showed stronger effects than those in Bulgaria.
Sector								Service firms had greater impact than manufacturing and construction firms.
Owner Characteristics								
Self-Assessment of Digitalization								Owners who overestimated their digital skills saw more impact than those who estimated accurately.
Locus of Control								Owners with low LOC experienced greater impac than those with high LOC.
Gender								Women-led or co-led businesses saw more impac than male-led firms.
CEO Experience								Owners with 10+ years of experience saw more impact on organization, less on management.

No statistically significant differentiation ($p > 0.10$).
Modest differentiation ($p \leq 0.10$ and effect 0-50 % above/below average).
Medium differentiation ($p \leq 0.05$ or effect 50-100 % above/below average).
Strong differentiation ($P \leq 0.01$ or effect >100 % above/below average).

Source: Digitrans team.

Firm characteristics strongly influenced intervention effectiveness, with consistent patterns across multiple outcome dimensions. Smaller firms (5-9 employees) consistently experienced larger improvements from the combined Financial Subsidy and Technical Assistance than larger firms, particularly in digital tool adoption and usage. Firms with lower baseline digital sophistication saw substantially greater impacts than more digitally advanced peers, especially from the combined support package, suggesting that programs that select the "best firms" are not necessarily those that have the largest impacts.

Geographic differences also emerged, with firms in capital cities experiencing larger gains than those in non-metropolitan areas, and Bulgarian and Slovak firms showing stronger differences than Polish firms. Capital-based firms achieved digital tool usage rates averaging 0.25 points higher on the GBF intensive index than their non-metropolitan counterparts when comparing the combined FS+TA intervention between locations – with no change between regions in the FS-only group. Location-based impact differences are also evident in firm capabilities – capital-located firms showed greater improvements in managerial practices, digital practices, and technological knowledge compared to firms outside capitals. However, there was no statistical difference in the share of firms implementing organizational changed. This differential may result from in-person group trainings being delivered in capitals in Bulgaria and Slovakia, with firms from these locations having slightly higher participation rates and thus greater opportunity to upgrade these practices and knowledge.

Owner characteristics proved equally important in determining intervention effectiveness, with behavioral factors playing a prominent role. Owners who overestimated their digital skills experienced significantly larger improvements than those with more accurate self-assessments, suggesting that overconfidence may create openness to adoption while creating implementation gaps that Technical Assistance effectively addresses. Owners with lower locus of control (believing external factors strongly influence outcomes) showed greater improvements from interventions than those with high internal locus of control. Gender also influenced outcomes, with women-led or co-led businesses demonstrating substantially higher impacts than male-led firms. Interestingly, more experienced business owners (10+ years) saw greater organizational changes but fewer managerial practice improvements, suggesting they leveraged interventions differently than less experienced peers.

The heterogeneity analysis reveals important interaction effects between firm characteristics and intervention components. The combined Financial Subsidy and Technical Assistance proved particularly valuable for firms facing capability barriers (smaller size, lower baseline digitalization, less experienced owners), while financial support alone was sometimes sufficient for firms with stronger initial capabilities. The differential effects across firm segments provide insights into the mechanisms through which interventions influenced business digitalization and highlight the varied responses across the diverse MSME population in the three pilot countries.

ANNEXES

ANNEX **1**

FINANCIAL SUBSIDY COST ELIGIBILITY

GENERAL ELIGIBILITY CRITERIA

Maximum Reimbursement Amount

- Maximum subsidy per firm: EUR 2,000 (or equivalent in local currency)
- Currency exchange: At the rate applicable on the date of purchase
- VAT treatment: VAT-inclusive amounts are eligible if the firm cannot reclaim VAT

Eligible Cost Period

- Pre-approval period: 6-12 weeks from signing Beneficiary Agreement
- Purchase period: Must be completed within the eligible cost period
- Reimbursement processing: Weeks 13-14 following Beneficiary Agreement signature

Eligible Cost Types

Eligible:

- Software subscription fees (monthly, annual, or multi-year)
- Software license purchases (perpetual or term-limited)
- Digital platform access fees
- Cloud-based service subscriptions (SaaS, PaaS)
- Mobile application subscriptions
- Training costs directly related to purchased digital tools (maximum 20% of total subsidy)
- Initial implementation and setup services (maximum 15% of total subsidy)

Not Eligible:

- Hardware purchases (computers, tablets, smartphones, servers)
- Internet connectivity costs
- General IT consulting not directly related to approved tools
- Maintenance and support beyond the initial subscription period
- Internal staff costs
- Costs incurred before pre-approval or after the eligible period

The eligibility of individual costs submitted by respective beneficiaries was verified each time by the Financial Management Specialist during the processing of the Pre-Approval Form.

TECHNOLOGY REQUIREMENTS

Focus on General Business Functions (GBFs)

Eligible digital tools must support one or more of the following General Business Functions:

- Administration - Business management, document management, accounting
- Planning - Resource planning, scheduling, project management
- Sourcing - Supplier management, procurement processes
- Marketing - Customer outreach, brand management, market research
- Sales - Customer relationship management, sales processes
- Payment Methods - Digital payment processing, invoicing
- Quality Control - Process monitoring, compliance management
- Internal Communication - Team collaboration, knowledge management

Sector-Specific Exclusions – Ineligible for support were digital tools and technologies designed exclusively for sector-specific functions and use cases. These included specialized tools such as route planning optimization software for the transportation sector, medical diagnostic systems for healthcare providers or production machinery control systems.

General IT Infrastructure Exclusions – Also ineligible for support were digital tools and technologies that served as general IT or digital infrastructure rather than direct business applications. This category encompassed digital products such as operating systems, server software, network management tools, and cybersecurity infrastructure.

Technology Sophistication Levels

Eligible tools must correspond to Sophistication Levels 2-5 according to the Firm-level Technology Adoption (FAT) Framework:

Level 5 (Most Sophisticated):

- Enterprise Resource Planning (ERP) systems
- Advanced AI and Big Data analytics tools
- Integrated supply chain management systems
- Automated inspection and control systems

Level 4:

- Specialized business software
- Supplier/Customer Relationship Management (SRM/CRM) systems
- E-commerce platforms
- Statistical process control systems

Level 3:

- Mobile applications and digital platforms
- Structured survey and feedback systems
- Online sales platforms
- Process control systems with digital support

Level 2 (Least Sophisticated):

- Standard software applications (Excel, Google Workspace, etc.)
- Basic online communication tools
- Simple digital process support tools
- Basic e-commerce solutions

GROUP-SPECIFIC ELIGIBILITY CRITERIA

Financial Subsidy Only Recipients (Group 1)

Eligible tools: Any digital business tools supporting General Business Functions at Sophistication Levels 2-5

Financial Subsidy + Technical Assistance Recipients (Group 2)

Eligible tools: Digital business tools supporting General Business Functions at Sophistication Levels 2-5 that align with Priority Recommendations from the Integrated Assessment

Additional requirements over Group 1:

- Technological Advisor endorsement required confirming alignment with Priority Recommendation
- Tools must support implementation of identified improvement areas
- Preference for tools that integrate with existing systems
- Must demonstrate potential for organizational impact beyond pure technology adoption

PRE-APPROVAL PROCESS REQUIREMENTS

Required Documentation

For all applicants:

- Pre-approval request form
- Detailed tool description and vendor information
- Cost breakdown including all fees and subscriptions

For Technical Assistance recipients (Group 2) - Additional requirements:

- Technological Advisor confirmation
- Alignment statement with Priority Recommendation
- Business justification for tool selection

Prohibited Uses and Restrictions

Sector-Specific Exclusions

- Tools exclusively designed for sector-specific functions (production machinery control, specialized industry software)

- Regulatory compliance tools required by law
- Tools primarily used for financial sector operations

Vendor and Service Restrictions

- Vendors must provide verifiable business registration, contact information, and references from previous clients
- Pricing must reflect current market rates with no artificial inflation for subsidized purchases
- Technology neutrality – no preference for specific vendors or technologies
- Subscription renewals beyond the program period are not reimbursable

The legitimacy of all proposed purchases and vendors was verified by the Financial Management Specialist during processing of each Pre-Approval Form to ensure protection against fraudulent providers and guarantee reliable service delivery.

QUALITY ASSURANCE AND COMPLIANCE

Verification Requirements

- Purchase documentation (invoices, receipts, contracts)
- Proof of tool activation and initial usage
- Vendor confirmation of service delivery
- Compliance with data protection and cybersecurity standards

Pre-Approval Modifications

- Changes to approved tools require new pre-approval
- Vendor substitutions allowed if maintaining equivalent functionality

Appeal Process

- Written appeals within 10 working days of decision
- Review by independent panel including World Bank representative
- Decision final within 15 working days of appeal submission

ANNEX **2**

TECHNICAL ASSISTANCE METHODOLOGY

OVERVIEW

The Technical Assistance component employed standardized methodologies across all three pilot countries to ensure consistent delivery and enable rigorous impact evaluation. This annex details the specific frameworks, tools, and procedures used to deliver the three core Technical Assistance activities: Integrated Assessment, Group Change Management Training, and Personalized Technological Advisory.

INTEGRATED ASSESSMENT METHODOLOGY

Analytical Framework

The Integrated Assessment utilized the Firm-level Adoption of Technology (FAT) framework, focusing on General Business Functions (GBFs) applicable across all sectors. The assessment evaluated both technological and managerial dimensions through a structured four-step process:

Step 1: Current State Analysis

- Comprehensive inventory of digital tools adopted and used across seven GBFs (administration, planning, sourcing, marketing, sales, payment, quality control)
- Assessment of managerial practices and organizational structures
- Documentation of existing business processes and decision-making processes

Step 2: Opportunity Identification

- Exploration of options for adopting new digital technologies across single or multiple GBFs
- Evaluation of possibilities for increasing usage of already-adopted sophisticated technologies
- Assessment of potential for expanding functionality of existing digital tools
- Analysis of opportunities to extend digital tool usage to new business functions

Step 3: Business Priority Alignment

- Integration of technological opportunities with company's key business priorities
- Assessment of resource constraints and implementation feasibility
- Evaluation of organizational readiness for proposed changes

Step 4: Recommendation Prioritization

- Selection of 3-5 key recommendations offering optimal cost-benefit balance
- Focus on gradual enhancements feasible within three-month timeframe
- Prioritization based on impact potential and implementation feasibility

Delivery Process

Preparatory Call (Week 1)

- 90-minute video conference with business owner and key managers
- Introduction of advisors and program objectives
- Initial business overview and expectation alignment
- Planning of on-site visit structure and participants

On-site Visit (Week 2)

- Half-day joint session (1 hour) with both advisors, owner, and key decision-makers
- Independent advisor sessions (3-4 hours) with different firm representatives
- Technological component focused on digital tool inventory and upgrade opportunities
- Managerial component examining organizational structures and business processes
- Concluding debriefing (30 minutes) with preliminary findings

Report Preparation (Weeks 2-3)

- Joint analysis by Technological and Managerial Advisors
- Standardized 10-page report following prescribed template
- Cost-benefit analysis for each recommendation
- High-level implementation roadmap with resource requirements

Report Presentation (Week 3-4)

- 2-hour presentation meeting with firm owners/decision-makers
- Discussion and validation of findings and recommendations
- Selection of single priority recommendation for Technical Assistance focus
- Designation of focal point for subsequent training and advisory activities

Quality Standards

- Maximum 14-day interval between on-site visit and results presentation
- All interviews conducted with confidentiality assurances and clear objectives
- Standardized data collection protocols across technological and managerial components
- Full documentation requirements of all firm interactions

GROUP CHANGE MANAGEMENT TRAINING METHODOLOGY

Design Thinking Framework

The training program employed Design Thinking methodology as the core framework for change management, structured around the double diamond process with four distinct phases:

Phase 1: Empathy and Understanding

- Territory mapping to identify stakeholders affected by priority recommendations
- Stakeholder interviews to understand current processes and pain points
- Journey mapping to document existing workflows and identify improvement opportunities

Phase 2: Insights and Ideas

- Empathy mapping to synthesize stakeholder interview findings
- Concept development to transform insights into actionable solutions
- Ideation workshops to generate multiple implementation approaches

Phase 3: Prototyping and Testing

- Service blueprint development incorporating all stakeholder roles
- Low fidelity prototyping of proposed organizational changes
- Feedback collection and iterative improvement cycles

Phase 4: Planning and Implementation

- Comprehensive implementation plan development with timelines and resource allocation
- Risk identification and mitigation strategy formulation
- Communication plan preparation for organizational change activation

Session Structure

Session 1: Establishing Change Management Foundations (4 hours)

- Design Thinking methodology introduction and adaptation for MSME context
- Visual mapping of priority recommendations across participating firms
- Territory mapping exercise to identify key stakeholders
- Stakeholder interview preparation and empathy research introduction

Session 2: Deepening Understanding and Generating Ideas (4 hours)

- Stakeholder interview insights sharing and empathy mapping
- Current state journey mapping with pain point identification
- Concept development workshops and initial ideation rounds
- Introduction to feedback collection methodologies

Session 3: Refining Solutions and Blueprinting (4 hours)

- Prototype refinement based on peer and stakeholder feedback
- Service blueprint development incorporating multiple stakeholder perspectives
- Systematic review of implementation feasibility and resource requirements
- Leadership and communication strategy development

Session 4: Planning and Activation (4 hours)

- Implementation plan finalization with clear timelines and responsibilities
- Risk assessment and mitigation strategy development

- Presentation preparation and communication plan activation
- Integration of Design Thinking into broader organizational change processes

Delivery Standards

The training program followed standardized operational parameters to ensure consistent quality and engagement across all cohorts and countries.

- Maximum 15 participants per cohort to ensure effective peer learning
- In-person delivery prioritized with flexible online alternatives when necessary
- Structured homework assignments between sessions to maintain engagement
- Individual and group activities supervised by experienced managerial advisors

PERSONALIZED TECHNOLOGICAL ADVISORY METHODOLOGY

Six-Step Implementation Framework

Step 1: Strategic Technology Evaluation

- Detailed assessment of available digital tools aligned with priority recommendations
- Utilization of authoritative market reports and benchmarking data
- Structured evaluation discussions to validate technology choices
- Delivery of comprehensive technology compendium as reference material

Step 2: Usage of Existing Sophisticated Technologies

- Analysis of current technology utilization gaps and optimization opportunities
- Capacity building on advanced functions of already-adopted systems
- Integration recommendations for existing technology portfolio
- Best practices guidance for maximizing current technology investments

Step 3: Priority Recommendation Implementation

- Customized implementation strategy development based on firm-specific needs
- Integration planning for new technologies with existing systems
- Step-by-step implementation roadmap with optimization objectives
- Implementation checklists and best practice documentation

Step 4: Vendor Analysis and Engagement

- Comprehensive vendor identification and comparative analysis
- Support for vendor negotiations and shortlisting processes
- Vendor comparison sheets with weighted selection criteria
- Local technology integrator database and evaluation framework

Step 5: Preparation for Contracting

- Contract negotiation support and best practice guidance
- Service-level agreement template provision and customization
- Procurement process facilitation and compliance assurance
- Legal and technical documentation review support

Step 6: Operational Transition

- Pre-launch preparation assistance including data migration planning
- Staff training coordination and support mechanism establishment
- Organizational process amendment guidance
- Post-implementation checklist and success measurement framework

Delivery Approach

- 16 hours of advisor time per firm over four weeks
- Minimum one meeting per week accommodating firm availability
- Balance of face-to-face and online interactions based on firm preferences
- Customized time allocation across six steps based on specific firm needs
- Preliminary timeline provided with flexibility for firm-specific adjustments

Code of Conduct

- Strict impartiality requirements with no vendor / technology preferences
- Conflict of interest reporting and mitigation protocols
- Professional integrity standards with beneficiary firm interest prioritization through mandatory signing of Non-Disclosure Agreements and adherence to written code of conduct covering data protection, confidentiality, and professionality standards.

All trainers and advisors completed mandatory face-to-face certification training covering methodology delivery, ethical standards, and confidentiality requirements, followed by written assessment and practical evaluation with test scenarios before authorization to work with beneficiary firms.

STANDARDIZATION AND QUALITY ASSURANCE

Cross-Country Consistency

- Identical methodologies implemented across Bulgaria, Poland, and Slovakia
- Standardized materials translated and localized for each country context
- Regular coordination meetings between implementing partners
- World Bank oversight ensuring methodology adherence

Quality Control Mechanisms

- Spot-checks on minimum 15% of individual calls and sessions
- Random back-checks with 10% of participants to confirm participation
- Interaction logs documenting all advisor-firm communications
- Performance analytics tracking service utilization and satisfaction rates
- Ongoing feedback surveys administered at multiple intervention stages

Adaptive Management

- Real-time feedback collection enabling mid-course corrections
- Structured lessons learned integration between cohorts
- Implementing partner coordination for best practice sharing

RESOURCE TEMPLATES AND MATERIALS

Integrated Assessment Resources:

- Company Information template for advisor preparation prior to first interaction with representatives of beneficiary firms
- Standardized interview questionnaires for technological and managerial components
- Report template with required sections and cost-benefit analysis framework
- Presentation materials for results communication

Training Resources

- Detailed training agenda with session-by-session breakdown
- Comprehensive presentation materials adapted to local contexts
- Design Thinking template library for practical exercises
- Case study examples from relevant MSME contexts

Advisory Resources

- Technology evaluation matrices and vendor comparison templates
- Implementation checklists for different technology categories
- Contract template library and service-level agreement examples
- Best practice guides for technology integration and change management

This standardized methodology enabled consistent, high-quality Technical Assistance delivery across all pilot countries while maintaining flexibility to address individual firm needs and local market conditions.

ANNEX **3**

PROCUREMENT PROCESS AND PARTNER SELECTION CRITERIA

PROCUREMENT OVERVIEW AND TIMELINE

The partner procurement process was conducted in parallel across Bulgaria, Poland, and Slovakia between February and May 2024, following standardized procedures to ensure comparability while allowing for country-specific adaptations. The procurement strategy aimed to identify partners capable of delivering high-quality services while maintaining the experimental integrity required for rigorous impact evaluation.

The procurement timeline was structured as follows:

- February 2024: Launch of Expressions of Interest (EoI) calls in all three countries
- March 2024: EoI submission deadline and initial evaluation
- April 2024: Request for Proposals (RFP) to shortlisted candidates
- May 2024: Final selection and contract negotiations
- June 2024: Contract signing and partner onboarding

The partner procurement process was organized and conducted entirely by the World Bank team in accordance with the World Bank Corporate Procurement Policy, ensuring consistency and objectivity across all three countries. The process was managed internally by World Bank staff without engagement of external evaluators or representatives from the partner agencies participating in the Digitrans program (BSMEPA, PARP, and MIRRI). This approach maintained independence in partner selection while ensuring alignment with World Bank standards for competitive procurement.

QUESTIONS AND ANSWERS PROCESS

To ensure transparency and equal access to information, applicants were provided with opportunities to seek clarifications through a structured Q&A process. During both the EoI and RFP stages, interested firms could submit questions through a public forum accessible to all potential applicants on the procurement portal.

The World Bank team prepared responses to all submitted questions, compiling them into joint document that was published and made available to all applicants between the EoI and full offer stages. The Q&A documents covered technical requirements, evaluation criteria, contractual terms, and operational expectations, providing applicants with the information necessary to prepare competitive proposals.

IMPLEMENTING PARTNER SELECTION CRITERIA

Organizational Requirements

Minimum Experience Thresholds:

- At least 3 years of documented experience providing business support services to MSMEs in the respective country
- Demonstrated track record in digital transformation or firm competitiveness enhancement projects
- Minimum 5 completed projects with small business clients in the past 3 years
- Annual revenue of at least EUR 200,000 to demonstrate organizational stability

Sectoral Expertise:

- Experience working across multiple sectors rather than specialization in single industries
- Understanding of general business functions applicable to diverse MSME portfolios
- Previous engagement with firms in the 5-35 employee range
- Knowledge of local business ecosystem and regulatory environment

Network and Reputation:

- Active participation in small business support networks or associations
- Positive references from at least 5 previous MSME clients
- Recognition by national business development institutions
- Demonstrated ability to reach firms outside traditional support program participants

Core Team Requirements - Each implementing partner was required to demonstrate access to qualified personnel meeting specific role requirements. All team members must be fluent in the local language, while key staff personnel must also demonstrate fluency in English. CVs of all key staff personnel were required to be included in submitted offers, and replacement of key personnel during the project duration required approval from the World Bank team.

Project Coordinator:

- Master's degree in Economics, Finance, Business Administration, or related field
- 5-10 years' experience in business advisory projects or management consulting
- Strong project management and organizational skills
- Fluency in both English and local language
- Experience with EU-funded projects preferred

Technological Advisor:

- Bachelor's degree in Computer Science, Information Technology, or related field
- Minimum 5 years' experience in technological consulting and digital transformation
- Familiarity with digital technology platforms relevant to small businesses
- Ability to convey complex technical concepts to non-technical audiences
- Experience with technology needs assessments

Managerial Advisor:

- Master's degree in Management, Business Administration, or related field
- Minimum 5 years' experience in management consulting or organizational development
- Certified experience in change management methodologies
- Proven track record in group training and facilitation
- Experience with managerial practice assessments

Financial Management Specialist:

- Bachelor's degree in Finance, Accounting, or Business Administration
- Minimum 3 years' experience in financial management of public support programs
- Knowledge of subsidy administration and compliance procedures
- Experience with digital financial management systems
- Understanding of EU financial regulations and reporting requirements

Outreach Specialist:

- Bachelor's degree in Marketing, Communications, or related field
- Minimum 3 years' experience in outreach and recruitment for business programs
- Demonstrated success in multi-channel marketing campaigns
- Experience with digital marketing tools and social media platforms
- Track record of achieving recruitment targets within specified timeframes

Infrastructure Requirements:

- Physical office space suitable for group training sessions (minimum 20 participants)
- IT infrastructure capable of supporting video conferencing and digital collaboration
- Project management systems for tracking multiple concurrent firm engagements
- Data security measures compliant with GDPR and national data protection laws
- Financial management systems capable of processing subsidy payments

Geographic Coverage:

- Ability to provide services across designated regions within each country
- Travel capabilities for on-site firm visits
- Network of local contacts or partners in secondary cities
- Understanding of regional business variations within the country

DATA COLLECTION PARTNER SELECTION CRITERIA

Survey Experience:

- Minimum 5 years' experience conducting business surveys
- Portfolio of at least 10 large-scale survey projects (500+ respondents)
- Experience with video-call interview methodologies
- Demonstrated ability to achieve response rates above 70% in business surveys
- Experience with panel surveys and longitudinal data collection

Quality Assurance Systems:

- Established protocols for enumerator training and supervision
- Real-time data quality monitoring capabilities
- Standardized interview recording and review processes
- Experience with Computer-Assisted Telephone Interviewing (CATI) systems
- Ability to conduct surveys in multiple languages if required

Technical Infrastructure:

- Secure data storage and transmission systems
- Video conferencing capabilities for remote interviews
- Real-time survey monitoring and reporting dashboards
- Integration capabilities with international data standards
- Backup systems and disaster recovery protocols

Staff Requirements:

- Pool of at least 15 ready to be trained enumerators available during data collection periods
- Supervisory staff with experience in business survey coordination
- Training capabilities for specialized business interview techniques
- Quality control mechanisms for ongoing performance monitoring
- Flexible staffing to accommodate varying interview schedules

EVALUATION METHODOLOGY

The World Bank evaluation team utilized a weighted scoring system addressing two primary assessment dimensions – quality of proposal and price. Each offer was independently reviewed and scored by three World Bank staff evaluators, with final scores calculated as the average of the three independent assessments to ensure objectivity and consistency in the evaluation process.

Quality (70%):

- Technical competence and demonstrated expertise in required service areas
- Team qualifications, credentials, and experience of proposed personnel
- Organizational capacity including financial stability and infrastructure
- Quality of proposed methodologies and innovation in solutions
- Geographic coverage and operational reach

Price (30%):

- Cost-effectiveness and value proposition relative to proposed budget
- Cost structure transparency and justification
- Payment schedule feasibility

All scoring decisions for each firm were documented with justifications and recorded in the World Bank procurement system to ensure transparency and auditability. Participating firms were informed of their right to request feedback on their proposals and were provided with clarifications and explanations of evaluation outcomes upon request.

ANNEX **4**

OUTREACH MATERIALS AND PILOT DESCRIPTIONS

This annex presents the core outreach materials and program descriptions used across the three pilot countries, demonstrating how the pilot instrument was communicated to potential beneficiaries. The materials showcase consistent messaging approach adapted to local contexts while maintaining unified value propositions focused on business benefits rather than technical features.

CORE MESSAGING

Universal Value Proposition – All countries employed a standardized value proposition structure emphasizing tangible business benefits:

Primary Message: "Transform your business through digitalization with expert support and financial assistance"

Supporting Pillars:

- Simple Process: 8-10 minute application, minimal bureaucracy
- Fast Implementation: Results within 30 days of application
- Professional Quality: Expert support throughout the journey
- Comprehensive Package: Combined financial and advisory support

Business-Centric Language Strategy – The outreach materials consistently avoided technical jargon in favor of business-focused terminology:

- Instead of: "Digital transformation program"
- Used: "Increase your profits and market reach"
- Instead of: "Technology adoption support"
- Used: "Tools to reduce costs and increase revenue"
- Instead of: "Capability building intervention"
- Used: "Practical training to overcome barriers"

PROGRAM DESCRIPTION

Value Proposition Structure – Total Support Package: Equivalent of €15,000 in comprehensive assistance

Key Messaging Elements: "DIGITRANS is a grant program supporting the use of digital technologies in your business. Managed by the World Bank, funded by the European Commission, tailored for micro and small enterprises."

Benefits Framework:

- Financial grant for software licenses and digital tools
- Digital assessment comparing digitalization level to industry peers
- Digital transformation plan with actionable recommendations
- Individual consulting for detailed implementation steps
- Professional training to overcome organizational barriers

Support Package Description

The materials emphasized a four-step progression:

1. Digital Assessment: "Find out where you stand compared to competitors"
2. Transformation Plan: "On-site visit to analyze your processes and identify automation opportunities"
3. Individual Consulting: "Detailed roadmap with priorities for maximum time and cost savings"
4. Professional Training: "Overcome organizational barriers preventing effective digital tool usage"

Localized Messaging Variations

- Partnership-Focused Approach: "We provide unbiased support in implementing your strategic growth and competitiveness plans. Our approach is oriented toward close collaboration, focused on building strong relationships with you."
- Solution-Oriented Approach: "Join us to accelerate your business's digital transformation and achieve competitive advantage. Companies meeting the criteria can receive comprehensive support including financial assistance and expert guidance."
- Innovation-Focused Approach: "Enter the digital era and bring your company into the age of digitalization. Increase competitiveness and efficiency of your business through proven methodologies."

ELIGIBILITY COMMUNICATION

Standardized Eligibility Framework

- All countries used consistent eligibility presentation focusing on inclusivity:
- Geographic Scope: National or regional coverage as appropriate
- Employment: 5-35 employees
- Business History: Minimum 12 months operational
- Legal Forms: All business structures accepted

Accessibility Messaging
"The program is designed for companies with fundamental level of digitalization"

Rather than providing technical definitions, materials used relatable examples:

- Companies using primarily manual processes
- Businesses relying on paper-based documentation
- Firms using basic tools like spreadsheet software for most functions
- Organizations wanting to "automate routine manual tasks"

Each country's website included detailed pages and documents specifying firm eligibility criteria, cost structures, and administrative requirements to ensure full compliance with national and EU regulatory frameworks. While these comprehensive specifications were necessary for legal and administrative purposes, the pilot teams made deliberate efforts to present this information in user-friendly formats. Key eligibility requirements and program benefits were summarized in accessible language on main pages, with detailed regulatory documentation available through clearly marked links for complete information.

APPLICATION PROCESS POSITIONING

Simplicity Emphasis – All countries consistently positioned the application as remarkably simple:

- "Application takes 8-10 minutes, no complicated applications or projects needed"
- "Filling out the application takes about 8 minutes, no project plans, budgets or other documents required"
- "Takes about 8 minutes to complete, form contains general company information and digital tools you use"

Speed and Transparency – Timeline commitments were clearly communicated:

- Application Processing: Contact within days of submission
- Decision Timeline: Results within 30 days of application
- Program Duration: 6 months total from application to completion

Professional Quality Assurance – "At every step of the program, professionals will communicate with you: from processing your application, through in-depth assessment of your digitalization level, individual digitalization consulting, to specialized training."

Program websites featured profiles of key advisors and trainers, including professional photographs, names, backgrounds, and relevant experience in digitalization and small business support. This transparency in team qualifications helped build credibility and trust with potential applicants by demonstrating they would work with qualified professionals.

PARTNERSHIP AND CREDIBILITY MESSAGING

Institutional Authority – All materials prominently featured the partnership structure:

- World Bank: International development expertise
- European Commission: EU policy alignment and funding
- National Partners: Local market knowledge and implementation

Trust-Building Elements

- Official government agency endorsement
- International best practices methodology
- Professional consultant team profiles
- Clear data protection and confidentiality commitments

Materials emphasized three core characteristics:

Partnership Approach: "We take time to understand your business, your industry, and your competition, to ensure that the digital strategies we develop are not only aligned with your goals, but also give you a competitive advantage."

Tailored Support: "We understand that there is no one-size-fits-all solution in today's business world. Your company is unique, as are its challenges and goals. That's why we focus on providing personalized support."

Comprehensive Solutions: "Our goal is to provide holistic digital strategy that not only meets your current needs, but also scales with your business as it grows."

Urgency Without Pressure - "Don't hesitate and fill out your application now! Applications will be evaluated continuously - the sooner you submit your application, the higher your chance of receiving support."

Multiple Contact Channels - Each implementation provided:

- Dedicated email addresses for inquiries
- Phone numbers for direct consultation
- Online application forms with technical support
- Clear next-steps communication

The outreach materials from this pilot suggest that consistent messaging approach can be adapted across different contexts while maintaining core value propositions. The pilot's recruitment success–achieving target application numbers within compressed timeframes across all three countries–indicates that the emphasis on tangible business benefits, simplified process communication, and institutional credibility resonated with the target audience. These findings may offer useful insights for future digitalization support programs, particularly regarding the effectiveness of business-focused language over technical terminology and the importance of clear, concrete value propositions in engaging small business owners.

ANNEX **5**

STRUCTURE AND CONTENT OF THE MEASUREMENT SURVEY

SURVEY OVERVIEW

The Digitrans measurement survey was a 60-minute structured questionnaire administered via video call by trained enumerators. The survey was conducted in two rounds: baseline (before intervention) and endline (3 months post-intervention). The instrument was developed based on the FAT framework, World Management Survey methodology, and prior Digitrans diagnostic surveys.

SURVEY DEVELOPMENT AND ADMINISTRATION

- Foundation: Built on 2023 Digitrans surveys, 2022 Polish FAT survey, and internationally validated instruments (WMS, CIS).
- Delivery Method: Video calls with trained enumerators across all three countries, with standardized protocols and real-time quality monitoring.
- Training: 16-hour enumerator training including program context, methodology, practice interviews, and certification testing.

BASELINE VS. ENDLINE SURVEY COMPARISON

Section	Baseline	Endline
1. Firm Characteristics	5 minutes – full of application data, employment details, business structure	3 minutes – Updates only on changes in employment and structure
2. Technology Adoption & Use	20 minutes – Complete extended FAT framework across 8 GBFs, AI adoption and usage	25 minutes – As in baseline with additional questions on changes implemented over last 6 months per GBF
3. Firm Performance	8 minutes – Revenue, growth, productivity, assets	10 minutes – As in baseline plus additional growth questions
4. Digital Practices	10 minutes – Systems, integration, information subindices, recent investment analysis	10 minutes – The same as in baseline
5. Managerial Practices	12 minutes – Monitoring, People, Change Management, Planning, Risk analysis	12 minutes – The same as in baseline
6. Financial Preferences	5 minutes – Time preference, investment scenarios, risk assessment	Not included
7. Business Perceptions	7 minutes – Locus of control, self-efficacy, growth orientation	Not included
8. Recent Digital Investments	Not included	
9. Digital Knowledge Test	Not included	






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